
CAMBRIDGE INTERNATIONAL MATHEMATICS**0607/43**

Paper 4 (Extended)

May/June 2017

MARK SCHEME

Maximum Mark: 120

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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Cambridge is publishing the mark schemes for the May/June 2017 series for most Cambridge IGCSE[®], Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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This document consists of **7** printed pages.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

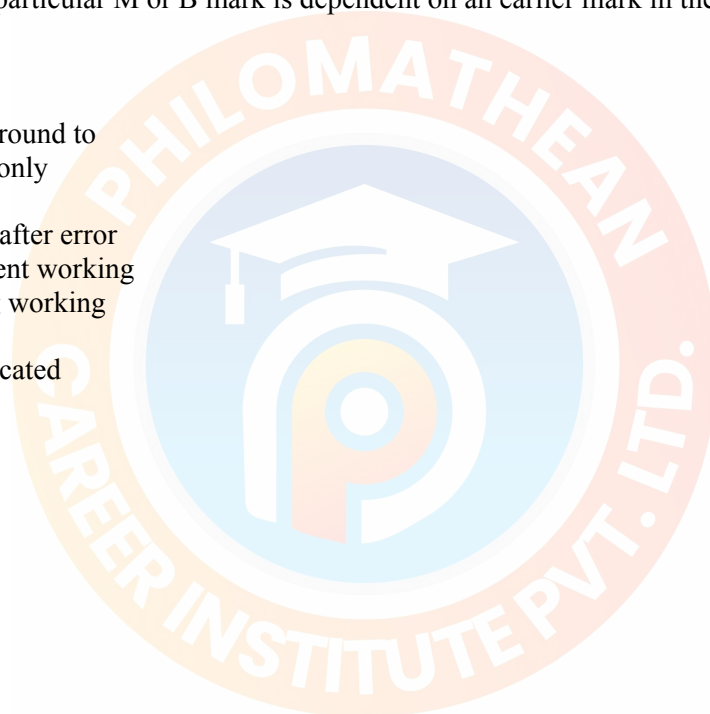
Types of mark

- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more ‘method’ steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation ‘**dep**’ is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfwf	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

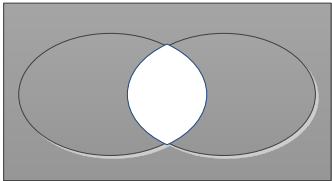


Question	Answer	Marks	Part Marks
1(a)	Image at (0, 5), (3, 5), (3, 3)	2	SC1 for translation $\begin{pmatrix} -2 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 7 \end{pmatrix}$
1(b)(i)	Image at (2, 2), (5, 2), (5, 4)	1	
1(b)(ii)	Image at (−4, −2), (−7, −2), (−7, −4)	1	
1(b)(iii)	Rotation 180 [centre] (−1, 0)	3	B1 for each
1(c)	Stretch [factor]2 x-axis oe invariant	3	B1 for each
2(a)(i)	44	2	M1 for [angle BAC or DEC =] $180 - 2 \times 68$, soi by angle $CDE = 44$ or M1 for angle $BAC = \text{their angle } CDE$
2(a)(ii)	isosceles	1	
2(b)	162	3	M2 for $180 - \frac{360}{20}$ or $\frac{180 \times (20 - 2)}{20}$ or M1 for $\frac{360}{20}$ or $180 \times (20 - 2)$
2(c)(i)	Angle sum of triangle oe	1	
2(c)(ii)(a)	similar	1	
2(c)(ii)(b)	5.4	2	M1 for $\frac{5}{3} = \frac{9}{QR}$ oe
3(a)(i)	6.5	1	
3(a)(ii)	4.5	1	
3(a)(iii)	3	1	
3(b)(i)	Positive	1	
3(b)(ii)	13	1	
3(b)(iii)	15.5	1	
3(b)(iv)	$7.32t - 55.3$	2	(7.322 to 7.323) $t - (55.25 \dots)$ B1 for $7.32t + k$ or $kt - 55.3$ or SC1 for $7.3t - 55$
3(b)(v)	Correct line (positive gradient and not below the x-axis)	2	B1 for positive gradient

Question	Answer	Marks	Part Marks
4(a)(i)	5 : 4	2	B1 for any other correct ratio
4(a)(ii)	41.68	2	M1 for $0.16 \times 260.5[0]$ oe
4(a)(iii)	12.5[0]	3	M2 for $11.25 \div 0.9$ oe or M1 for recognising 11.25 as 90%
4(a)(iv)	300 nfw	3	M2 for $\frac{200 \times 2 \times 25}{100} + 200$ oe or M1 for $\frac{200 \times 2 \times 25}{100}$ oe (implied by 100 nfw)
4(a)(v)	311.72	3	M2 for 190×1.02^{25} oe or M1 for 190×1.02^n oe where $n > 1$
4(b)	17	3	B2 for 16.5 or 16.52 to 16.53 or M2 for $\frac{\log\left(\frac{300}{120}\right)}{\log 1.057}$ or appropriate sketch or $120 \times 1.057^n = 300$ and at least 2 trials which reach from 250 to 350 or M1 for $120 \times 1.057^n [= 300]$
5(a)	804 or 804.2 to 804.4	3	M1 for $\frac{1}{3}\pi \times 8^2 \times 16$ M1 for $\frac{4}{3}\pi \times 4^3$
5(b)	450 or 449.5 to 449.6...	3	M2 for $\pi \times 8 \times \sqrt{8^2 + 16^2}$ or M1 for $\sqrt{8^2 + 16^2}$ or $\pi \times 8 \times \text{their } l$

Question	Answer	Marks	Part Marks
5(c)	8.94 or 8.944...	4	P is point of contact between slant edge and circle. B2 for $PV = 8$ nfw or M1 for $\frac{8}{4} = \frac{16}{PV}$ oe M1 for $OV^2 = 4^2 + PV^2$ OR B2 for $l = \sqrt{320}$ oe or M1 for $l^2 = 8^2 + 16^2$ M1 for $\frac{8}{4} = \frac{l}{OV}$ soi OR x is semi-vertical angle of cone M1 for $\tan x = \frac{8}{16}$ oe M2 for $\frac{4}{\sin x}$ or M1 for $\frac{4}{OV} = \sin x$
6(a)	Correct sketch 	2	B1 for correct shape
6(b)	(2.17, 0.488) or (2.171..., 0.4877...)	2	B1 for each
6(c)	$0.488 \leq f(x) \leq 1.51$ or $0.4877... \leq f(x) \leq 1.505...$	2	FT their 0.488 B1 for $0.488 \leq f(x)$ oe or $f(x) \leq 1.51$ oe
6(d)	0.502 or 0.5015... 5.83 or 5.827...	2	B1 for each
6(e)	$0.502 < x < 5.83$ or $0.5015... < x < 5.827...$	1	FT their (d)
6(f)(i)	15.[0] or 15.00... 25.[0] or 25.00... 35.[0] or 35.00...	1	
6(f)(ii)	[an] asymptote oe	1	

Question	Answer	Marks	Part Marks
7(a)	9.77 or 9.766...	3	M2 for $\frac{8}{\cos 35}$ oe or M1 for $\cos 35 = \frac{8}{AB}$ oe
7(b)	60.6 or 60.61...	3	M2 for $\frac{6^2 + 9^2 - 8^2}{2 \times 6 \times 9}$ or M1 for $8^2 = 6^2 + 9^2 - 2 \times 6 \times 9 \cos C$
8(a)	10	1	
8(b)	4	2	M1 for $[h(1) =] \frac{1}{2}$ or for $[gh(x) =] 3 + 2\left(\frac{1}{x+1}\right)$
8(c)	$5x^2 + 12x + 11$	3	M1 for $(3 + 2x)^2 + 1 + x^2 + 1$ B1 for $9 + 6x + 6x + 4x^2$ or better for $(3 + 2x)^2$
8(d)	$\frac{1}{x} - 1$ or $\frac{1-x}{x}$ oe final answer	3	M1 correct first step M1 correct second step
8(e)(i)	-1	2	M1 for $3 + 2x = 1$
8(e)(ii)	5	1	
9(a)	15, 7, 12 correctly placed	2	B1 for two correctly placed or M1 for $41 - (40 - 6)$ seen oe or correct equation
9(b)(i)	7	1	FT <i>their</i> Venn diagram
9(b)(ii)	28	1	FT <i>their</i> Venn diagram
9(c)	15	1	FT <i>their</i> Venn diagram
9(d)	$\frac{462}{1560}$ oe	2	M1 for $\frac{22}{40} \times \frac{21}{39}$
9(e)(i)	$\frac{7}{19}$	1	FT <i>their</i> Venn diagram
9(e)(ii)	$\frac{168}{342}$ oe	3	M2 for $\frac{\text{their } 7}{19} \times \frac{\text{their } 12}{18} + \frac{\text{their } 12}{19} \times \frac{\text{their } 7}{18}$ oe or M1 for one of these products
9(f)	8	3	M2 for $\frac{\text{their } 7 + n}{40 + n} = \frac{5}{16}$ oe or M1 for at least two trials

Question	Answer	Marks	Part Marks
9(g)		1	
10(a)(i)	3.0875	2	M1 for 2.75, 3.125, 3.5 soi
10(a)(ii)	Correct histogram	3	B1 correct widths B1 for two correct heights
10(b)(i)	$\frac{200}{x} - \frac{200}{x+10} = \frac{20}{60}$ oe	B2	B1 for $\frac{200}{x}$ or $\frac{200}{x+10}$
	$60 \times 200(x+10) - 60 \times 200x = 20x(x+10)$ oe	M1	i.e. correctly clearing fractions or all over common denominator
	$x^2 + 10x - 6000 = 0$	A1	completion with at least one interim line and without any errors or omissions
10(b)(ii)	2 h 45 min	4	B2 for 72.6 or 72.62... or M1 for correct use of formula or correct sketch M1 for $200 \div \text{their positive } x$, implied by 2.75.....
11(a)(i)	$-a + b$ oe	1	
11(a)(ii)	$-\frac{1}{4}a + \frac{1}{4}b$ oe	1	FT <i>their</i> (i)
11(a)(iii)	$\frac{3}{4}a + \frac{1}{4}b$ oe	2	B1 for correct unsimplified answer or a correct route
11(b)	(6.5, 1.5)	3	FT <i>their</i> (a)(iii) B2 for $\begin{pmatrix} 6.5 \\ 1.5 \end{pmatrix}$ or M1 for $\frac{3}{4} \times \begin{pmatrix} 8 \\ 0 \end{pmatrix} + \frac{1}{4} \times \begin{pmatrix} 2 \\ 6 \end{pmatrix}$ OR B2 for (5, 3) at <i>M</i> or $[\overrightarrow{OM} =] \begin{pmatrix} 5 \\ 3 \end{pmatrix}$ or B1 for (<i>k</i> , 3) or (5, <i>k</i>) at <i>M</i> or $[\overrightarrow{OM} =] \begin{pmatrix} k \\ 3 \end{pmatrix}$ or $\begin{pmatrix} 5 \\ k \end{pmatrix}$

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GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

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- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

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- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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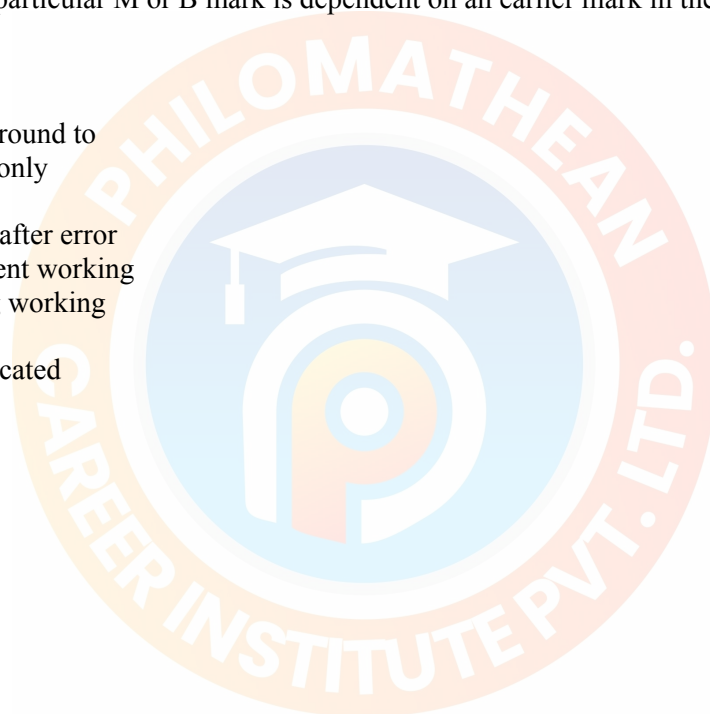
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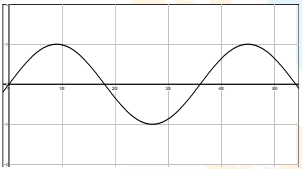
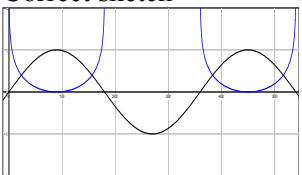
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nfw	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied



Question	Answer	Marks	Partial Marks
1(a)(i)	Translation $\begin{pmatrix} 1 \\ 7 \end{pmatrix}$	2	B1 for each
1(a)(ii)	Reflection $x = -\frac{1}{2}$ oe	2	B1 for each
1(b)	Triangle drawn at (3, 2), (3, 4), (–3, 2)	2	B1 for enlargement factor 2 with wrong centre, or correct centre with wrong positive factor (not 1)
1(c)	Triangle drawn at (2, 1), (8, 1), (8, 2)	2	B1 for stretch factor 2 with x -axis invariant. or stretch factor 2 translated horizontally
2(a)	$\frac{8000}{5+7+8} \times 8 [= 3200]$	M2	M1 for $8000 \div (5 + 7 + 8)$ If 0 scored SC1 for $\frac{3200}{8} \times 20 = 8000$ oe
2(b)	15.4 or 15.38...	3	M2 for $\frac{65-55}{65} [\times 100]$ or $\frac{55}{65} \times 100$ or $1 - \frac{55}{65}$ or M1 for $65 - 55$ or $\frac{55}{65}$
2(c)	500	2	M1 for $\frac{2500 \times 2.5 \times 8}{100}$ oe
2(d)	501.42	3	M2 for 2400×1.024^8 oe (2901 or 2901.4[0] or 2901.42...) or M1 for 2400×1.024^n oe where $n > 1$
2(e)	84	3	M2 for $79.80 \div \left(1 - \frac{5}{100}\right)$ oe or M1 for recognising 79.80 is 95%
3(a)	$2 \times 3 - 7 = -1$ oe	1	Correct substitution
3(b)(i)	$\left(-\frac{1}{2}, 0\right)$ oe	1	
3(b)(ii)	(0, 4)	1	
3(c)	$[y =] 3x - 1$	3	M1 for gradient = $\frac{11-2}{4-1}$ oe or better M1 for substituting (1, 2) or (4, 11) into $y = (\text{their } m)x + c$

Question	Answer	Marks	Partial Marks
4(a)(i)	$\begin{array}{r l} 1 & 7\ 8\ 9\ 9 \\ \hline 2 & 0\ 1\ 1\ 1\ 1\ 2\ 2\ 3\ 4\ 5 \\ & \text{e.g. } 2 3 = 23 \end{array}$	3	B1 for each row B1 for key
4(a)(ii)	21 19	2	B1 for each
4(a)(iii)	102.8 to 102.9	2	M1 for $\frac{4}{14}$ oe or $\frac{360}{14}$ oe
4(b)(i)	2.4	1	
4(b)(ii)	0.9	2	B1 for 3 or 2.1 seen
4(b)(iii)	20	2	M1 for 180 seen
4(c)(i)	253.125 or 253.13 or 253.1 or 253	2	M1 for evidence of at least two mid-values 150, 225, 325 soi by e.g. 50625
4(c)(ii)	Correct histogram	4	B1 for bars with correct widths B1 for first bar with height 0.25 B1 for second bar with height 2 B1 for third bar with height 0.5 If 0 scored SC1 for three correct frequency densities seen
5(a)(i)	Correct sketch 	2	B1 for sine graph with different amplitude and/or period but must go through (0, 0) or for correct sine graph but only one cycle
5(a)(ii)	$-1 \leq f(x) \leq 1$	1	
5(b)(i)	Correct sketch 	2	i.e. Correct shape with 2 branches above x -axis and gap of at least 120 between the branches and only slightly crossing either axis. B1 for 2 branches above x -axis but gap less than 120 between the branches and only slightly crossing either axis or one branch correct
5(b)(ii)	logarithms of negative numbers do not exist oe	1	
5(b)(iii)	(90, 0), (450, 0)	2	B1 for each
5(b)(iv)	$x = 0, x = 180, x = 360, x = 540$	2	B1 for 2 or 3 correct

Question	Answer	Marks	Partial Marks
5(c)(i)	23.5 or 23.51 to 23.52	1	
5(c)(ii)	$23.5 < x < 156.5$ $383.5 < x < 516.5$	2	B1 for each Allow 23.51 to 23.52, 156.48 to 156.49 Allow 383.51 to 383.52, 516.48 to 516.49
5(c)(iii)	Any integer less than – 1	1	
6(a)(i)(a)	$2\pi r = 12$ oe	M1	
	1.9096 to 1.9099	A1	
6(a)(i)(b)	458 or 457.9 to 458.5	2	M1 for $\pi \times 1.91[0]^2 \times 40$
6(a)(ii)	1070	4	B3 for volume of other cylinder 1530 or 1527 to 1529. ... or M2 for $\pi \times \left(\frac{40}{2\pi}\right)^2 \times 12$ or M1 for $40 \div (2\pi)$ oe
6(b)	40	4	M3 for $\sqrt[3]{\frac{4.8 \times 100^3}{75}}$ oe or M1 for $\sqrt[3]{\frac{\text{figs } 48}{\text{figs } 75}}$ oe or $\sqrt[3]{\frac{\text{figs } 75}{\text{figs } 48}}$ oe and M1 for 4.8×100^3 or $75 \div 100^3$ oe
7(a)	$\frac{6}{30}$ oe	2	M1 for $\frac{3}{6} \times \frac{2}{5}$ oe
7(b)	$\frac{12}{30}$ oe	3	M2 for $\frac{2}{6} \times \frac{3}{5} \times 2$ oe or M1 for $\frac{2}{6} \times \frac{3}{5}$ oe
7(c)	$\frac{28}{30}$ oe	2	M1 for $1 - \frac{2}{6} \times \frac{1}{5}$ oe
8(a)	16 000	2	M1 for $12960 \div \left(1 - \frac{10}{100}\right)^2$ oe or B1 for 14400

Question	Answer	Marks	Partial Marks
8(b)	7 nfw	3	B2 for 6.58 or 6.578 to 6.579 or M2 for $\frac{\log\left(\frac{6480}{12960}\right)}{\log 0.9}$ oe or appropriate sketch or at least two trials with $n > 3$ or M1 for $12960 \times \left(1 - \frac{10}{100}\right)^n = 6480$ oe if 0 scored, SC1 for answer 9 nfw, coming from 16000
9(a)	12.2 or 12.21 to 12.22	2	M1 for $\sin 70 = \frac{[]}{13}$ oe
9(b)	15.5 or 15.49...	2	M1 for $\tan 50 = \frac{BD}{13}$ oe
9(c)	5.32 or 5.316 to 5.319...	4	B1 for [angle $DBC =$] 20 M1 for $(\text{their } BD)^2 + 15^2 - 2 \times \text{their } BD \times 15 \cos(\text{their } DBC)$ A1 for 28.26 to 28.30...
9(d)	art 195	3	M2 two of $0.5 \times 13 \times 13 \times \sin 40$ oe $0.5 \times 13 \times \text{their } BD$ oe $0.5 \times 15 \times \text{their } BD \times \sin(\text{their } 20)$ or M1 for one of above
10(a)	7.64 or 7.636...	3	M2 for $\left(\frac{4 \times \frac{30}{60} + 5}{\frac{30}{60} + \frac{25}{60}}\right)$ oe or M1 for $4 \times \frac{30}{60} + 5$ or $\frac{30}{[60]} + \frac{25}{[60]}$
10(b)(i)	$\frac{10x + 60y}{10 + z}$ or $\frac{10(x + 6y)}{10 + z}$	3	M2 for $\left(\frac{x \times \frac{10}{60} + y}{\frac{10}{60} + \frac{z}{60}}\right)$ oe or M1 for total distance = $x \times \frac{10}{60} + y$ or total time = $\frac{10}{[60]} + \frac{z}{[60]}$
10(b)(ii)	5	2	M1 for correct substitution of $x = 3, z = 20$ and average speed = 11 in <i>their</i> formula which must contain x, y and z . or B1 for 5.5 oe or 330 seen

Question	Answer	Marks	Partial Marks
10(c)(i)	$\frac{3t}{60} + \frac{7(t+10)}{60}$ oe	M1	
	$\frac{t}{[60]} + \frac{t+10}{[60]}$	M1	The two M1 s may be seen together in a correct fraction
	Correct simplification to $\frac{5t+35}{t+5}$ seen	A1	dep on M1M1 At least one line of working and no errors
10(c) (ii)	15	2	M1 for $(5t + 35) = (5\frac{1}{2})(t + 5)$ oe or better
11(a)	5	1	
11(b)	0.1oe , 1, 10, 100	2	B1 for 3 correct or all correct seen and spoilt.
11(c)	6.5 oe	2	M1 for $2x - 1 = 12$
11(d)	$2x - 2$ or $2(x - 1)$	3	B2 for correct unsimplified answer OR M1 for substituting $x - 2$ for x M1 for adding 3 to a function in x oe OR M1 for $y = 2x + c$ ($c \neq -1$) leading to answer with gradient 2 M1 for substituting coords of valid point into $y = 2x + c$
11(e)	$\log x$	2	M1 for $\log y = x$ or $x = 10^y$
11(f)	23, 113	3	B2 for 23 or B1 for $[g(x) =] 45$ soi

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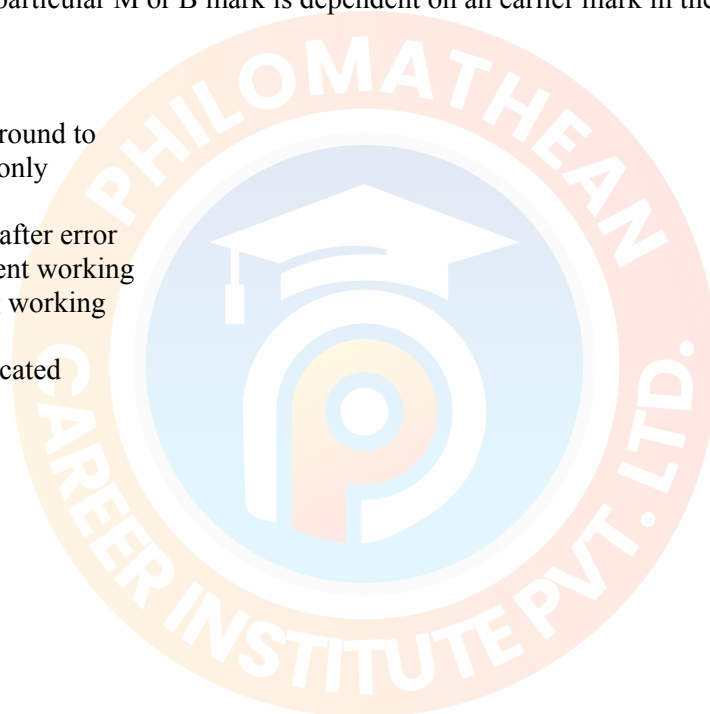
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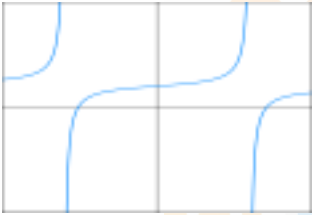
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When a part of a question has two or more ‘method’ steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation ‘**dep**’ is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfw	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied



Question	Answer	Marks	Partial Marks
1(a)	0.744 or 0.7437 to 0.7438	1	
1(b)(i)	130.5 final answer	1	
1(b)(ii)	100 [.00] final answer	1	
1(c)	17.66	2	M1 for 0.23×76.8 oe
1(d)	1000	3	M2 for $\frac{8-3}{3+4+6+8} \times 4200$ oe or M1 for $\frac{4200}{3+4+6+8}$ [$\times 3$ or 8] oe
1(e)	Any irrational number less than 10	1	e.g. π , $\sqrt{12}$, e ; and not with decimal or fractional equivalent
1(f)	$2.29[1] \times 10^{-1}$ final answer	2	B1 for figs 229[1]
2(a)	Correct sketch 	3	B1 for each branch
2(b)	$y = 1$, $x = 3$, $x = -3$	3	B1 for each
2(c)	-2.87 or -2.874 to -2.873 1.15 or 1.149 to 1.150 2.72 or 2.723 to 2.724	3	B1 for each If 0 scored SC1 for -2.9 , 1.1 and 2.7
3(a)(i)	$8\sqrt{x}$ oe	2	M1 for $y = k\sqrt{x}$
3(a)(ii)	16	1	
3(a)(iii)	$\frac{y^2}{64}$ or $\left(\frac{y}{8}\right)^2$ or $\frac{y^2}{8^2}$	2	FT only if wrong k , k numeric and $k \neq 1$ M1 for $\frac{y}{\text{their } k} = \sqrt{x}$ or $y^2 = (\text{their } k\sqrt{x})^2$ $k \neq 1$ or better SC1 for answer $\frac{y^2}{k^2}$ or $\left(\frac{y}{k}\right)^2$

Question	Answer	Marks	Partial Marks
3(b)	2	3	M2 for $p = \frac{12}{q+2}$ oe or $p = \frac{12}{4+2}$ oe or M1 for $p = \frac{k}{q+2}$ OR M2 for $p = \frac{3(2+2)}{4+2}$ or M1 for $p(4+2) = 3(2+2)$
4(a)(i)	198	2	M1 for 3 or more 50, 120, 160, 215, 275 soi
4(a)(ii)	$\frac{306}{9900}$ oe or 0.0309 or 0.030 90 to 0.030 91	2	M1 for $\frac{18}{100} \times \frac{17}{99}$
4(a)(iii)	$\frac{2850}{6642}$ oe or 0.429 or 0.4290 to 0.4291	3	M2 for $\frac{57}{82} \times \frac{25}{81} + \frac{25}{82} \times \frac{57}{81}$ oe or M1 for $\frac{57}{82} \times \frac{25}{81}$ or $\frac{25}{82} \times \frac{57}{81}$
4(b)(i)	0.04, 0.35, 0.55, 0.5, 0.5	2	B1 for 3 correct
4(b)(ii)	Correct histogram	4	FT <i>their</i> fully completed table in (b)(i) with linear scale B1 for suitable scale (must include all heights) B1 for correct column widths B2FT for columns with all heights correct or B1FT for 3 or 4 columns with correct heights
5(a)	20	2	M1 for $16^2 + (30-18)^2$
5(b)	36.9 or 36.86 to 36.87	2	M1 for $\tan[\] = \frac{18}{24}$ oe
5(c)	100	3	M2 for $2 \times (\text{their (a)} + \sqrt{18^2 + 24^2})$ oe or M1 for $18^2 + 24^2$ or $RS = 30$ or $PQ = 30$ seen
5(d)	576	3	M2 for $(40 \times 30) - 2 \times (0.5 \times 18 \times 24) - 2 \times (0.5 \times 16 \times 12)$ oe or M1 for any correct and relevant area
5(e)	Correct explanation	2	B1 for partial explanation e.g. ratio of two sides the same, with names or numbers given.
6(a)	Correct triangle. (-6, -1), (-4, -1), (-6, 3)	2	B1 for $\begin{pmatrix} -7 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -3 \end{pmatrix}$

Question	Answer	Marks	Partial Marks
6(b)	Correct triangle. (-1, 4), (-1, 6), (-5, 4)	2	B1 for correct rotation about any centre or for correct centre but 90° clockwise
6(c)	Rotation 90° clockwise oe [Centre] (-6, 4)	3	B1 for each
6(d)	Correct triangle. (3, -1), (7, -1), (7, -9)	2	B1 for correct enlargement with wrong centre
6(e)	Enlargement [centre] (3, 1) [SF] -0.5	2	B1 for each
7(a)	90	2	M1 for $55 + 5k$, $k = 7$ or 8
7(b)	11	2	M1 for $55 + 5(n - 1) = 105$ or better or $\frac{105 - 55}{5} [+5]$ soi by 10
7(c)	82.5	2	M1 for $\frac{42}{28} = \frac{[]}{55}$ oe
7(d)(i)	$28 + 2a = 44$ oe or $44 - 28$ oe seen	M1	
	$2a = 16$ or $\frac{44 - 28}{2}$ oe [= 8]	A1	
7(d)(ii)	56 900 or 56 900 to 56 920	2	M1 for $\frac{\pi}{3} \times (3 \times 14^2 + 3 \times 14 \times 8 + 8^2) [\times 55]$
7(d)(iii)	192 000 or 192 000 to 192 200	3	M2 for <i>their</i> (d)(ii) $\times \left(\frac{42}{28}\right)^3$ or M1 for $\left(\frac{42}{28}\right)^3$ or $\left(\frac{28}{42}\right)^3$ OR M2 for $\frac{\pi}{3} \times \text{their}(\text{c}) \times (3 \times 21^2 + 3 \times (8 \times 1.5) \times 21 + (8 \times 1.5)^2)$ or B1 for $a = 12$
7(d)(iv)	$[h =] \frac{3V}{\pi(3r^2 + 3ar + a^2)}$	2	M1 for $3V = \pi h(3r^2 + 3ar + a^2)$ or $\frac{V}{\pi(3r^2 + 3ar + a^2)} = \frac{h}{3}$ or $\pi h = \frac{3V}{3r^2 + 3ar + a^2}$
8(a)	[Angle between] tangent and radius/diameter	1	
8(b)(i)	108	2	M1 for $ADO = 36$ soi

Question	Answer	Marks	Partial Marks
8(b)(ii)	54	2	M1 for $\frac{\text{their}(\mathbf{b})(\mathbf{i})}{2}$ or $90 - 36$ or $\frac{180 - 72}{2}$
8(b)(iii)	90	1	
8(b)(iv)	18	1	
8(b)(v)	48	1	
9(a)	$[\cos A] = \frac{11^2 + 9.1^2 - 8.2^2}{2 \times 11 \times 9.1}$	M2	M1 for $8.2^2 = 11^2 + 9.1^2 - 2 \times 11 \times 9.1 \times \cos[]$
	46.98 to 46.99	A1	
9(b)	$[\sin B] = \frac{11}{8.2} \times \sin 47.0$	M2	M1 for $\frac{8.2}{\sin 47} = \frac{11}{\sin B}$
	78.8 or 78.74 to 78.84	A1	If 0 scored then SC1 for correct answer from cosine rule or other method
9(c)	36.6 or 36.54 to 36.60...	2	M1 for $0.5 \times 9.1 \times 11 \times \sin 47.0$ or M1 for $0.5 \times 9.1 \times 8.2 \times \sin(\text{their}(\mathbf{b}))$ or M1 for $0.5 \times 8.2 \times 11 \times \sin(180 - 47 - \text{their}(\mathbf{b}))$
9(d)	6.65 or 6.66 or 6.647 to 6.656...	2	M1 for $9.1 \times \sin 47.0$ oe or $\text{their}(\mathbf{c}) \div (0.5 \times 11)$
10(a)(i)	Points correctly plotted	3	B2 for 5 or 6 correct points B1 for 3 or 4 correct points
10(a)(ii)	Positive	1	
10(b)(i)	17.1 or 17.09...	1	
10(b)(ii)	21.2	1	
10(c)(i)	$y = 14.2 + 0.411x$	2	B1 for $14.2 + kx$ or $a + 0.411x$ If 0 scored, SC1 for $14 + 0.41x$
10(c)(ii)	22.4 or 22.39 to 22.42	1	FT $\text{their}(\mathbf{c})(\mathbf{i})$
10(c)(iii)	cm/ml oe	1	
11(a)(i)	-1	1	
11(a)(ii)	4	2	M1 for $2x - 7 = 1$ or better
11(b)	$\frac{x+7}{2}$ oe	2	M1 for $y + 7 = 2x$ or $\frac{y}{2} = x - \frac{7}{2}$ or $x = 2y - 7$ Allow $f(x)$ for y

Question	Answer	Marks	Partial Marks
11(c)(i)	$2\sqrt{x} - 7$ final answer	1	Allow $x^{\frac{1}{2}}$ for $\sqrt{x}$
11(c)(ii)	36	3	M2 for $\sqrt{x} = \frac{5+7}{2}$ or better e.g. $\sqrt{x} = 6$ or M1 for $2\sqrt{x} - 7 = 5$ or <i>their (c)(i) = 5</i>
11(d)(i)	$\frac{1}{\sqrt{2x-7}}$ oe final answer	2	M1 for $\sqrt{2x-7}$ If 0 scored SC1 for answer $\frac{1}{\text{their(c)(i)}}$
11(d)(ii)	$x > 3.5$	2	M1 for $2x - 7 > 0$



CAMBRIDGE INTERNATIONAL MATHEMATICS**0607/43**

Paper 4 (Extended)

May/June 2018

MARK SCHEME

Maximum Mark: 120

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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Cambridge International is publishing the mark schemes for the May/June 2018 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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This document consists of **8** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

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- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

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GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

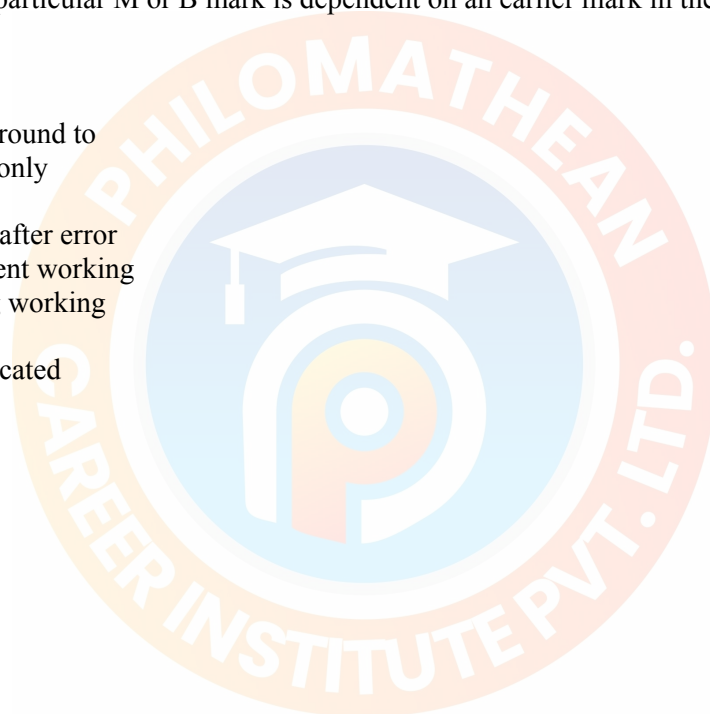
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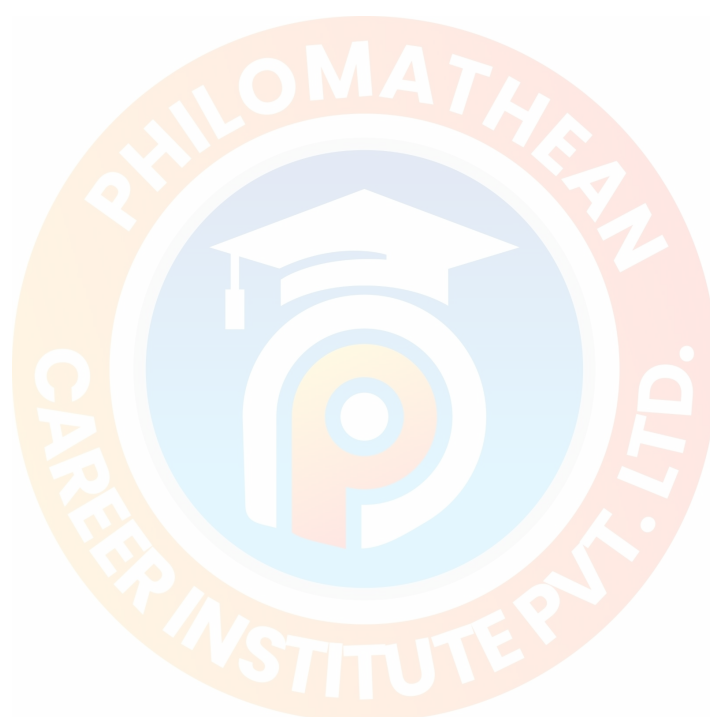
Question	Answer	Marks	Partial Marks
1(a)	10	1	
1(b)	6 nfw	2	B1 for 17 or 11 seen
1(c)	10	1	
1(d)	15	1	
1(e)	14.5 or 14.51 to 14.52	2	M1 for Σxf soi by e.g. 450
1(f)	It is the smallest oe	1	
2(a)	30.51	2	M1 for $22.6 \times \left(1 + \frac{35}{100}\right)$ oe
2(b)	40.9 or 40.93 to 40.94	3	M2 for $\frac{43 - \text{their } 30.51}{\text{their } 30.51} [\times 100]$ oe or M1 for $43 - \text{their } 30.51$ or $\frac{43}{\text{their } 30.51}$
2(c)	80	3	M2 for $92 \div \left(1 + \frac{15}{100}\right)$ oe or M1 for $92 = 115\%$ oe
2(d)	7 nfw	3	M2 for $\frac{\log(\frac{32}{45})}{\log 0.95}$ soi by 6.64 to 6.65 or trials as far as $n = 5$ or M1 for 45×0.95^n oe soi
3(a)	Triangle at (4, -4), (5, -4), (5, -6)	2	B1 for translation $\begin{pmatrix} 3 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -1 \end{pmatrix}$
3(b)	Triangle at (5, 0), (7, 0), (7, 3)	2	B1 for any stretch in with x -axis invariant or correct stretch translated vertically
3(c)	Rotation 90° clockwise oe (3, -1)	3	B1 for each
4(a)	6 points plotted correctly	3	B2 for 4 or 5 correct or B1 for 2 or 3 correct
4(b)(i)	Negative	1	
4(b)(ii)	D	1	

Question	Answer	Marks	Partial Marks
4(c)(i)	$p = -967d + 6300$	2	or $(-967.4 \text{ to } -967.3)d + 6297 \text{ to } 6298$ or B1 for $-967d + k$ or $kd + 6300$
4(c)(ii)	3980 or 4000 or 3975 to 3980	1	FT
4(c)(iii)	[Too] far outside range of data oe	1	
5(a)	Correct sketch 	3	B1 for each branch
5(b)	$(0.0295, -0.833)$	2	or $(0.02948 \text{ to } 0.02949, -0.8329\dots)$ B1 for each
5(c)	$x = -3, x = 2, y = 2$	3	B1 for each
5(d)	$-0.833 < k \leq 2$	2	FT their (b) B1 for each inequality
5(e)(i)	$-5.13, 2.81$	2	$-5.131\dots, 2.812 \text{ to } 2.813$ B1 for each
5(e)(ii)	$-5.13 < x < -3,$ $2 < x < 2.81$	2	$-5.131\dots, 2.812 \text{ to } 2.813$ B1 for each FT their (c) and (e)(i)
6(a)	$\left(\begin{array}{l} 72.5 \text{ or } 72.50\dots \\ 33.8 \text{ or } 33.80 \text{ to } 33.81 \end{array} \right)$	4	B2 for 72.5 or 72.50... or M1 for $\frac{[\dots]}{80} = \sin 65$ oe seen (80sin65 oe) B2 for 33.8 or 33.80 to 33.81 or M1 for $\frac{[\dots]}{80} = \cos 65$ oe seen (80cos65 oe)
6(b)	$\left(\begin{array}{l} 187 \text{ or } 187.1 \text{ to } 187.2 \\ -46.5 \text{ or } -46.49\dots \end{array} \right)$	5	M2 for <i>their</i> $72.5 + 140\cos 35$ oe or M1 for $\frac{[\dots]}{140} = \cos 35$ oe seen (140cos35 oe) M2 for <i>their</i> $33.8 - 140\sin 35$ oe or M1 for $\frac{[\dots]}{140} = \sin 35$ oe seen (140 sin35 oe)
6(c)(i)	$(\text{their } 187)^2 + (\text{their } [-] 46.5)^2$	M1	
	193 or 192.6 to 192.9	B1	

Question	Answer	Marks	Partial Marks
6(c)(ii)	$\tan[x] = \frac{\text{their } 46.5}{\text{their } 187}$ oe soi by 13.9...	M1	
	284 or 283.9 to 284.0	B2	M1 for $270 + \text{their } x$ oe
7(a)	0.278 or 0.2781 to 0.2782	3	M2 for $0.5 \times 0.6 \times 0.8 \times \sin 105 \times 1.2$ oe or M1 for $0.5 \times 0.6 \times 0.8 \times \sin 105$
7(b)	3.48 to 3.49	5	M2 for $\sqrt{0.6^2 + 0.8^2 - 2 \times 0.6 \times 0.8 \times \cos 105}$ or M1 for $0.6^2 + 0.8^2 - 2 \times 0.6 \times 0.8 \times \cos 105$ A1 for 1.12 or 1.117... M1 for $\text{their } 1.117 \times 1.2 + 2 \times \text{their area of } ABC + 0.6 \times 0.12 + 0.8 \times 1.2$
7(c)	0.0348 to 0.0349	3	FT $\text{their } (b) \div 100$ M2 for $\text{their } (b) \div \sqrt[3]{\frac{2170}{2.17}} \times \left(\sqrt[3]{\frac{2170}{2.17}} \right)^2$ oe or M1 for $\frac{2170}{2.17}$ or $\frac{2.17}{2170}$ or $(\sqrt[3]{k})^2$ implied by $\text{their } (b) \div 1000$
8(a)	Fully correct tree diagram	3	B1 for each column correct of 0.9 and 0.1 correctly placed (L & R) 0.8 and 0.2 correctly placed (S) 0.7 and 0.3 correctly placed (W)
8(b)	0.504	2	M1 for $0.9 \times 0.8 \times 0.7$
8(c)	0.398	4	M3 for $0.9 \times 0.8 \times 0.3 + 0.9 \times 0.2 \times 0.7 + 0.1 \times 0.8 \times 0.7$ or M2 for 2 of above products or M1 for 1 of above products
9(a)	25 25 16 36 25 41 61	3	B1 for each row
9(b)(i)	225 196	2	B1 for each
9(b)(ii)	361 400	2	B1 for each
9(c)	$n^2 + (n - 1)^2$ oe	2	M1 for 2nd differences all 4 or quadratic expression

Question	Answer	Marks	Partial Marks
10(a)	$\frac{930}{x}$	1	
10(b)(i)	$\frac{930}{x} - \frac{930}{x+5}$ soi	M1	
	$1860(x+5) - 1860x = x(x+5)$ oe	M1	FT dep on 1st M1 with $\frac{1}{2}$ as separate term in equation
	Completion to $x^2 + 5x - 9300 = 0$ with no errors	A1	
10(b)(ii)	94[.0] or 93.96 to 93.97 –99[.0] or –98.97 to –98.96	3	M1 for $\frac{5 \pm \sqrt{5^2 - 4(1)(-9300)}}{2}$ or sketch of parabola B1 for 1 correct
10(b)(iii)	9 [h] 24 or 23 to 24 [min]	2	M1 for $930 \div (\text{their } 94 + 5)$ oe If 0 scored SC1 for 9 h 53min to 9 h 54 min
11(a)(i)	Angle X is common oe $\angle XCB = \angle XAD$ (angles in same segment) oe	2	B1 for each If 0 scored SC1 for 2 pairs of angles without reasons
11(a)(ii)	$\frac{XA}{XC} = \frac{XD}{XB}$ oe	1	
11(b)	8	2	M1 for $12 \times 7 = XA \times 6$ soi (implied by 14)
11(c)(i)	$\frac{49}{36}$ oe	1	
11(c)(ii)	$\frac{64}{25}$ oe	1	
12(a)	$-\frac{3}{4}$ or –0.75	1	
12(b)	$\frac{4-x}{3}$ oe final answer	2	M1 for $x = 4 - 3y$ or $y + 3x = 4$ or $y - 4 = -3x$ or $\frac{y}{3} = \frac{4}{3} - x$
12(c)	–17	2	B1 for $[f(3)] = 7$ or M1 for $4 - 3(2x + 1)$ soi

Question	Answer	Marks	Partial Marks
12(d)	$9x - 8$ final answer	2	M1 for $4 - 3(4 - 3x)$
12(e)	3	2	M1 for $2^x - 1 = 7$ or $\log_2(x + 1)$ oe
12(f)	$\frac{5 - x}{(2x + 1)(4 - 3x)}$ oe final answer	3	M1 for $4 - 3x + 2x + 1$ M1 for common denominator $(2x + 1)(4 - 3x)$



CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/41

Paper 4 (Extended)

May/June 2019

MARK SCHEME

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Published

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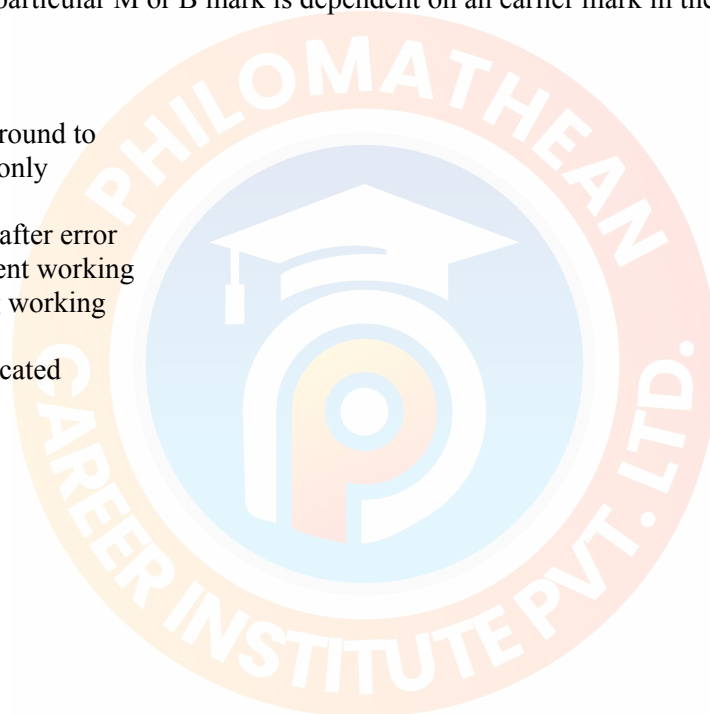
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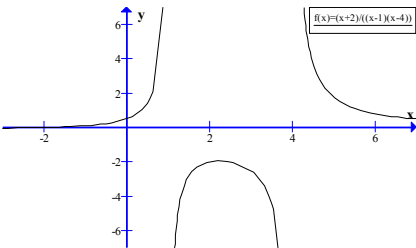
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soi	seen or implied



Question	Answer	Marks	Partial Marks
1(a)	535.5[0] final answer	2	M1 for $630 \times \left(1 - \frac{15}{100}\right)$ oe
1(b)	\$1120	3	M2 for $952 \div \left(1 - \frac{15}{100}\right)$ oe or M1 for 85% associated with 952
1(c)	12 nfw	4	M3 for $n \log \left(1 - \frac{5}{100}\right) = \log \left(\frac{\frac{1}{2}(630)}{\text{their } 535.50}\right)$ oe soi by 10.3 or 10.4 or 10.34 to 10.36... or correct trials as far as 10 and 11 or suitable sketch(es) e.g. $y = 535.5 \times 0.95^x$ and $y = 315$ or M2 for $\left(1 - \frac{5}{100}\right)^n = \left(\frac{\frac{1}{2}(630)}{\text{their } 535.50}\right)$ oe or at least 3 correct trials or final answer 11 nfw or M1 for $\text{their } 535.5 \times \left(1 - \frac{5}{100}\right)^n = \frac{1}{2}(630)$ soi oe
2(a)	Reflection $y = -1$	2	B1 for each
2(b)	Triangle at (6, -3), (11, -3), (10, -1)	2	B1 for translation $\begin{pmatrix} k \\ -3 \end{pmatrix}$ or $\begin{pmatrix} 6 \\ k \end{pmatrix}$
2(c)(i)	63.4 or 63.43 to 63.44	3	B2 for $\tan [\theta] = \frac{4}{2}$ oe or B1 for correct angle clearly identified and no other angle seen.
2(c)(ii)	5	3	M2 for $\frac{\sqrt{125}}{5}$ or $\frac{10}{\sqrt{20}}$ or $\frac{5}{\sqrt{5}}$ or M1 for $\sqrt{10^2 + 5^2}$ or $\sqrt{4^2 + 2^2}$ or $\sqrt{1^2 + 2^2}$ or $\sqrt{125}$ or $\sqrt{20}$ or $\sqrt{5}$
3(a)	1, 2, 3, 4, 6, 8, 12, 24	B2	B1 for 7 correct and 1 incorrect or 6 or 7 correct and none incorrect or 8 correct and 1 extra
	$(3 + 1) \times (1 + 1) = 8$	B1	soi by $4 \times 2 = 8$

Question	Answer	Marks	Partial Marks
3(b)	$360 = 2^3 \times 3^2 \times 5$	B2	M1 for two steps in a factor ladder or tree oe or listing all factors of 360 with no extras or omissions.
	$(3 + 1) \times (2 + 1) \times (1 + 1)$	M1	soi by $4 \times 3 \times 2$ FT dep on factors being prime
	24	B1	
4(a)	27.7 or 27.70 to 27.71	2	M1 for at least 3 midpoints soi
4(b)	Correct cf curve	5	Curve/polygon through (10, 0), (20, 16), (25, 44), (30, 76), (35, 100), (40, 114), (50, 120) or B4 for curve through 5 or 6 points or 7 points with no curve or B3 for 'correct curve' through all other consistent points in interval or B2 for all correct cfs or B1 for 4 or 5 correct cfs. If 0 scored SC1 for any cumulative frequency diagram.
4(c)(i)	26 to 28	1	Dep on increasing curve FT
4(c)(ii)	9 to 11.5	2	Dep on increasing curve FT B1 for $lq = 22$ to 23.5 or $uq = 32.5$ to 33.5
4(c)(iii)	10 to 15	2	Dep on increasing curve FT B1 for 105 to 110 seen
4(d)(i)	5.6, 6.4, 4.8, 2.8, 0.6	2	B1 for 3 or 4 correct
4(d)(ii)	Correct histogram	3	B2 FT for bars with <i>their</i> heights or B1FT for 3 or 4 bars with <i>their</i> heights or bars with all correct widths
5(a)	72	2	M1 for $\frac{12}{60} \times 360$
5(b)	$\frac{1}{4}$ oe	1	
5(c)(i)	$\frac{4}{59}$ oe	3	M2 for $\frac{12}{60} \times \frac{10}{59} + \frac{10}{60} \times \frac{12}{59}$ oe or M1 for $\frac{12}{60} \times \frac{10}{59}$ or $\frac{10}{60} \times \frac{12}{59}$ soi $\frac{2}{59}$

Question	Answer	Marks	Partial Marks
5(c)(ii)	$\frac{303}{590}$ oe	3	M2 for $1 - \left(\frac{42}{60} \times \frac{41}{59}\right)$ oe or $\frac{18}{60} \times \frac{42}{59} + \frac{42}{60} \times \frac{18}{59} + \frac{18}{60} \times \frac{17}{59}$ or M1 for $\frac{18}{60} \times \frac{42}{59}$ or $\frac{42}{60} \times \frac{18}{59}$ or $\frac{18}{60} \times \frac{17}{59}$ or $\frac{42}{60} \times \frac{41}{59}$
6(a)(i)	$y = \frac{18}{\sqrt{x}}$ oe	2	M1 for $y = \frac{k}{\sqrt{x}}$ oe
6(a)(ii)	3.29 or 3.286...	1	FT wrong k only
6(a)(iii)	1.44 oe	2	M1 for $\sqrt{x} = \frac{\text{their}18}{15}$ or $225 = \frac{(\text{their}18)^2}{x}$
6(b)	$z = 3\left(\frac{18}{\sqrt{x}} + 5\right)$ oe	2	M1 for $z = K(\text{their}(a(i)) + 5)$ $K \neq 1$ or for $z = 3(y + 5)$
7(a)	Vector $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ drawn Vector $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$ drawn Vector $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$ drawn	3	B1 for each with arrows If 0 scored SC1 for all three without arrows or all incorrect arrows
7(b)	$[\mathbf{p}] = -3\mathbf{b}$ oe $[\mathbf{q}] = 3\mathbf{a} + 3\mathbf{b}$ oe $[\mathbf{r}] = 2\mathbf{b} - \mathbf{a}$ oe	3	B1 for each
8(a)	Correct Pythagoras statement leading to $11^2 - 6^2$ or $121 - 36$ or 85	M2	or M1 for $[BD]^2 + 6^2 = 11^2$ oe
	9.219...	A1	9.219... implies M1 A1
8(b)	43.8 or 43.80... nfw	3	M2 for $\cos[ABD] = \frac{9.22^2 + 13^2 - 9^2}{2 \times 9.22 \times 13}$ or better or M1 for $9^2 = 9.22^2 + 13^2 -$ $2 \times 9.22 \times 13 \cos [ABD]$ oe

Question	Answer	Marks	Partial Marks
8(c)	69.1 or 69.13 to 69.14... nfw	3	M1 for $0.5 \times 9.22 \times 6$ oe M1 for $0.5 \times 9.22 \times 13 \times \sin$ (<i>their</i> 43.8) oe
8(d)	17.7 or 17.69...	3	M1 for $6^2 + 13^2 -$ $2 \times 6 \times 13 \cos (90 + \text{their } 43.8)$ A1 for 313 or 312.9 to 313.0
9(a)	$(45 + 4x)(20 + 2x) = 2208$	M1	
	$900 + 90x + 80x + 8x^2$	B1	For expansion
	Completion to $4x^2 + 85x - 654 = 0$ with no errors or omissions	A1	
9(b)	$\frac{-85 \pm \sqrt{85^2 - 4(4)(-654)}}{2 \times 4}$	M1	or $(x - 6)(4x + 109)$ or sketch of parabola ($+x^2$) with one positive zero and one negative
	6, -27.25 oe	B2	B1 for each
9(c)	Length = 69 Height = 32	B2	B1FT for each
10(a)(i)	11	1	
10(a)(ii)	-23	2	M1 for $5 - 2(3 \times 4 + 2)$ soi or $5 - 2(3x + 2)$
10(a)(iii)	$\frac{1}{8}$ oe	3	M1 for $5 - 2x = 2(3x + 2)$ oe M1FT for $5 - 4 = 6x + 2x$ or better
10(a)(iv)	$\frac{5-x}{2}$ oe final answer	2	M1 for $2x + y = 5$ or better or $x = 5 - 2y$ or $\frac{y}{2} = \frac{5}{2} - x$
10(a)(v)	$17 - 6x$ oe final answer	2	M1 for $3(5 - 2x) + 2$
10(a)(vi)	$\frac{5x+16}{(5-2x)(3x+2)}$ or $\frac{5x+16}{10+11x-6x^2}$ final answer	3	M1 for common denominator $(5 - 2x)(3x + 2)$ oe M1 for $3(3x + 2) + 2(5 - 2x)$ oe
10(b)	x	1	
11(a)	Correct sketch 	3	B1 for each branch

Question	Answer	Marks	Partial Marks								
11(b)	(2.24, −1.94)	2	or (2.242 to 2.243, −1.943 to −1.942) B1 for each co-ordinate								
11(c)	$x = 1, x = 4, y = 0$	3	B1 for each								
11(d)(i)	1.34 or 1.344 to 1.345 2.79 or 2.789... 5.87 or 5.866...	3	B1 for each If 0 scored, SC1 for 1.3, 2.8 and 5.9								
11(d)(ii)	$x < 1$ $1.34 < x < 2.79$ $4 < x < 5.87$	3	B1 for each FT dep on two solutions to (i) between 1 and 4. FT dep on solution to (i) > 4								
12(a)	<table border="1"><tr><td>(5)</td><td>(6)</td></tr><tr><td>37</td><td>37</td></tr><tr><td>24</td><td>54</td></tr><tr><td>(61)</td><td>91</td></tr></table>	(5)	(6)	37	37	24	54	(61)	91	5	B1 for each
(5)	(6)										
37	37										
24	54										
(61)	91										
12(b)	$[p =] -3$ $[q =] 1$	2	B1 for each								

CAMBRIDGE INTERNATIONAL MATHEMATICS**0607/42**

Paper 4 (Extended)

May/June 2019

MARK SCHEME

Maximum Mark: 120

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **8** printed pages.**Cambridge Assessment**
International Education

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

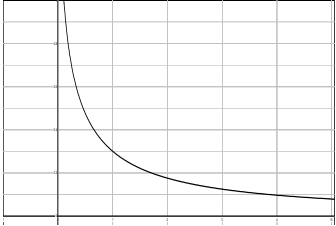
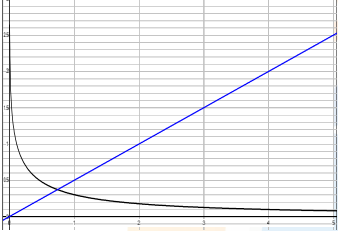
- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more ‘method’ steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation ‘**dep**’ is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfw	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	$\frac{11}{11+14} \times 50$ or $\frac{11}{25} \times 50$ oe	M1	
1(b)	16 : 22 oe isw	2	M1 for 22 – 6 and 50 – 22 – 6 oe If 0 scored, SC1 for 22 : 16 oe
1(c)	8.5[0]	1	
1(d)(i)	5.4[0]	2	M1 for 0.9×6 oe
1(d)(ii)	4.1[0] nfw	3	M2 for $\frac{3.69}{0.9}$ oe or M1 for associating 3.69 with 90%

Question	Answer	Marks	Partial Marks
2(a)	Correct sketch 	2	Must not cross axes B1 for correct shape
2(b)	$y = 0, x = 0$	2	B1 for each If 0 scored, SC1 for answers x -axis and y -axis
2(c)	0.462 or 0.4624 to 0.4625	1	
2(d)	Correct sketch 	1	
2(e)	0.742 or 0.7415 to 0.7416	1	
2(f)	Region that is below $y = 0.5$ and above other two graphs.	1	
3(a)	0.3 0.8 0.05 and 0.95	3	B1 B1 B1
3(b)(i)	0.155 oe	3	M2 for $0.7 \times 0.2 + \text{their } 0.3 \times \text{their } 0.05$ or M1 for 0.7×0.2 or $\text{their } 0.3 \times \text{their } 0.05$
3(b)(ii)	31	1	FT $200 \times \text{their (i)}$
4(a)	12 cao final answer	3	B2 for 11.98 to 12.02 or M1 for $\pi \times 3^2 \times l + 2 \times \frac{2}{3} \times \pi \times 3^3 [=144\pi]$ oe
4(b)	3.53 or 3.528 to 3.529...	2	M1 for $144\pi \times 7.8$ soi by figs 353 or 3528 to 3529

Question	Answer	Marks	Partial Marks
4(c)	2.95 or 2.96 or 2.950 to 2.963...	3	M2 for $\frac{144\pi - 20 \times 2.8^3}{144\pi} [\times 100]$ or $\frac{20 \times 2.8^3}{144\pi} \times 100$ oe or M1 for $144\pi - 20 \times 2.8^3$ or $\frac{20 \times 2.8^3}{144\pi}$ oe
4(d)	1.5 oe cao final answer	3	B2 for 1.498 to 1.502 or M2 for $3 \times \sqrt[3]{\frac{18\pi}{144\pi}}$ oe or M1 for $\sqrt[3]{\frac{18\pi}{144\pi}}$ or $\sqrt[3]{\frac{144\pi}{18\pi}}$ oe or better or for $\left(\frac{3}{x}\right)^3 = \frac{144\pi}{18\pi}$ oe
5(a)	224	3	M2 for $200 + \frac{200 \times 1.5 \times 8}{100}$ oe or M1 for $\frac{200 \times 1.5 \times 8}{100}$ oe implied by 24
5(b)	223.53	3	M2 for $200 \times \left(1 + \frac{1.4}{100}\right)^8$ oe M1 for $200 \times \left(1 + \frac{1.4}{100}\right)^k$ oe k integer > 1 If 0 scored, SC1 for 23.5 or 23.52 to 23.53
5(c)	3 nfw cao	2	M1 for trials with 1.5% and 1.4% beyond <i>their</i> 224 and <i>their</i> 223.53 respectively, implied by 11, or appropriate equation or graph sketch implied by 10.79..., 2.79...
6(a)	Correct reflection at (1, -1), (4, -1), (4, -3), (3, -3)	1	
6(b)	Correct translation at (5, -4), (2, -4), (2, -6), (3, -6)	2	B1 for translation $\begin{pmatrix} 6 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -3 \end{pmatrix}$
6(c)	Rotation 90° clockwise oe (0, 1)	3	B1 for each
6(d)	Correct stretch at (-1, -2), (-4, -2), (-4, -6), (-3, -6)	2	B1 for stretch factor 2 displaced vertically

Question	Answer	Marks	Partial Marks
7(a)	15.1	1	
7(b)(i)	Correct sketch 	2	Must pass through origin and cross x -axis reasonably close to $x = 4$, not 4.5. B1 for correct shape
7(b)(ii)	$[h =]20.4$ or 20.40 to 20.41 $[t =]2.04$ or 2.040 to 2.041	2	B1 for each or both correct reversed answers
7(b)(iii)	4.08 or 4.081 to 4.082	1	
7(b)(iv)	1.4[0] or 1.401 to 1.403	3	B1 for 2.74 or 2.741 to 2.742 B1 for 1.34 or 1.339 to 1.340
8(a)	$7n$ oe	1	
8(b)	$13 - 3n$ oe	2	B1 for $k - 3n$ or $13 - kn$ oe
8(c)	2^{n+2} oe	2	B1 for $[c \times] 2^{n+k}$ where c is a power of 2 and k is any integer (including 0) seen
8(d)	$n^2 + n$ oe	2	B1 for quadratic expression or for second differences = 2 seen
9(a)	$50 < t \leq 55$	1	Allow e.g. 50 to 55
9(b)	50	2	M1 for at least three of 30, 45, 52.5, 65 soi
9(c)	Correct histogram	4	B1 for each correct column height. B1 for all widths correct If 0 scored, SC1 for 7, 24, 1.5 seen
9(d)(i)	$\frac{30}{240}$ oe	1	
9(d)(ii)	$\frac{22350}{57360}$ oe	2	M1 for $\frac{150}{240} \times \frac{149}{239}$
9(d)(iii)	40 ... 50	2	M1 for $\frac{k}{240} \times \frac{k-1}{239}$ attempted

Question	Answer	Marks	Partial Marks
10(a)	$3p + r = 67$ oe $2p + 3r = 96$ oe	B2	B1 for each Accept words in equations, using + and =.
	correctly eliminating one variable	M1	
	[pencil =] 15	B1	
	[ruler =] 22	B1	If M0 scored in addition to B0 (for answers) scored then award SC1 for answers satisfying one of <i>their</i> two original equations in 2 variables
10(b)(i)	$\frac{1}{2}(x+1)x = \left(\frac{5x}{4} + 3\right)(x-4)$ oe	M2	M1 for $\frac{1}{2}(x+1)x$ or $\left(\frac{5x}{4} + 3\right)(x-4)$
	$\frac{5x^2}{4} - 5x + 3x - 12$ oe	B1	i.e. correct expansion for rectangle
	$3x^2 - 10x - 48 = 0$ reached with no errors or omissions including at least one line of working	A1	Dependent on B1
10(b)(ii)	$(3x+8)(x-6)$	2	B1 for $3x(x-6) + 8(x-6)$ or $x(3x+8) - 6(3x+8)$ or $(3x+a)(x+b)$ with $ab = -48$ or $a+3b = -10$
10(b)(iii)	21	2	FT $\frac{1}{2} \times (\text{their positive } x) \times (\text{their positive } x+1)$ if $x > 4$ M1 $\frac{1}{2} \times (\text{their positive } x) \times (\text{their positive } x+1)$ if $x > 4$
11(a)	118 or 117.7 to 117.8	3	M2 for $\frac{102}{\cos 30}$ oe or M1 for $\frac{102}{DC} = \cos 30$ oe or $102 = DC \times \cos 30$ oe
11(b)	106 or 106.2...	3	M1 for $110^2 + 102^2 - 2 \times 110 \times 102 \times \cos 60$ A1 for 11 284
11(c)	7860 or 7858 to 7870	3	M1 for $0.5 \times 102 \times \text{their } DC \times \sin 30$ oe (3000 or 3010 or 3001 to 3009) M1 for $0.5 \times 102 \times 110 \times \sin 60$ oe (4860 or 4858...)

Question	Answer	Marks	Partial Marks
11(d)	236 or 236.2 to 236.4...	4	B2 for 56.3 or 56.4 or 56.25 to 56.44... or M2 for $\frac{102 \sin 60}{\text{their } AB}$ oe or M1 for $\frac{\sin 60}{\text{their } AB} = \frac{\sin BAD}{102}$ oe and M1 for $180 + \text{their angle } BAD$ oe
12(a)	10	1	
12(b)	9.5 oe	2	M1 for $10 - \frac{1}{x}$ so i.e.g. $10 - \frac{1}{2}$
12(c)	$x^2 - 20x + 101$	3	M1 for $(10 - x)^2 + 1$ B1 for $100 - 10x - 10x + x^2$ oe
12(d)	f(x) and h(x)	2	B1 for each
12(e)	$\frac{10 - 2x}{x(10 - x)}$ oe	3	M1 for common denominator $x(10 - x)$ oe B1 for $(10 - x) - x$ oe seen
12(f)(i)	5	1	
12(f)(ii)	$3\sqrt{3}$ oe or $3^{\frac{3}{2}}$ or 5.2[0] or 5.196...	1	
12(f)(iii)	3^x	2	M1 for $x = \log_3 y$ or $x = 3^y$

CAMBRIDGE INTERNATIONAL MATHEMATICS**0607/43**

Paper 4 (Extended)

May/June 2019

MARK SCHEME

Maximum Mark: 120

Published

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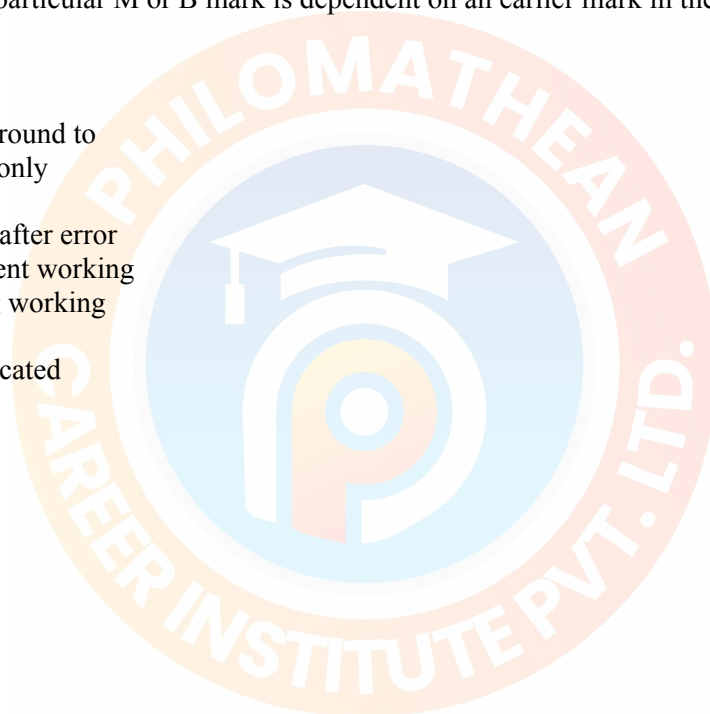
Types of mark

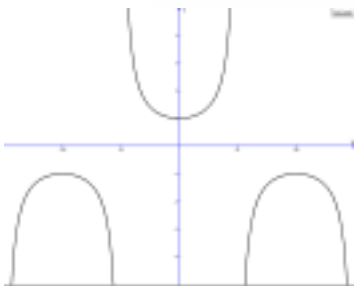
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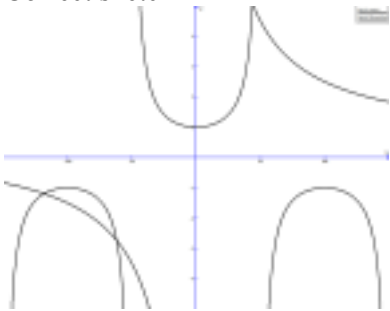
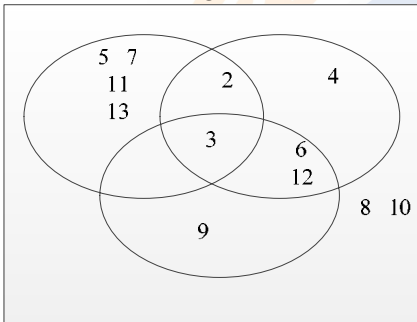
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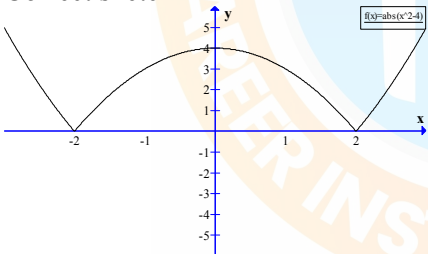
Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfw	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied



Question	Answer	Marks	Partial Marks
1(a)	2.5	2	M1 for $7 \times 3 \times h = 52.5$ or better
1(b)	4500	2	M1 for $\frac{1}{3} \times 500 \times 27$ oe
2(a)(i)	59.4	1	
2(a)(ii)	57.2	1	
2(b)	$[y =] 21.8 + 0.596x$	2	B1 for $[y =] 21.8 + kx$ or $[y =] k + 0.596x$ or $22 + 0.6[0]x$
2(c)(i)	58 or 57.5 to 57.8	1	FT <i>their (b)</i>
2(c)(ii)	25 or 24.8 or 24.75 to 24.78	1	FT <i>their (b)</i>
2(d)	60 Data within range oe	1	Both needed
3(a)(i)	$\frac{19}{120}$ oe	1	
3(a)(ii)	$\frac{3}{4}$ oe	1	
3(b)	$\frac{13}{60}$ oe	2	M1 for $\frac{k}{16+19+12+13}$
3(c)	$\frac{13}{145}$ or $\frac{2184}{24360}$ oe	3	M2 for $\frac{14}{30} \times \frac{13}{29} \times \frac{12}{28}$ or M1 for $14 \times 13 \times 12$ oe seen or $30 \times 29 \times 28$ oe seen
4(a)	Correct sketch 	3	B1 for correct shape B1 for max and min approx. correct B1 for asymptotes approx. correct
4(b)	$f(x) \leq -1$ and $f(x) \geq 1$	2	B1 for each

Question	Answer	Marks	Partial Marks
4(c)(i)	Correct sketch 	2	B1 for each branch
4(c)(ii)	–213 or –212.9 to –212.8 –111 or –111.5 to –111.4 78.6[...]	3	B1 for each
4(c)(iii)	$x = 0$ $y = 0.5$	2	B1 for each
5(a)	2, 3, 5, 7, 11, 13	1	
5(b)	Correct Venn diagram 	3	B2 for 1 or 2 errors/omissions or B1 for 3 or 4 errors/omissions
5(c)	8, 9, 10	1	FT <i>their</i> Venn diagram
5(d)	4	1	FT <i>their</i> Venn diagram
6(a)(i)	(1, 5)	1	
6(a)(ii)	(–y, –x)	2	B1 for each co-ordinate
6(a)(iii)	Reflection $y = -x$	2	B1 for each
6(b)	Enlargement Scale factor 2 Centre (0, 0)	3	B1 for each
6(c)	Stretch x -axis invariant SF 0.5	2	B1 for each

Question	Answer	Marks	Partial Marks
7(a)(i)	2318.55	3	M2 for $2000 \times \left(1 + \frac{3}{100}\right)^5$ or M1 for $2000 \times \left(1 + \frac{3}{100}\right)^k$, $k > 1$ If 0 scored, SC1 for 318.5... or 319 or 320
7(a)(ii)	24	3	B2 for 23.4 or 23.44 to 23.45 or M2 for $n = \frac{\log\left(\frac{4000}{2000}\right)}{\log 1.03}$ oe or M1 for $2000 \times 1.03^n = 4000$ oe
7(b)	2309.37	3	M2 for $2000 \times \left(1 + \frac{0.24}{100}\right)^{60}$ or M1 for $2000 \times \left(1 + \frac{0.24}{100}\right)^k$, $k > 1$
7(c)	0.247 or 0.2466...	3	M2 for $\sqrt[12]{1 + \frac{3}{100}}$ implied by 1.00246[6..] or M1 for $x^{12} = 1 + \frac{3}{100}$ oe
8(a)	Correct sketch 	3	B1 for no part of graph below x-axis B1 for symmetry about y-axis
8(b)	$x = 0$	1	
8(c)	-2, 2	1	Accept $x = -2$, $x = 2$ but not $(-2, 0)$ or $(2, 0)$
8(d)(i)	4	1	
8(d)(ii)	$0 < k < 4$ cao	2	B1 for 0 and 4 seen or $k < 4$ or $k > 0$
9(a)(i)	27	1	
9(a)(ii)	-5	2	M1 for $5 - 25 = 7x - 3x$ or better

Question	Answer	Marks	Partial Marks
9(a)(iii)	1.05 or 1.054... –1.3[0] or –1.304...	4	M3 for $\frac{-2 \pm \sqrt{2^2 - 4 \times 8 \times -11}}{2 \times 8}$ or correct sketch which would lead to solution. or M2 for $\frac{b}{2a}$ correct or $b^2 - 4ac$ correct or M1 for $8x^2 + 2x - 11$ or $-8x^2 - 2x + 11$ or sketch of $8x^2$ or $11 - 2x$
9(b)(i)	$x \leq -2$ oe	2	M1 for $6 - 10 \geq 2x$ or $-2x \geq 10 - 6$ or $3 - x \geq 5$ or better If 0 scored SC1 for $x \geq -2$ or $x = -2$
9(b)(ii)	$2 < x < 2\frac{1}{3}$ oe	3	M2 for $x = 2$ and $x = \frac{7}{3}$ or correct sketch which would lead to solution. or M1 for $1 > 3(x - 2)$ or better or sketch of $y = \frac{1}{x - 2}$ or B1 for $x < 2\frac{1}{3}$ or for $x > 2$
9(c)	Correctly equating one set of coefficients oe	M1	
	Correct method to eliminate one variable	M1	
	$[x=] 4$	B1	
	$[y=] -3$	B1	
			If 0 scored SC1 for correct substitution into one of original equations and evaluation to find other variable.
9(d)	$\frac{13}{16}$ or 0.8125	3	M1 for $\log 2^4$ or better M1 for correct use of $\log p - \log q = \log \frac{p}{q}$ or use of $\log p + \log q = \log pq$
10(a)	$\begin{pmatrix} 6 \\ 3 \end{pmatrix}$	1	
10(b)	6.71 or 6.708... or $\sqrt{45}$ oe	2	M1 for $(7-1)^2 + (5-2)^2$ oe

Question	Answer	Marks	Partial Marks
10(c)	$k - 5 = \sqrt{(\text{their}(\mathbf{b}))^2 - 3^2}$	M2	M1 for $(k-5)^2 + (10-7)^2 = (\text{their}(\mathbf{b}))^2$ oe Reverse process scores 0.
	$k - 5 = 6$	A1	
10(d)	$[y =] -x + 12$ oe	4	M1 for grad $AC = \frac{11-2}{10-1}$ oe M1 for grad perp = $-\frac{1}{\text{their grad}}$ B1 for midpoint (5.5, 6.5)
10(e)	(4, 8)	3	M2 for $\begin{pmatrix} 10 \\ 11 \end{pmatrix} - \begin{pmatrix} 6 \\ 3 \end{pmatrix}$ or $\begin{pmatrix} 7 \\ 5 \end{pmatrix} - \begin{pmatrix} -3 \\ 3 \end{pmatrix}$ oe or M1 for $CD = \begin{pmatrix} 6 \\ 3 \end{pmatrix}$ or $BD = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$ oe
11(a)	$\tan 41 = \frac{FB}{5.5}$ oe	M1	e.g. $\frac{\sin 49}{5.5} = \frac{\sin 41}{FB}$
	$= 4.7810.. [= 4.781]$	A1	
11(b)	37.6 or 37.63 to 37.64	2	M1 for $\tan[FCB] = \frac{4.781}{6.2}$ oe
11(c)	$[CD =] \frac{4.781}{\tan 18} - 6.2$ oe	M2	M1 for $\tan 18 = \frac{4.781}{BD}$ oe
	8.5144...	A1	
11(d)	14.6 or 14.58 to 14.60	3	B2 for 212.7... or M1 for $[AD^2 =] 5.5^2 + (8.514 + 6.2)^2 - 2 \times 5.5 \times (8.514 + 6.2) \times \cos 78$
11(e)	39.5 or 39.6 or 39.54 to 39.6[0]	2	M1 for $0.5 \times 5.5 \times (8.514 + 6.2) \times \sin 78$
12(a)(i)	6.4	3	M2 for $y = 1.6\sqrt{x+1}$ or M1 for $y = k\sqrt{x+1}$ OR M2 for $y = \frac{8\sqrt{16}}{\sqrt{25}}$ or M1 for $\frac{8}{\sqrt{25}} = \frac{y}{\sqrt{16}}$

Question	Answer	Marks	Partial Marks
12(a)(ii)	99	2	FT M1 for $\sqrt{x+1} = \frac{16}{\text{their } 1.6}$ oe only FT $\sqrt{x+1}$
12(b)(i)	−7	1	
12(b)(ii)	38	1	
12(b)(iii)	−0.1 oe	1	
12(b)(iv)	120	2	B1 for row of 0 6 12 18 reached or M1 for $(n-2)^3 - (n-2)$ or $(n-1)(n-2)(n-3)$ oe





Cambridge IGCSE™

CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/41

Paper 4 (Extended)

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This document consists of **9** printed pages.

Generic Marking Principles

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- the standard of response required by a candidate as exemplified by the standardisation scripts.

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Marks must be awarded **positively**:

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- marks are awarded when candidates clearly demonstrate what they know and can do
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- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

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GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
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2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

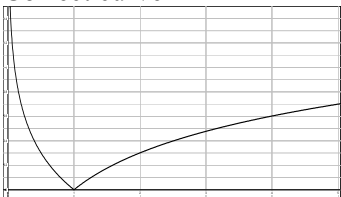
- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

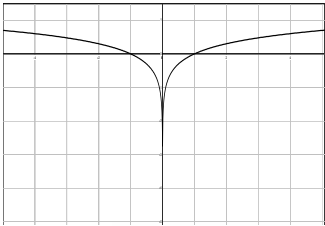
When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation '**dep**' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

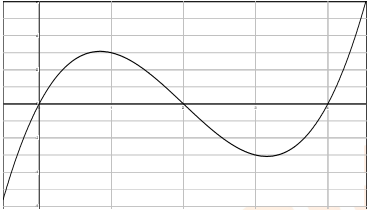
Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfw	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	Image at $(-1, 1)$, $(-3, 1)$, $(-3, 2)$	1	
1(a)(ii)	Image at $(-4, 4)$, $(-2, 4)$, $(-2, 5)$	2	B1 for translation $\begin{pmatrix} -5 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 3 \end{pmatrix}$
1(a)(iii)	Image at $(0, 2)$, $(4, 2)$, $(4, 4)$	2	B1 for enlargement factor 2, different centre
1(b)(i)	Rotation $(-2, 0)$ 90° clockwise oe	3	B1 for each
1(b)(ii)	Stretch -2 x -axis oe invariant	3	B1 for each
2(a)(i)	5	1	
2(a)(ii)	7.9	1	
2(a)(iii)	8	1	
2(a)(iv)	9	1	
2(b)(i)	1.5	1	
2(b)(ii)	1	1	
2(b)(iii)	1.4	1	
2(b)(iv)	18	2	B1 for 102 stated
2(b)(v)(a)	20, 22, 18	3	B1 for each
2(b)(v)(b)	1.71 or 1.712 to 1.713	2	M1 for mid-values soi
3(a)	$(9, 4)$	2	B1 for each
3(b)	$5\sqrt{2}$ cao	3	M1 for $(10-3)^2 + (2-1)^2$ or better A1 for 50 If 0 scored, SC1 for simplification of a surd if seen and if possible.
3(c)	$(6, 2.5)$	2	B1 for each

Question	Answer	Marks	Partial Marks
3(d)	$\left(0, \frac{4}{7}\right)$ oe $(-4, 0)$	5	M1 for gradient = $\frac{1}{7}$ M1 for subst (3, 1) or (10, 2) into $y = (\text{their } m)x + c$ A1 for $y = \frac{1}{7}x + \frac{4}{7}$ B1 FT for $\left(0, \frac{4}{7}\right)$ B1 FT for $(-4, 0)$
4(a)	$(n+3)^2$ oe	2	B1 for any quadratic
4(b)	$n^3 + 2$	2	B1 for any cubic
4(c)	2^{7-n} oe	2	B1 for 2^{a+b} oe or $\left(\frac{1}{2}\right)^{a+b}$ oe $a \neq 0$
5(a)(i)	$1 - 11p$ final answer	2	B1 for $1 + kp$ or for $k - 11p$
5(a)(ii)	$21g^2 + 71gh - 22h^2$ final answer	3	B2 for 3 from $21g^2, 77gh, -6gh, -22h^2$ or B1 for 2 from $21g^2, 77gh, -6gh, -22h^2$
5(b)(i)	$2x^2y^2(y - 2x)$ final answer	2	B1 for any correct partially factorised expression
5(b)(ii)	$(7t + 3u)(7t - 3u)$ final answer	2	B1 for $(at + b)(ct + d)$ with two of $ac = 49, bc + ad = 0, bd = -9$ correct or for $(7t + 3)(7t - 3)$ or $(7 + 3u)(7 - 3u)$
5(b)(iii)	$(2d - 1)(3d + 2)$ final answer	2	B1 for $(pd + q)(rd + s)$ with two of $pr = 6, ps + qr = 1, qs = -2$ correct or for $2d(3d + 2) - (3d + 2)$ or $3d(2d - 1) + 2(2d - 1)$
6(a)(i)	Correct curve 	2	B1 for correct curve from 0 to 1 or for correct curve from 1 to 5
6(a)(ii)(a)	0.631 or 0.6309 to 0.6310 1.58 or 1.584 to 1.585	2	B1 for each

Question	Answer	Marks	Partial Marks
6(a)(ii)(b)	0.106 or 0.1063... 2.45 or 2.446...	4	M2 for $y = 1 - \frac{x}{4}$ sketched or M1 for sketch of line through (0, 1) approx or line with negative but inaccurate gradient B1 for each answer
6(b)(i)	Correct curve 	2	B1 for one branch correct, not crossing y-axis
6(b)(ii)	-1.58 or -1.585 to -1.584 1.58 or 1.584 to 1.585	2	B1 for each
6(c)	$x \geq 1$ oe	1	
7(a)	100	2	M1 for $\frac{500 \times 2.5 \times 8}{100}$ oe
7(b)	104 or 104.4 to 104.5	4	B3 for 604 or 604.4 to 604.5 or M2 for $500 \times \left(1 + \frac{2.4}{100}\right)^8$ oe or M1 for $500 \times \left(1 + \frac{2.4}{100}\right)^n$ with $n > 1$ oe
7(c)	34	4	M3 for $[n =] \frac{\log 2}{\log(1.021)}$ oe or at least two trials with $n > 30$ or graph leading to solution oe (implied by 33.4 or 33.35...) or M2 for $1.021^n = 2$ oe or suitable graph e.g. $y = 1.021^x$ or 3 correct trials or B1 for 1.021^n oe seen
7(d)	3[.00] or 2.999...	3	M2 for $\sqrt[3]{2.5}$ oe or sketch of graph leading to answer or M1 for $(\dots)^{31} = 2.5$ oe or sketch of a relevant graph
8(a)(i)	0.3 0.9 and 0.1 0.2 and 0.8	3	B1 for each

Question	Answer	Marks	Partial Marks
8(a)(ii)	0.69 oe	3	M2 for $0.7 \times 0.9 + 0.3 \times 0.2$ or M1 for one correct product
8(b)(i)	25	1	
8(b)(ii)	$\frac{42}{870}$ oe	3	M2 for $\frac{7}{30} \times \frac{6}{29}$ or M1 for $\frac{p}{q} \times \frac{p-1}{q-1}$
9(a)	Correct curve 	2	B1 for correct shape but inaccurate x intercepts
9(b)	$[-0.5 <] x < 0$	1	
	$2 < x < 4$	2	B1 for $2 < x$ oe or for $x < 4$ oe
9(c)	3.08 or 3.079...	2	M1 for identifying local maximum point
10(a)	253	2	M1 for $2(5^x) + 3$ or $f(125)$
10(b)	$\frac{x-3}{2}$ oe final answer	2	M1 for $x = 2y + 3$ or for $y - 3 = 2x$ or for $\frac{y}{2} = x + \frac{3}{2}$
10(c)	-2.5 oe	2	M1 for $25 = 5^2$ and $\sqrt{5} = 5^{\frac{1}{2}}$ or better
10(d)	$\log_5 x$ oe	2	M1 for $x = 5^y$ or for $\log_5 y = x$ or for $x = \frac{\log y}{\log 5}$

Question	Answer	Marks	Partial Marks
11(a)	9.91 or 9.912...	7	B3 for $[AC =] \sqrt{133}$ or 11.5 or 11.53... or M1 for $11^2 + 12^2 - 2 \times 11 \times 12 \times \cos 60$ A1 for 133 B2 for area = 57.2 or 57.15 to 57.16 or M1 for $\frac{1}{2} \times 12 \times 11 \times \sin 60$ M1 for $\frac{1}{2} \times \text{distance} \times \text{their } AC = \text{their Area}$ OR B3 for $[AC =] \sqrt{133}$ or 11.5 or 11.53... or M1 for $11^2 + 12^2 - 2 \times 11 \times 12 \cos 60$ A1 for 133 M2 for $\sin ACB = \frac{12 \sin 60}{\text{their } AC}$ or for $\sin BAC = \frac{11 \sin 60}{\text{their } AC}$ or M1 for $\frac{12}{\sin ACB} = \frac{\text{their } AC}{\sin 60}$ or for $\frac{11}{\sin BAC} = \frac{\text{their } AC}{\sin 60}$ M1 for $11 \sin ACB$ or $12 \sin BAC$
11(b)(i)	$\frac{112}{\frac{1}{3} \times 8 \times 6}$ oe leading to 7	M2	M1 for $\frac{1}{3} \times 8 \times 6 \times h = 112$ oe
11(b)(ii)	8.60 or 8.602...	3	M2 for $7^2 + 5^2$ or M1 for diagonal of base = 10 or half = 5

Question	Answer	Marks	Partial Marks
11(b)(iii)	59.5 or 59.48 to 59.51	3	M2 for $2 \times \tan^{-1} \frac{4}{7}$ oe or M1 for $\tan \dots = \frac{4}{7}$ OR M2 for $[\cos \dots =]$ $\frac{(\sqrt{65})^2 + (\sqrt{65})^2 - 8^2}{2 \times \sqrt{65} \times \sqrt{65}} \text{ oe}$ or M1 for correct implicit version of cosine rule





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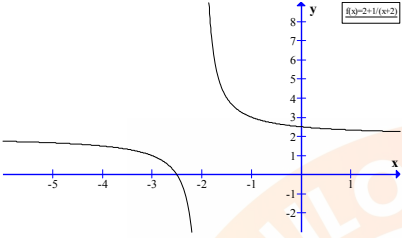
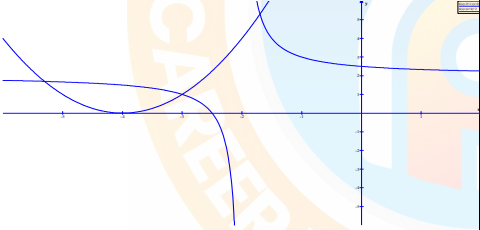
When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation '**dep**' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfw	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	6	1	
1(b)	10	1	
1(c)	6	1	
1(d)	4	2	B1 for LQ = 4 or UQ = 8
1(e)	5.55	2	M1 for attempt at $\sum fx$ (3 or more terms correct)
1(f)	$\frac{1}{120} = \frac{13}{1560}$ or 0.0083 recurring or 0.008333...	3	M2 for two of $\left(\frac{1}{40} \times \frac{2}{39}\right) + \left(\frac{2}{40} \times \frac{1}{39}\right) + \left(\frac{9}{40} \times \frac{1}{39}\right)$ or M1 for any one of above products
2(a)(i)	Translation $\begin{pmatrix} -6 \\ -10 \end{pmatrix}$	2	B1 for each
2(a)(ii)	Enlargement SF 2 Centre (0, 9)	3	B1 for each
2(b)(i)	(-2, -5)	1	
2(b)(ii)	(2, -5)	1	
2(b)(iii)	Reflection x-axis oe	2	B1 for each
2(b)(iv)	Reflection y-axis oe	2	B1 for each
3(a)(i)	All four points correct	2	B1 for two or three correct points
3(a)(ii)	positive	1	
3(b)	87	1	
3(c)(i)	$[y =] 35.9 + 0.811x$	2	B1 for $35.9 + kx$, or for $a + 0.811x$, If 0 scored SC1 for $36 + 0.81x$
3(c)(ii)	141	1	FT <i>their</i> (i)
3(c)(iii)	Outside data range oe	1	
4(a)(i)	1690 or 1687 or 1687.1 to 1687.3	3	M2 for $18 \times 12^2 - 18 \times \pi \times 4^2$ or M1 for either term correct
4(a)(ii)	$1.69[0] \times 10^3$	1	FT <i>their</i> (i)

Question	Answer	Marks	Partial Marks
4(b)	1500 or 1503 to 1504	4	M1 for $[4 \times] 18 \times 12$ M1 for $[2 \times] (12^2 - \pi \times 4^2)$ M1 for $\pi \times 2 \times 4 \times 18$
4(c)(i)	56.5 or 56.6 or 56.54 to 56.56	1	
4(c)(ii)	2.38 or 2.380 to 2.382	2	M1 for $\frac{3 \times \text{their}(c)}{4 \times \pi}$
5(a)(i)	$\frac{12}{50}$ oe	1	
5(a)(ii)	$\frac{16}{50}$ oe	1	
5(a)(iii)	$\frac{7}{50}$ oe	1	
5(b)	$\frac{42}{600}$ oe	2	M1 for $\frac{7}{25}$ soi
5(c)	$\frac{24}{42}$ oe	3	M2 for $\left(\frac{4}{7} \times \frac{3}{6}\right) + \left(\frac{3}{7} \times \frac{4}{6}\right)$ or M1 for $\frac{4}{7} \times \frac{3}{6}$ or $\frac{3}{7} \times \frac{4}{6}$
5(d)	$\frac{792}{117600}$ or $\frac{33}{4900}$ oe	4	M3 for $\left(\frac{2}{50} \times \frac{12}{49} \times \frac{11}{48}\right) + \left(\frac{12}{50} \times \frac{2}{49} \times \frac{11}{48}\right)$ $+ \left(\frac{11}{50} \times \frac{12}{49} \times \frac{2}{48}\right)$ oe or M2 for any two products or M1 for any one of above products
6(a)	10 000	3	M2 for $\frac{7392}{(1-0.16)(1-0.12)}$ oe or M1 for $\div(1-0.16)$ or $\div(1-0.12)$ oe or M1 for 88% is 'equivalent' to 7392

Question	Answer	Marks	Partial Marks
6(b)	5	4	$\log \frac{5000}{7392}$ M3 for $[k =] \frac{\log \frac{5000}{7392}}{\log 0.92}$ oe or correct trials as far as 4 and 5 or M2 for $0.92^k = \frac{5000}{7392}$ oe or at least 3 correct trials or M1 for $7392 \times 0.92^k = 5000$ oe
7(a)(i)		2	B1 for correct ‘hyperbolic shape’ B1 for intersects with axes correct (approx.)
7(a)(ii)	$(-2.5, 0)$ $(0, 2.5)$	2	B1 for each
7(a)(iii)	$x = -2$ $y = 2$	2	B1 for each
7(b)		2	B1 for correct ‘quadratic shape’ B1 for min point at $(-4, 0)$ (approx.)
7(c)	$[x =] -5.30$ $[x =] -3$ $[x =] -1.70$	3	B1 for each correct answer
7(d)	$-5.30 \leq x \leq -3$ and $-2 < x \leq -1.70$	2	B1 for each

Question	Answer	Marks	Partial Marks
8(a)(i)	138	1	
8(a)(ii)	318	1	FT <i>their</i> (i) + 180
8(b)	48.8 or 48.78...	2	M1 for $\tan[x] = \frac{6.13}{5.37}$
8(c)	31.1 or 31.08...	3	M2 for $\frac{6.13 \times 5.37}{2} + \frac{1}{2} \times 6.13 \times 6.42 \times \sin 48$ or M1 for $\frac{6.13 \times 5.37}{2}$ or $\frac{1}{2} \times 6.13 \times 6.42 \times \sin 48$
8(d)	5.11 or 5.111...	3	B2 for 26.1... or M1 for $6.13^2 + 6.42^2 - 2 \times 6.13 \times 6.42 \times \cos 48$
8(e)	201 or 200.9 to 201.1...	4	B3 for 69[.0] or 68.89 to 68.90 or M2 for $\sin C = \frac{\sin 48}{\text{their}(d)} \times 6.42$, $[C = 69.0]$ or M1 for $\frac{\sin C}{6.42} = \frac{\sin 48}{\text{their}(d)}$
9(a)(i)	-2	1	
9(a)(ii)	3	2	M1 for $f\left(\frac{1}{3}\right)$ oe
9(b)	$-\frac{2}{3}$ oe	2	M1 for $g\left(-\frac{1}{2}\right)$ oe
9(c)	$\frac{1}{3}$ and $\frac{7}{3}$	3	M2 for $4 - 3x = \pm 7$ or M1 for $f(x) = \pm 3$ If 0 scored B1 for each answer
9(d)	$3(3x - 5)(x - 1)$	3	B2 for $9x^2 - 12x - 12x + 15$ or M1 for $(4 - 3x)^2 - 1$

Question	Answer	Marks	Partial Marks
10	6.29 or 6.288 to 6.293	6	M5 for $\frac{12 \times \tan 20 \times \tan 50}{\tan 50 - \tan 20}$ or M4 for $BC = \frac{12 \tan 20}{\tan 50 - \tan 20}$ or M3 for $BC \tan 50 = (12 + BC) \tan 20$ or M2 for $\tan 20 = \frac{DC}{12 + BC}$ and $\tan 50 = \frac{DC}{BC}$ or M1 for $\tan 20 = \frac{DC}{12 + BC}$ or $\tan 50 = \frac{DC}{BC}$
11(a)(i)	-2	2	M1 for $2x = 1 - 5$ or $\frac{5}{2} + x = \frac{1}{2}$
11(a)(ii)	2	2	M1 for $-\frac{10}{x} = 1 - 6$ oe or $6x - 10 = x$
11(a)(iii)	-13.5	3	M1 for correct expansion $3 - 6x = 2 - 4x + 28$ M1 for correct collection of <i>their</i> terms $3 - 30 = 6x - 4x$ M1 for $\frac{\text{their}(3 - 30)}{\text{their}(6 - 4)}$
11(b)(i)	$6x^2 + 3x - 7 = 0$	B1	
	Correct sketch OR $\frac{-3 \pm \sqrt{3^2 - 4 \times 6 \times -7}}{2 \times 6}$	M2	M1 for any U-shaped parabola OR for $\frac{b}{2a}$ or $\sqrt{b^2 - 4ac}$ correct
	0.859, -1.359	B1	
11(b)(ii)	0.927, -0.927	2	FT <i>their</i> (b)(i) B1 for each
11(c)	1.41 or 1.414... cao	4	M3 for $5x^2 = 10$ or M2 for $\log 5x^2 [=1]$ or M1 for $\log x^2 + \log 5 [=1]$



Cambridge IGCSE™

CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/43

Paper 4 (Extended)

May/June 2020

MARK SCHEME

Maximum Mark: 120

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam. The answer column of the mark scheme shows the proposed basis on which Examiners would award marks for this exam. Where appropriate, this column also provides the most likely acceptable alternative responses expected from students. Examiners usually review the mark scheme after they have seen student responses and update the mark scheme if appropriate. In the June series, Examiners were unable to consider the acceptability of alternative responses, as there were no student responses to consider.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of 8 printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

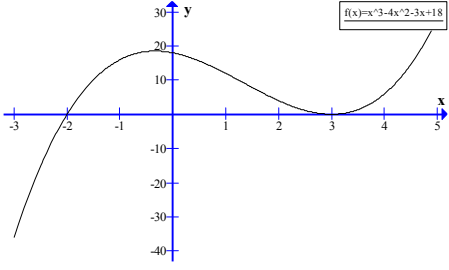
- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation '**dep**' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt	answers which round to
cao	correct answer only
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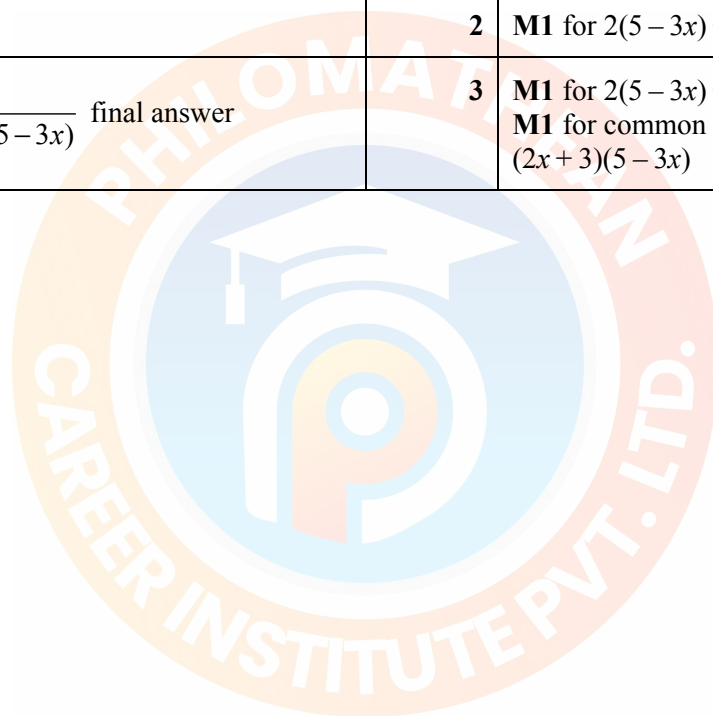
Question	Answer	Marks	Partial Marks
1(a)	$-5, -9$	1	
	$19 - 4n$ oe	2	B1 for $k - 4n$ or $19 - kn$ oe
1(b)	32, 64	1	
	2^{n-1} oe	2	B1 for $2^{(an+b)}$ oe $a \neq 0$
1(c)	54, 70	1	
	$n^2 + 3n$ oe	2	B1 for $an^2 + bn + c$ $a \neq 0$
2(a)	Five points plotted correctly	2	B1 for 3 or 4 points plotted correctly
2(b)	Positive	1	
2(c)(i)	$0.957x - 9.76$	2	or 0.9574..., -9.764 to -9.765 B1 for $0.957x - c$ or $mx - 9.76$
2(c)(ii)	36 or 36.2 or 36.19...	1	FT <i>their</i> (i)
3(a)(i)	5500	3	M2 for $5000 + \frac{5000 \times 2.5 \times 4}{100}$ oe or M1 for $\frac{5000 \times 2.5 \times 4}{100}$ oe
3(a)(i)	12	2	M1 for $\frac{5000 \times 2.5 \times n}{100} = 6500 - 5000$ oe
3(b)(i)	5412.16	3	M2 for $5000 \times \left(1 + \frac{2}{100}\right)^4$ or M1 for $5000 \times \left(1 + \frac{2}{100}\right)^n$, $n > 1$
3(b)(ii)	14	4	M3 for $[n =] \frac{\log\left(\frac{6500}{5000}\right)}{\log 2}$ soi by 13.2 or 13.24 to 13.25 or answer 13 or correct trials as far as 13 and 14 or M2 for $1.02^n = \left(\frac{6500}{5000}\right)$ or at least 3 correct trials or suitable graph or M1 for $5000 \times 1.02^n = 6500$ soi.

Question	Answer	Marks	Partial Marks
4(a)	Correct Sketch 	2	With maximum in second quadrant and minimum on positive x -axis B1 for cubic graph for +ve x^3
4(b)	-1.51 or -1.508 to -1.507 1.24 or 1.244... 4.26 or 4.263 to 4.264	3	B1 for each
4(c)(i)	(-0.333, 18.5) or (-0.3333..., 18.51 to 18.52)	2	B1 for each coordinate
4(c)(ii)	(3, 0)	1	
4(d)	$k < 0$, $k > 18.5$	2	B1FT for each
5(a)(i)	Reflection in the line $y = 3$	1	
5(a)(ii)	Translation $\begin{pmatrix} -5 \\ 4 \end{pmatrix}$	2	B1 for each
5(a)(iii)	Enlargement [centre] (2, -1) [scale factor] $\frac{1}{3}$	2	B1 for each
5(b)(i)	Triangle at (-6, 0), (-2, 0), (-2, -2)	2	B1 for rotation 90° clockwise about (-1, 0) or 90° anticlockwise about another centre
5(b)(ii)	Triangle at (2, 2), (2, 4), (3, 4)	2	B1 for enlargement scale factor $-\frac{1}{2}$, wrong centre or scale factor $\frac{1}{2}$, centre (1, 3)
5(b)(iii)	Stretch [Stretch factor] 3 Invariant line y -axis oe	3	B1 for each
6a)(i)	74	1	
6a)(ii)	18	2	B1 for 64 or 82
6a)(iii)	38	2	B1 for 82

Question	Answer	Marks	Partial Marks
6(b)(i)	Correct graph	3	B1 for minimum at $(10, h)$ where $h < 30$, lq and median correct B1 for uq correct B1 for maximum correct
6(b)(ii)	Answer in range 50 to 60	2	B1 for 74 or 14 to 24
6(b)(iii)	[A is] steeper oe	1	
7(a)	$2 \times 5 \times \cos 30$	M2	or M1 for $\frac{x}{5} = \cos 30$ oe
	8.660...	A1	
7(b)	241 or 240.9... to 241.2...	4	M1 for $3 \times 8.66 \times 5$ M1 for $3 \times \frac{120}{360} \times \pi \times 5^2$ M1 for $\frac{3}{4} \times 8.66^2 \times \sin 60$
7(c)	2890 to 2895	1	FT $12 \times \text{their (b)}$
8(a)(i)	5	1	
8(a)(ii)	4	1	
8(a)(iii)	3.46	2	M1 for $\frac{\sum fx}{100}$
8(b)(i)	$\frac{20}{9900}$ oe	2	M1 for $\frac{5}{100} \times \frac{4}{99}$ oe
8(b)(ii)	$\frac{896}{9900}$	3	M2 for $\frac{16}{100} \times \frac{28}{99} + \frac{28}{100} \times \frac{16}{99}$ oe or M1 for one of the above products
8(b)(iii)	$\frac{9558}{9900}$	2	M1 for $1 - \frac{19}{100} \times \frac{18}{99}$ oe
9(a)	$x = 3$ oe	1	
9(b)	$\left(\frac{1}{2}, 1\right)$ oe	1	
9(c)	$(-2, 0)$	2	B1 for each coordinate

Question	Answer	Marks	Partial Marks
9(d)	$y = \frac{1}{2}x + \frac{3}{4}$ oe	4	3 term equivalent M1 for gradient of $AC = \frac{-4-6}{3-(-2)}$ M1 for $m = \frac{-1}{\text{their gradient}}$ M1 for substituting <i>their</i> (b) into <i>their</i> $y = mx + c$
10(a)	$\frac{1}{2} \times 4x(2x+4) =$ $\frac{1}{2}(2x+1)(4x+5)\sin 30$	M2	M1 for either area
	$\sin 30 = \frac{1}{2}$ and eliminating fractions	M1	
	Expanding brackets	M1	FT
	Completion to $8x^2 + 18x - 5 = 0$ with no errors	A1	
10(b)	$(4x-1)(2x+5) = 0$	M1	or $x = \frac{-18 \pm \sqrt{18^2 - 4 \times 8 \times (-5)}}{2 \times 8}$ or sketch of parabola (U shaped) with one +ve and one -ve zero.
	$\frac{1}{4}, -2\frac{1}{2}$ oe	A2	A1 for each. If 0 scored, SC1 for $\frac{1}{4}, -2\frac{1}{2}$
10(c)	2.25	2	M1 for substituting <i>their</i> positive solution in either area formula.
11(a)	165 or 165.4...	3	M1 for $80^2 + 120^2 -$ $2 \times 80 \times 120 \times \cos 110$ A1 for 27 366 to 27 367
11(b)	43[.0] or 42.97 to 43.11	3	M2 for $\frac{120 \sin 110}{\text{their(a)}}$ or M1 for $\frac{\sin ABC}{120} = \frac{\sin 110}{\text{their(a)}}$
11(c)	283	2	FT 240 + <i>their</i> (b) B1 for 27 or 50 or 130 correctly identified at C

Question	Answer	Marks	Partial Marks
11(d)	1525	5	M1 for $\cos(\text{their } (b)) = \frac{x}{80}$ A1 for 58.4 or 58.5 or 58.40 to 58.54 M1 for $\text{their } 58.5 \div 37$ M1 for correctly converting <i>their</i> time to hours and mins.
12(a)	11	1	
12(b)	1.4 oe	2	M1 for $2x + 3x = 5 - 3 + 5$
12(c)	$\frac{5-x}{3}$ oe	2	M1 for $x = 5 - 3y$ or $y - 5 = -3x$ oe or $\frac{y}{3} = \frac{5}{3} - x$ oe
12(d)	$13 - 6x$	2	M1 for $2(5 - 3x) + 3$
12(e)	$\frac{19}{(2x+3)(5-3x)}$ final answer	3	M1 for $2(5 - 3x) + 3(2x + 3)$ M1 for common denominator $(2x + 3)(5 - 3x)$





Cambridge IGCSE™

CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/41

Paper 4 (Extended)

May/June 2021

MARK SCHEME

Maximum Mark: 120

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

MARK SCHEME NOTES

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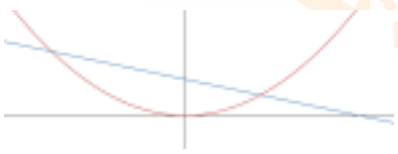
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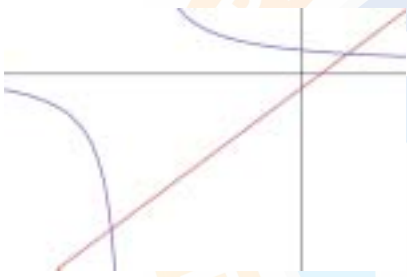
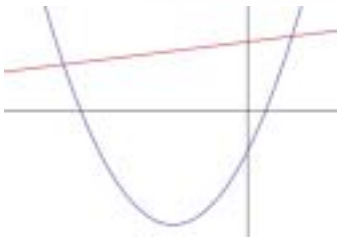
Question	Answer	Marks	Partial Marks
1(a)	Negative	1	
1(b)	48.9[0]	1	
1(c)(i)	$-120x + 11\,700$	2	B1 for $-120x + k$ or $-kx + 11\,700$
1(c)(ii)	6300 or 6294 to 6320	1	FT <i>their</i> (i)
2(a)	Correct triangle (-8, -1) (-7, -1) (-8, 3)	2	B1 for correct translation in y or for correct translation in x
2(b)	Translation $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$	2	B1 for each
2(c)	Correct triangle (4, 4) (4, 5) (8, 5)	2	M1 for correct rotation, wrong centre or for rotation anticlockwise 90° , about O
2(d)	Correct triangle (4, -4) (4, -5) (8, -5)	2	M1 for correct orientation, incorrect position or for reflection in $y = -x$
2(e)	Reflection $y = 0$ oe	2	B1 for each
3(a)	38	1	
	$5n + 8$ oe	2	M1 for $5n + c$ or $kn + 8$, $k \neq 0$
3(b)	26	1	
	$n^2 - 10$ oe	2	M1 for any quadratic expression or 2nd differences of 2
3(c)	6534	1	
	$1089n$ oe	1	
3(d)	-64	1	
	$-(-2)^n$ or $2 \times (-2)^{n-1}$ oe	2	M1 for an expression that gives powers of 2 with alternating signs. If 0 scored, SC1 for $2 \times -2^{n-1}$
4(a)	39.8 or 39.81 to 39.82	2	M1 for at least 5 correct mid-points soi
4(b)	[41], 73, 117, 167, 232, 280, [300]	1	
	In parts (c), (d) and (e), marks can only be earned with an increasing curve		
4(c)	Correct curve (10, 41) (20, 73) (30, 117) (40, 167) (60, 232) (80, 280) (100, 300)	3	M1 for horizontal plot correct M1 for at least 6 vertical plots from their table correct

Question	Answer	Marks	Partial Marks
4(d)(i)	35 to 38	1	
4(d)(ii)	35 to 39	2	B1 for [UQ =] 56 to 59 or [LQ =] 20 to 21
4(e)	46 to 50	2	B1 for 195 or 105 seen
5(a)(i)	−5	1	
5(a)(ii)	25	2	M1 for $g(-2) = 5$ or $(3-x)^2$ soi
5(b)	4	2	M1 for $2x - 1 = 7$
5(c)	$5 - 2x$ oe Final answer	1	
5(d)	$\frac{3}{7}$ oe	3	B2 for $6x - 3 + x - 2x^2 + 2x^2 [= 0]$ or better or B1 for $6x - 3 + x - 2x^2$ or M1 for $(2x - 1)(3 - x) + 2x^2 [= 0]$
5(e)	$3 - x$ Final answer	2	M1 for $x = 3 - y$ or $y + x = 3$
5(f)(i)	Correct sketch 	2	B1 for any quadratic graph
5(f)(ii)	$x = 0$ cao	1	
5(f)(iii)	Correct sketch 	1	
5(f)(iv)	$-2.3[0] < x < 1.3[0]$ or -2.303 to $-2.302 < x < 1.302$ to 1.303	2	For x , do not allow $f(x)$ or y for full marks B1 for either inequality or for $1.3[0]$ and $-2.3[0]$ y range included scores 0
6(a)(i)	6300	3	M2 for $5000 + 5000 \times 6.5 \times 4 \div 100$ oe or M1 for $5000 \times 6.5 \times 4 \div 100$ oe implied by 1300

Question	Answer	Marks	Partial Marks
6(a)(ii)	16	3	B2 for 15.4 or 15.38... or M2 for $\frac{5000 \times 100}{5000 \times 6.5}$ oe or M1 for $\frac{5000 \times 6.5 \times n}{100}$ oe
6(b)(i)	5849.29 or 5850	2	M1 for $5000 \times \left(1 + \frac{4}{100}\right)^4$ oe
6(b)(ii)	18	4	B3 for 17.7 or 17.67... as answer or M3 for $\log \frac{10000}{5000} = n \log \left(1 + \frac{4}{100}\right)$ oe or correct trials including 17 and 18 or good sketch indicating value between 17 and 18 or M2 for $\frac{10000}{5000} = \left(1 + \frac{4}{100}\right)^n$ oe or at least 3 correct trials with $n > 4$ or sketch that could lead to solution or M1 for $10000 = 5000 \times \left(1 + \frac{4}{100}\right)$ oe or at least 2 trials with $n > 4$ or suitable graph
6(c)	Correct sketch 	M3	M2 for suitable graphs, e.g. $y = 1.4^x$ and $y = 1 + 0.065x$ or M1 for one suitable graph, e.g. $y = 1.04x$ or $y = 1 + 0.0656x$
	24	B1	
7(a)	Correctly equating one set of coefficients	M1	or making x or y the subject of one equation
	Correct method to eliminate one variable	M1	May be intersection of two straight line graphs
	$[x =] 2$ $[y =] -3$	A2	A1 for each If 0 scored for whole question, SC1 for answers that satisfy one equation
7(b)(i)	-5	2	M1 for $3x = 4 - 19$ oe
7(b)(ii)	4	2	M1 for $5x - 3x = 15 - 7$ oe

Question	Answer	Marks	Partial Marks
7(b)(iii)	-8	2	M1 for $28 = -4(x + 1)$ oe or $\frac{28}{4} = -(x + 1)$ oe
7(c)	$\sqrt{\frac{p^3}{8q}}$ oe final answer	3	M1 for $\log p^3$ or $\log x^2$ or better M1 for correct use of $\log a + \log b = \log ab$ or $\log a - \log b = \log \frac{a}{b}$
8(a)	$\frac{1}{3}$ oe	1	
8(b)	0	1	
8(c)	$\frac{1}{6}$ oe	2	M1 for $\frac{1}{6} \times \frac{1}{4} + \frac{1}{6} \times \frac{1}{4} + \frac{1}{6} \times \frac{1}{4} + \frac{1}{6} \times \frac{1}{4}$ oe or for $\frac{k}{6 \times 4}$ where $k < \text{their } (6 \times 4)$ or for table of outcomes with correct 4 identified
8(d)	$\frac{5}{12}$ oe	3	M2 for $\frac{4}{6} \times \frac{1}{4} + \frac{2}{6} \times \frac{3}{4}$ or M1 for either of these products seen OR M2 for table of outcomes with correct 10 identified or M1 for table of outcomes with 8 or 9 correct identified
8(e)	$\frac{1}{3}$ oe	2	M1 for at least 6 of (2, 4) (3, 3) (4, 2) (4, 5) (5, 4) (6, 3) (7, 2) (7, 5) identified
9(a)	106 or 106.4 to 106.5	3	M2 for $\sqrt{32^2 + 12^2}$ and $\sqrt{20^2 + 20^2}$ oe or M1 for $32^2 + 12^2$ or $20^2 + 20^2$ oe
9(b)(i)	632	3	M2 for $0.5 \times 20 \times 20$ and $0.5 \times 32 \times 12$ and 20×12 oe or M1 for $0.5 \times 20 \times 20$ or $0.5 \times 32 \times 12$ or $0.5 \times 12 \times 12$ or $0.5 \times 12 \times 12$
9(b)(ii)	6	2	M1 for $\sqrt{\frac{\text{their } 632}{158}}$ or $\sqrt{\frac{158}{\text{their } 632}}$
9(c)	69.4 or 69.42 to 69.45	2	M1 for $\tan[x] = \frac{32}{12}$ oe

Question	Answer	Marks	Partial Marks
10(a)(i)	42	1	
10(a)(ii)	71	1	
10(a)(iii)	109	1	FT 180 – <i>their</i> (ii)
10(a)(iv)	29	1	
10(a)(v)	38	1	
10(b)	74.2 or 74.21 to 74.25...	5	M1 for $[AB =] 2 \times 11 \cos 19$ oe M2 for $[AD =] \frac{\text{their}AB \times \sin(\text{their}42)}{\sin(\text{their}109)}$ or $[BD =] \frac{\text{their}AB \times \sin(\text{their}29)}{\sin(\text{their}109)}$ or M1 for $\frac{AD}{\sin \text{their}42} = \frac{\text{their}AB}{\sin \text{their}109}$ or $\frac{BD}{\sin \text{their}29} = \frac{\text{their}AB}{\sin \text{their}109}$ and M1 for $[\text{Area} =] 0.5 \times \text{their}AB \times \text{their}AD \times \sin(\text{their}29)$ or $[\text{Area} =] 0.5 \times \text{their}AB \times \text{their}BD \times \sin(\text{their}42)$ or $[\text{Area} =] 0.5 \times \text{their}AD \times \text{their}BD \times \sin(\text{their}109)$
11(a)	Correct sketch 	M2	or M1 for exponential graph
	1.43 or 1.430 to 1.431	B1	

Question	Answer	Marks	Partial Marks
11(b)	Algebraic method		
	$(6x-1)(2x+3)=[5+x]$ or better	M1	
	$12x^2 + 15x - 8 = 0$	A1	
	$x = \frac{-15 \pm \sqrt{15^2 - 4 \times 12 \times -8}}{2 \times 12}$	M1	Correct use of formula or correct sketch of parabola
	0.403 or 0.4032...	B1	
	-1.65 or -1.653...	B1	
11(b)	Graphical method (1)		
	Correct sketch of $y = 6x - 1$	M1	
	Correct sketch of $y = \frac{5+x}{2x+3}$ 	M2	or M1 for hyperbolic graph
	0.403 or 0.4032...	B1	
	-1.65 or -1.653...	B1	
11(b)	Graphical method (2)		
	Correct sketch 	M3	M2 for parabola $y = (6x-1)(2x+3)$ oe or M1 for parabola and M1 for $y = 5+x$
	0.403 or 0.4032...	B1	
	-1.65 or -1.653...	B1	



Cambridge IGCSE™

CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/42

Paper 4 (Extended)

May/June 2021

MARK SCHEME

Maximum Mark: 120

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **9** printed pages.

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GENERIC MARKING PRINCIPLE 1:

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- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

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- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

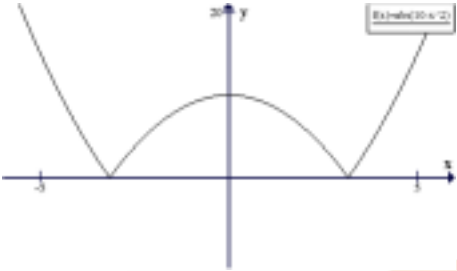
- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation '**dep**' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfww	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

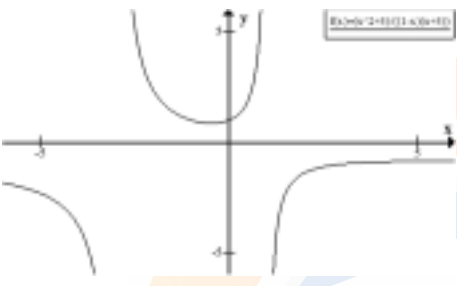
Question	Answer	Marks	Partial Marks
1(a)	32.4	3	M2 for $\frac{250 - 6.5 \times 26}{250} [\times 100]$ oe or $\frac{6.5 \times 26}{250} \times 100$ or M1 for $250 - 6.5 \times 26$ soi by 81 or $\frac{6.5 \times 26}{250}$
1(b)	320	2	M1 for $(\dots) \times \left(1 + \frac{24}{100}\right) = 396.8$ or better
1(c)	23	4	B3 for 22.49... or 22.5 or 22 as answer or M3 for $n \log \left(1 - \frac{3}{100}\right) = \log \left(\frac{200}{396.8}\right)$ oe or correct trials as far as 22 and 23 or sketch indicating value between 22 and 23 or M2 for $\left(1 - \frac{3}{100}\right)^n = \frac{200}{396.8}$ oe or at least 3 correct trials or a sketch that could lead to solution e.g. $y = 0.97^x$ and $y = 200$ or M1 for $396.8 \times \left(1 - \frac{3}{100}\right)^n = 200$ soi. or at least 2 correct trials
2(a)(i)	Triangle at $(-2, 4)$, $(0, 4)$, $(0, 5)$	2	B1 for rotation 90° clockwise about $(-1, 2)$ or rotation anticlockwise 90° about wrong centre
2(a)(ii)	Stretch [Factor] 2 Invariant [line] $y = -1$ oe	3	B1 for each
2(b)	Translation $\begin{pmatrix} 8 \\ 0 \end{pmatrix}$	2	B1 for each
3(a)	$100 < m \leq 120$	1	
3(b)	$60 < m \leq 80$	1	
3(c)	Any correct statement	1	

Question	Answer	Marks	Partial Marks
3(d)	Correct labelled pie chart (labels indicating masses)	4	B3 for pie chart with all angles correct or B2 for pie chart with two angles correct or B1 for 2 correct angles calculated B1 for correct labels on sectors.
4(a)	Correct sketch 	2	B1 for correct middle section
4(b)	± 4 ± 2	2	B1 for 2 correct solutions
4(c)	$x < -4$ $-2 < x < 2$ $x > 4$	3	B1 for each
4(d)	0 $[k] > 10$	2	B1 for each
5(a)	$20^2 + 10^2$	M1	
	22.360 to 22.361	A1	
5(b)	$\tan 35 = \frac{10}{AB}$ oe	M1	$\frac{\sin 35}{10} = \frac{\sin 55}{AB}$, i.e correct implicit
	14.281...	A1	
5(c)	17.4 or 17.43...	2	M1 for $\sin 35 = \frac{10}{AP}$ oe or $14.28^2 + 10^2$
5(d)	30.5 or 30.52...	3	M1 for $20^2 + 14.28^2 - 2 \times 20 \times 14.28 \times \cos 125$ A1 for 931.5 to 931.6...
5(e)	99.2 to 99.5	3	M2 for $[\cos =]$ $\frac{22.36^2 + (their 17.4)^2 - (their 30.5^2)}{2 \times 22.36 \times (their 17.4)}$ or M1 for $(their 30.5)^2 = 22.36^2 + (their 17.4)^2$ $- 2 \times 22.36 \times (their 17.4) \times \cos APB$

Question	Answer	Marks	Partial Marks
6(a)(i)	31.9	1	
6(a)(ii)	0.55	2	M1 for [UQ =] 32.1 or [LQ =] 31.55 seen
6(a)(iii)	32.15	2	B1 for 128 seen
6(b)(i)	Lower median (average) oe	1	
6(b)(ii)	Smaller IQR oe	1	
7(a)	$\frac{200}{x}$	1	
7(b)	$\frac{200}{x} + \frac{50}{x-40} = \frac{7}{2}$ oe	M1	
	$400(x-40) + 100x = 7x(x-40)$ oe	M2	FT only an equation of the correct form with equivalent difficulty M1FT for $\frac{200(x-40) + 50x}{x(x-40)}$ or better
	Completion to $7x^2 - 780x + 16000 = 0$ with no errors or omissions	A1	With at least one intermediate step
7(c)	2 [h] 22 [min]	5	B4 for 2.37 or 2.371 to 2.372 or 2h 22 to 22.3... min or 142 to 142.3... or B3 for 27.1 or 27.10 to 27.11 and 84.3 or 84.3... , or M2 for $\frac{-(-780) \pm \sqrt{(-780)^2 - 4 \times 7 \times 16000}}{2 \times 7}$ or sketch of parabola (positive x^2) with two positive zeros or M1 for $\sqrt{(-780)^2 - 4 \times 7 \times 16000}$ or $\frac{-(-780) \pm \sqrt{p}}{2 \times 7}$ and M1 for $200 \div \text{their solution from their quadratic if } > 40$
8(a)(i)	$0.05 = \frac{k}{\sqrt{25}}$ oe	M1	
	$k = 0.25$ and $y = \frac{1}{4\sqrt{x}}$	A1	

Question	Answer	Marks	Partial Marks
8(a)(ii)	$[\pm]\frac{1}{12}$ oe	1	
8(a)(iii)	$\frac{1}{16y^2}$ or $\frac{1}{(4y)^2}$ oe	2	M1 for $4y\sqrt{x} = 1$ or better or for $y^2 = \frac{1}{16x}$
8(a)(iv)	$\frac{1}{4}$ oe cao	1	
8(b)	3	2	B1 for 2^3 soi
9(a)	204 or 204.2...	3	M2 for $\pi \times 5 \times \sqrt{(5^2 + 12^2)}$ or M1 for $5^2 + 12^2$ (implied by 13)
9(b)(i)	$\frac{r}{12-r} = \frac{5}{\text{their } 13}$ oe	M1	$\frac{r}{13-5} = \frac{5}{12}$
	$r(\text{their } 13) = 5(12-r)$	M1	M1 dep on first M1 for $12r = 5(13-5)$
	Completion to $r = 3.\dot{3}$ or $3\frac{1}{3}$ or $\frac{10}{3}$ or 3.333... with no errors	A1	
9(b)(ii)	159 or 159.0 to 159.5	3	M1 for $\frac{1}{3} \times \pi \times 5^2 \times 12$ M1 for $\frac{4}{3} \times \pi \times 3.33^3$
10(a)	(0.7) 0.25 oe 0.05 oe	B1	
	0.4 oe 0.2 oe (0.55) 0.25 oe	B1	
10(b)(i)	0.13 oe	3	M2 for $0.6 \times \text{their } 0.05 + \text{their } 0.4 \times$ $\text{their } 0.25$ oe or M1 for one of above products

Question	Answer	Marks	Partial Marks
10(b)(ii)	0.63 oe	3	M2 for $their (b)(i) + 0.6 \times 0.7 + their 0.4 \times their 0.2$ or M1 for $0.6 \times 0.7 + their 0.4 \times their 0.2$ OR M2 for $0.6 \times their 0.75 + their 0.4 \times their 0.45$ oe or M1 for one of above products OR M2 for $1 - 0.6 \times their 0.25 - their 0.4 \times 0.55$ or M1 for $0.6 \times their 0.25$ and $their 0.4 \times 0.55$
10(c)	0.36[0] or 0.3601 to 0.3602 oe	3	M2 for $5 \times 0.7^4 \times 0.3$ oe or M1 for $0.7^4 \times 0.3$ oe
11(a)	$8 - -2 = 10, 3 : 2 = 6 : 4,$ $x = -2 + 6 = 4$ oe 4 to $-1 = 5, y = 4 - 3 = 1$ oe	M2	M1 for each coordinate
11(b)	$y = 2x - 7$ oe final answer	4	B3 for $2x - 7$ as final answer OR M1 for gradient of $AB = \frac{-1-4}{8-(-2)}$ M1 for $m = \frac{-1}{their\left(-\frac{1}{2}\right)}$ M1 for $1 = (their2) \times 4 + c$ or $y - 1 = their2(x - 4))$
11(c)	$2 \times 6 - 7 = 5$ oe	1	
11(d)(i)	$5\sqrt{5}$ or $\sqrt{125}$ final answer	2	M1 for $(8 - (-2))^2 + ((-1) - 4)^2$ oe
11(d)(ii)	25 [.0] cao nfw	3	M1 for $(6 - 4)^2 + (5 - 1)^2$ M1 dep on first M1 for $\frac{1}{2} \times their(d)(i) \times their \sqrt{20}$
12(a)	-10	1	
12(b)	$\frac{1}{4}$ oe	3	M2 for $5 = 8 - 12x$ oe or M1 for $\frac{5}{2-3x} = 4$

Question	Answer	Marks	Partial Marks
12(c)	$\frac{2-x}{3}$ oe	2	M1 for $3x + y = 2$ or $x = 2 - 3y$ or $\frac{y}{3} = \frac{2}{3} - x$ or better
12(d)	$\frac{5}{-4+9x}$ oe final answer	2	M1 for $\frac{5}{2-3(2-3x)}$
12(e)	$\frac{9x^2-12x-1}{2-3x}$ oe final answer	3	M1 for $\frac{(2-3x)(2-3x)-5}{2-3x}$ B1 for $4-6x-6x+9x^2$
13(a)	Correct sketch 	3	B1 for each branch
13(b)	$x = 1,$ $x = -3$	2	B1 for each
13(c)	-3.79 or $-3.791...$ -1 0.791 or 0.7912 to 0.7913	3	B1 for each If 0 scored SC1 for $y = 2x + 3$ sketched and cutting both axes



Cambridge IGCSE™

CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/43

Paper 4 (Extended)

May/June 2021

MARK SCHEME

Maximum Mark: 120

Published

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MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

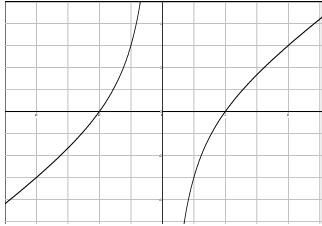
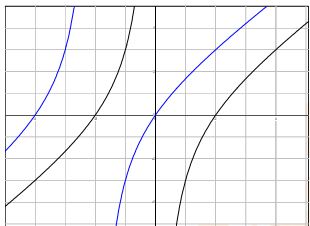
Types of mark

- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation '**dep**' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

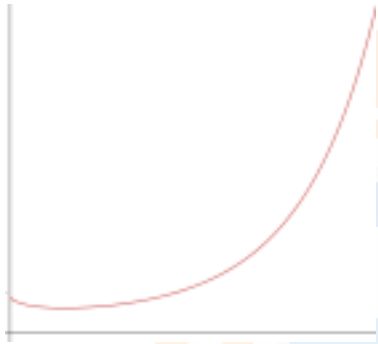
- awrt answers which round to
- cao correct answer only
- dep dependent
- FT follow through after error
- isw ignore subsequent working
- nfwf not from wrong working
- oe or equivalent
- rot rounded or truncated
- SC Special Case
- soi seen or implied

Question	Answer	Marks	Partial Marks
1(a)	Correct sketch 	2	B1 for each branch or B1 for correct but branches joined
1(b)	-2, 2	2	B1 for each
1(c)	-1.24 or -1.236... 3.24 or 3.236...	2	B1 for each
1(d)(i)	Correct sketch 	2	B1 for each branch or B1 FT their $f(x)$ translated in x direction
1(d)(ii)	Translation $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$	2	B1 for each
2(a)	192.5[0]	2	M1 for $55 \times \frac{250}{100}$ oe or better
2(b)(i)	590	3	M2 for $500 + \frac{500 \times 1.5 \times 12}{100}$ oe or M1 for $\frac{500 \times 1.5 \times 12}{100}$
2(b)(ii)	7.81	3	B2 for 597.8..... or 598 seen OR M2 for $500 \left(1 + \frac{1.5}{100} \right)^{12}$ – their (b)(i) oe or M1 for $500 \left(1 + \frac{1.5}{100} \right)^{12}$ oe

Question	Answer	Marks	Partial Marks
2(c)	34	4	B3 for 33.4 or 33.35... OR M3 for $n \log \left(1 + \frac{2.1}{100} \right) = \log 2$ oe or for trials reaching 33 and 34 or good sketch indicating value between 33 and 34 or M2 for $\left(1 + \frac{2.1}{100} \right)^n = 2$ oe or for at least 3 correct trials or for suitable graph or M1 for $\left(1 + \frac{2.1}{100} \right)^n$ oe soi by two trials For M2 and M1 oe includes use of a sum of money
2(d)	8400	3	M2 for $5158.65 \div \left(\frac{100-15}{100} \right)^3$ oe or M1 for $5158.65 \div \left(\frac{100-15}{100} \right)^n$, including $n = 1$
3(a)	$y = -7x + 35$ oe final answer	4	B3 for $-7x + 35$ as final answer OR M1 for gradient of $AB = \frac{0 - -1}{5 - -2}$ oe M1 for gradient of $BC = \frac{-1}{\text{their gradient of } AB}$ (m) M1 for substitution of $(5, 0)$ in $y = (\text{their } m)x + c$ oe
3(b)	3	2	M1 for use of $14 = 2 \times 7$ oe e.g. $\begin{pmatrix} x \\ 14 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ -7 \end{pmatrix}$ or $14 = \text{their } (-7p + 35)$
3(c)	$(-4, 13)$	2	FT $\text{their } p - 7$ for x-coordinate B1 for each.

Question	Answer	Marks	Partial Marks
3(d)	100 nfw	4	M3 for $\sqrt{200} \times \sqrt{50}$ oe or M2 for $\sqrt{7^2 + 1^2}$ oe or $\sqrt{(-2)^2 + 14^2}$ oe or M1 for $(5 - -2)^2 + (0 - -1)^2$ oe or $(-4 - -2)^2 + (14 - 0)^2$ oe OR M3 for $9 \times 15 - 2 \times \frac{1}{2} \times 2 \times 14 - 2 \times \frac{1}{2} \times 7 \times 1$ or M1 for 9×15 and M1 for $\frac{1}{2} \times 2 \times 14$ or $\frac{1}{2} \times 7 \times 1$
4(a)	990 or 989.6 to 989.7...	5	M2 for $\pi \times 5 \times \sqrt{5^2 + 12^2}$ or M1 for $\sqrt{5^2 + 12^2}$ implied by 13 M1 for $2\pi \times 5 \times 20$ M1 for $\frac{1}{2} \times 4\pi \times 5^2$
4(b)(i)	296	3	M2 for $\frac{2050\pi}{3} \div \left(\frac{4}{3}\pi \times 1.2^3\right)$ implied by final answer 296.5 to 296.6 or M1 for $\frac{4}{3}\pi \times 1.2^3$ implied by 7.24 or 7.238 to 7.239...
4(b)(ii)	0.197 or 0.1972 to 0.1975	3	M2 for $\left(\frac{2050\pi}{3} - \text{their } 296 \times \frac{4}{3} \times \pi \times 1.2^3\right) \div \frac{2050\pi}{3}$ oe or for $\left(\text{their } 296 \times \frac{4}{3} \times \pi \times 1.2^3\right) \div \frac{2050\pi}{3} \times 100$ oe or M1 for $\frac{2050\pi}{3} - \text{their } 296 \times \frac{4}{3} \times \pi \times 1.2^3$ or for $\left(\text{their } 296 \times \frac{4}{3} \times \pi \times 1.2^3\right) \div \frac{2050\pi}{3}$
5(a)	178 or 178.4 to 178.5	2	M1 for evidence of mid-values
5(b)	[x =] 7 [y =] 18	3	B2 for $y = 18$ or M1 for $([x \times 0] + y [\times 1] + 30 \times 2 + 45 \times 3) \div 100 = 2.13$ oe If 0 scored SC1 for $\text{their } x + \text{their } y = 25$
6(a)(i)	Positive	1	
6(a)(ii)	$1.03x + 1.1[0]$ or $1.027...x + 1.095...$	2	B1 for $1.03x + k$ or $kx + 1.1[0]$ or $1.027...x + k$ or $kx + 1.095...$

Question	Answer	Marks	Partial Marks
6(a)(iii)	73	1	FT <i>their</i> (a)(ii)
6(b)(i)	23	1	
6(b)(ii)	54	1	
6(b)(iii)	46.5	1	
7(a)	17	3	M2 for $x + 5 + 8x + x + 5 = 180$ oe or M1 for angle $A = x + 5$
7(b)	10.1 or 10.10...	4	B3 for correct sketch indicating roots or for $\frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-92)}}{2(1)}$ oe or B2 for $p^2 - 2p + p - 2 [= 90]$ or better or M1 for $(p + 1)(p - 2) [= 90]$
7(c)	7	3	M2 for $2(y - 4) = y - 1$ or better or M1 for $\frac{y - 4}{y - 1} = \sin 30$ If 0 scored SC1 for $\sin 30 = 0.5$
7(d)	2.04 oe	6	B4 for $(5w - 1)(w + 3)$ or correct sketch indicating roots or $\frac{-14 \pm \sqrt{14^2 - 4(5)(-3)}}{2(5)}$ or B3 for $5w^2 + 14w - 3 = 0$ and M1 for correct calculation of area of triangle with <i>their</i> positive w OR M1 for $(w + 1)^2 + (2w + 3)^2 = 13$ B1 for $w^2 + w + w + 1$ oe or $4w^2 + 6w + 6w + 9$ oe and M1 for correct calculation of area of triangle with <i>their</i> positive w
8(a)	$\frac{18^2 + 13^2 - 16^2}{2 \times 18 \times 13}$	M2	M1 for $16^2 = 18^2 + 13^2 - 2 \times 18 \times 13 \cos(\dots)$
	59.574 to 59.575	A1	
8(b)	164 or 163.8 to 163.9	3	M1 for $\frac{1}{2} \times 18 \times 7$ oe M1 for $\frac{1}{2} \times 18 \times 13 \times \sin 59.57$ oe

Question	Answer	Marks	Partial Marks
8(c)	15.5 or 15.52...	2	M1 for $\sin 59.57 = \frac{\text{distance}}{18}$ oe
8(d)	191 or 190.5 to 190.6	6	M2 for $7^2 + 13^2 - 2 \times 7 \times 13 \cos(90 + 59.57)$ or B1 for [angle $BCD =$] 149.57 M2 for $\frac{7 \sin(90 + 59.57)}{\text{their } BD}$ or M1 for $\frac{\text{their } BD}{\sin(90 + 59.57)} = \frac{7}{\sin DBC}$ M1 for $180 + \text{their } DBC$ oe
9(a)	Correct sketch 	2	B1 for correct shape but cutting either axis or without minimum
9(b)	(0.368, 0.692) or (0.3678 to 0.3679, 0.6922...)	2	B1 for each
9(c)(i)	0.237 or 0.2369 to 0.2370 2.31 or 2.311....	3	M1 for correct line sketched B1 for one correct
9(c)(ii)	$[0 <] x \leq 0.237$ or 0.2369 to 0.2370 $x \geq 2.31$ or $2.311...$	2	B1 for each
10(a)(i)	0.055 oe	3	M2 for $0.9 \times 0.06 + 0.1 \times 0.01$ oe or M1 for 0.9×0.06 oe or 0.1×0.01 oe
10(a)(ii)	0.945 oe	1	FT 1 – <i>their</i> (a)(i)
10(a)(iii)	189	1	FT $200 \times \text{their (a)(ii)}$
10(b)	0.15 oe	2	M1 for $0.8 \times p = 0.12$ oe
11(a)(i)	8	1	
11(a)(ii)	29	2	M1 for $g(3) = 9$ or $3(x^2) + 2$
11(a)(iii)	8	3	B1 for $h(g(3)) = 2^9$ or 512 B1 for $g(h(3)) = 2^6$ or 64

Question	Answer	Marks	Partial Marks
11(a)(iv)	$\frac{x-2}{3}$ oe final answer	2	M1 for $x = 3y - 2$ or $y - 2 = 3x$ or $\frac{y}{3} = x + \frac{2}{3}$
11(a)(v)	$\log_2 x$ or $\frac{\log x}{\log 2}$	2	M1 for $x = 2^y$ or $x \log 2 = \log y$ oe
11(b)(i)	$4\frac{1}{2}$ oe	2	B1 for $[\log_3 81 =]4$ or $\left[\log_9 \left(\frac{1}{3}\right) =\right] -\frac{1}{2}$
11(b)(ii)	125	2	B1 for $25 = b^{\frac{2}{3}}$ oe





Cambridge IGCSE™

CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/41

Paper 4 (Extended)

May/June 2022

MARK SCHEME

Maximum Mark: 120

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **9** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

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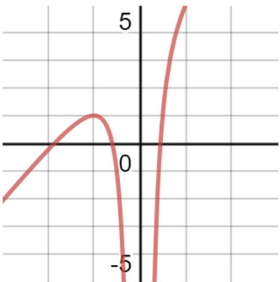
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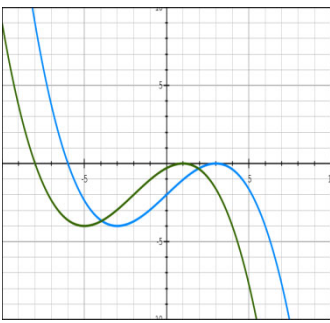
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- nfwf not from wrong working
- oe or equivalent
- rot rounded or truncated
- SC Special Case
- soi seen or implied

Question	Answer	Marks	Partial Marks
1(a)	correct triangle (-4, 1) (-3, 1) (-3, 1.5)	2	B1 for translation $\begin{pmatrix} k \\ 2 \end{pmatrix}$ or $\begin{pmatrix} -2 \\ k \end{pmatrix}$ or translation through $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$
1(b)	correct triangle (-1, 1.5) (-1, 2) (-2, 2)	2	B1 for reflection in either $y = k$ or $x = 0.5$
1(c)	Rotation 180° (1, -1.5)	3	B1 for each
1(d)	correct triangle (3, 1) (3, 2) (1, 1)	2	B1 for any enlargement scale factor -2
2(a)(i)	31	1	
2(a)(ii)	260	2	B1 for 40 seen
2(a)(iii)	27	2	M1 for $\frac{70}{100} \times 300$ soi or $\frac{30}{100} \times 300$
2(b)	33.1	2	M1 for at least three mid-values soi
2(c)(i)	$y = 0.618x + 13.6$	2	B1 for $0.618x + k$ or $kx + 13.6$ or $0.62x + 14$
2(c)(ii)	positive	1	
2(c)(iii)	32 or 32.0 to 32.6	1	FT <i>their (c)(i)</i> if linear eqn
3(a)	correct sketch 	3	B2 for correct branches but joined or touching y-axis B1 for one correct branch
3(b)	$x = 0$	1	
3(c)	(-1, 1)	1	

Question	Answer	Marks	Partial Marks						
3(d)	$-2.31 \leq x < 0$ and $0 < x \leq 0.388$	4	B3 for $-2.3 \leq x \leq 0.388$ or $-2.31 \leq x \leq 0.39$ or B2 for $-2.3 \leq x \leq 0.39$ or $-2.31 \leq x$ or $x \leq 0.388$ or B1 for -2.31 or 0.388 seen or for correct sketch						
4(a)	180	2	M1 for $\frac{216}{5+1}[\times 5]$						
4(b)	741	2	M1 for $\frac{171}{3}$						
4(c)	1505	3	M2 for $1400 + 3(1400 \times \frac{2.5}{100})$ oe or M1 for $3(1400 \times \frac{2.5}{100})$ oe						
4(d)(i)	3247.30 or 3250	2	M1 for $3000 \times \left(1 + \frac{2}{100}\right)^4$ oe						
4(d)(ii)	15	4	B3 for 14.5 or 14.52 to 14.53 seen or M3 for $n \log(1.02) = \log\left(\frac{4000}{3000}\right)$ oe or for correct trials reaching 14 and 15 or good sketch indicating value between 14 and 15 or M2 for $1.02^n = \frac{4000}{3000}$ or at least three correct trials for $n > 4$ or suitable graph or M1 for $3000 \times 1.02^n = 4000$ soi by at least 2 correct trials for $n > 4$						
5(a)	<table border="1"><tr><td>14</td><td>18</td><td>$4n + 2$ oe</td></tr><tr><td>6</td><td>12</td><td>$n^2 - n$ oe</td></tr></table>	14	18	$4n + 2$ oe	6	12	$n^2 - n$ oe	6	B2 for all four numbers correct or B1 for at least two correct B2 for $4n + 2$ oe or M1 for $4n + k$ B2 for $n^2 - n$ oe or M1 for any quadratic or for second differences of 2 seen
14	18	$4n + 2$ oe							
6	12	$n^2 - n$ oe							
5(b)	$n^2 + 3n + 2$ or $(n + 1)(n + 2)$	1	FT <i>their</i> grey + white if both in terms of n						

Question	Answer	Marks	Partial Marks
5(c)	94	4	M1 for <i>their</i> $k^2 + 3k + 2 = 600$ oe M1 for correct method for solving <i>their</i> quadratic M1 for substituting <i>their</i> positive integer (from a quadratic) k into <i>their</i> $4n+2$
5(d)	$\frac{343}{1728}$ oe	2	M1 for $\left(1 - \frac{5}{12}\right)^3$ oe
5(e)	$\frac{56}{153}$ oe	3	FT <i>their</i> 18 from (a) for M marks only M2 for $\frac{4}{18} \times \frac{14}{17} + \frac{14}{18} \times \frac{4}{17}$ oe M1 for one product
6(a)	26.5 or 26.47 to 26.48	4	B1 for $\angle AOC = 150$ or $\angle BOC = 75$ soi M1 for $\frac{\text{their} 150 \times \pi \times 5^2}{360}$ or $\frac{\text{their} 75 \times \pi \times 5^2}{360} \times 2$ M1 for $\frac{1}{2} \times 5 \times 5 \times \sin[\text{their} 150]$ or $\frac{1}{2} \times 5 \times 5 \times \sin[\text{their} 75 \times 2]$
6(b)	5.5[0] or 5.49 to 5.51...	5	M2 for $\frac{(360 - 40)}{360} \times 2 \times \pi \times 12 = 2\pi r$ oe or $\frac{(360 - 40)}{360} \times \pi \times 12^2 = \pi r 12$ oe or M1 for $\frac{(360 - 40)}{360} \times 2 \times \pi \times 12$ or $\frac{(360 - 40)}{360} \times \pi \times 12^2$ If minor sector SC1 for radius = $\frac{4}{3}$ oe AND M2 for $12^2 - \text{their}(\text{radius})^2$ oe dependent on at least M1 or SC1 or M1 for $h^2 + \text{their}(\text{radius})^2 = 12^2$ oe dependent on at least M1 or SC1
7(a)	2300	2	M1 for $2 \times 23 \times 50$ oe

Question	Answer	Marks	Partial Marks
7(b)(i)	4.6	3	M2 for $1.15 \times \sqrt[3]{\frac{18400}{287.5}}$ oe or M1 for $\frac{h^3}{1.15^3} = \frac{18400}{287.5}$ or $\sqrt[3]{\frac{18400}{287.5}}$ or $\sqrt[3]{\frac{287.5}{18400}}$ seen
7(b)(ii)	2 : <i>their</i> (b)(i) or <i>their</i> (a) : 18400 soi	M1	FT <i>their</i> figures
	showing comparison of length ratio or volume ratio	M1	
	not similar	A1	Dep on M1M1
8(a)	27.6 or 27.58 to 27.59	3	M2 for $((-11) - 8)^2 + (7 - (-13))^2$ oe or M1 for $((-11) - 8)$ or $(7 - (-13))$ oe
8(b)	$y = -\frac{1}{4}x + 8\frac{1}{4}$ oe	6	B5 for answer $-\frac{1}{4}x + 8\frac{1}{4}$ OR B2 for (5, 7) or B1 for (5, <i>k</i>) or (<i>k</i> , 7) M1 for $\frac{11 - -5}{6 - 2}$ oe (=m ₁) M1 for grad = $-\frac{1}{\text{their } m_1}$ M1 for substituting <i>their</i> (5, 7) into $y = (\text{their } m)x + c$
9(a)(i)	13	1	
9(a)(ii)	$-8x - 10$ oe final answer	2	M1 for $2 - 4(2x + 3)$ or better seen
9(a)(iii)	$\frac{2-x}{4}$ oe final answer	2	M1 for correct first step $y + 4x = 2$ or $\frac{y}{4} = \frac{1}{2} - x$ or $x = 2 - 4y$ $y - 2 = -4x$
9(a)(iv)	3.52 or 3.523 to 3.524 or $\log_3 48$ or $\frac{\log 48}{\log 3}$	2	M1 for $\log 48 = \log 3^x$ or better or suitable sketch

Question	Answer	Marks	Partial Marks
9(b)(i)		1	
9(b)(ii)	$2m(x)$	1	
10(a)	$16xy^2$ Final answer	2	M1 for $\frac{4x^2y}{3} \times \frac{12y}{x}$ or better
10(b)	$\frac{-x^2 + 6x - 7}{2(x - 3)}$ Final answer	3	B1 for $2[\times 1] - (x - 3)(x - 3)$ B1 for $2(x - 3)$ as denominator
10(c)	$2^2 \times a + 2 \times b - 5 = -3$ oe $3^2 \times a + 3 \times b - 5 = 4$ oe	M2	M1 for either
	correctly equating one set of coefficients or making a or b subject of one equation	M1	FT
	correct method for eliminating one variable or correctly substituting in other equation	M1	FT
	$[a=] 2$ $[b=] -3$	B2	B1 for each
11(a)	349 or 350 or 349.4 to 349.5...	4	M3 for $[0.9 \times 2.1 + 2 \times \frac{1}{2}(2.1 \times 2.1 \tan 20)] \times 100$ oe or $\frac{1}{2}(0.9 + (2 \times 2.1 \tan 20 + 0.9)) \times 2.1 \times 100$ oe or M2 for area of cross section $0.9 \times 2.1 + 2[\frac{1}{2}(2.1 \times 2.1 \tan 20)]$ oe or $\frac{1}{2}(0.9 + (2 \times 2.1 \tan 20 + 0.9)) \times 2.1$ oe oe or M1 for $\tan 20 = \frac{x}{2.1}$ If 0 scored SC1 for triangle marked/drawn with 20 or 70

Question	Answer	Marks	Partial Marks
11(b)	19 h 50 (or 51) min	4	B3 for 19.8 or 19.84... or M2 for $\frac{300 \times 1000}{4.2 \times 60 \times 60}$ oe or M1 for 300×1000 or $4.2 \times 60 \times 60$ or for <i>their</i> volume divided by <i>their</i> rate
12(a)	160 or 159.5...	2	M1 for $\frac{1}{2} \times 16 \times 22 \times \sin 115$ oe
12(b)	84[.0] or 84.02 to 84.03	6	B3 for 32.2 or 32.21.. soi OR M2 for $\sqrt{16^2 + 22^2 - 2 \times 16 \times 22 \times \cos 115}$ or M1 for $16^2 + 22^2 - 2 \times 16 \times 22 \times \cos 115$ AND M2 for $\frac{9 \sin 80}{\text{their}(32.21)}$ dependent on cosine rule used or M1 for $\frac{\sin ABD}{9} = \frac{\sin 80}{\text{their}(32.21)}$



Cambridge IGCSE™

CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/42

Paper 4 (Extended)

May/June 2022

MARK SCHEME

Maximum Mark: 120

Published

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Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **10** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

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Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

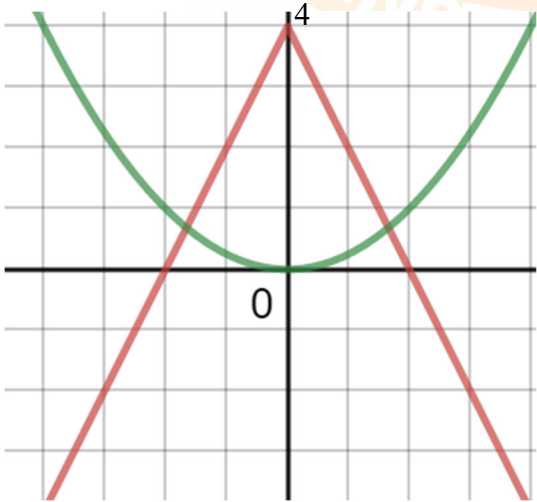
Types of mark

- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation '**dep**' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

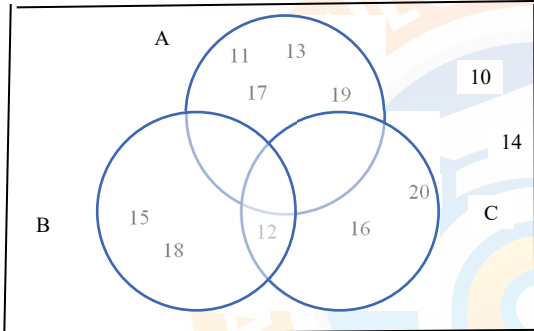
Abbreviations

- awrt answers which round to
- cao correct answer only
- dep dependent
- FT follow through after error
- isw ignore subsequent working
- nfwf not from wrong working
- oe or equivalent
- rot rounded or truncated
- SC Special Case
- soi seen or implied

Question	Answer	Marks	Partial Marks
1(a)	Reflection $x = 1$ oe	2	B1 for each
1(b)	Correct triangle (5, -4) (5, -7) (7, -7)	2	B1 for correct rotation with incorrect centre or 90 rotation anticlockwise about correct centre
1(c)	Reflection $y = x - 1$ oe	2	B1 for each
2(a)(i)	Correct points	2	B1 for 4 or 5 points correct
2(a)(ii)	Positive	1	
2(b)	31.2	1	
2(c)(i)	$y = 3.01x + 15$ [.0]	2	B1 for $y = 3x + 15$ or $y = 3.01x + k$ or $y = kx + 15$ [.0]
2(c)(ii)	Marks per hour oe	1	
2(d)(i)	45	1	FT <i>their</i> (c)(i) if linear and answer is positive
2(d)(ii)	7	1	FT <i>their</i> (c)(i) if linear and answer is positive
3(a)	$2x - 60$ or $2(x - 30)$ final answer	1	
3(b)	$330 - 3x$ or $3(110 - x)$ final answer	3	M2 for $360 - 90 - x -$ (<i>their</i> (a)) oe or B1 for $OAQ = 90$
3(c)	$240 - 3x$ or $3(80 - x)$ final answer	3	M2 for $180 - 90 - (180 -$ <i>their</i> (b)) oe or B1 for $DCB = 90$
4(a)	Correct sketch 	3	B1 for inverted 'v' B1 for symmetrical about y-axis
4(b)	-2, 2	1	

Question	Answer	Marks	Partial Marks
4(c)	Correct sketch	2	Must touch x -axis at origin and without serious curl backs B1 for a u-shaped parabola
4(d)	$x = 0$	1	
4(e)	-1.66 or -1.657 to -1.656 1.66 or 1.656 to 1.657	2	B1 for each
4(f)	Correct region shaded	1	Dependent on at least B2 in (a) and at least B1 in (c)
5(a)(i)	$\frac{1240}{5} \times 3 [= 744]$	1	
5(a)(ii)	434	3	M2 for $[x =] 992 - \frac{744 \times 3}{4}$ oe or M1 for $\frac{744 \times 3}{4}$ oe (558) (186×3) or $\frac{3}{7} \times 1302$ seen or for 992 seen
5(b)(i)	73.6[0]	2	M1 for $92 - \frac{20 \times 92}{100}$ oe or B1 for 18.4
5(b)(ii)	165[.00]	2	M1 for $x \times \frac{100 - 20}{100} = 132$ oe or better
5(c)(i)	894.78	3	M2 for $\frac{1240 \times (100 - 18) \times (100 - 12)}{100 \times 100}$ oe or M1 for $\frac{1240 \times (100 - 18)}{100}$ or $\frac{(their\ 1016.8) \times (100 - 12)}{100}$

Question	Answer	Marks	Partial Marks
5(c)(ii)	2033 nfw	4	B3 for 9.3 or 9.288 to 9.29... OR 8.3 or 8.29 as final value or M3 for $n \log 0.88 = \log \frac{310}{1240 \times 0.82}$ or $n \log 0.88 = \log \frac{310}{\text{their } 894.78}$ or correct trials reaching 9 and 10 or 8 and 9 or good sketch indicating value between 9 and 10 or between 8 and 9 or M2 for $0.88^n = \frac{310}{1240 \times 0.82}$ or $0.88^n = \frac{310}{\text{their } 894.78}$ or at least 3 correct trials or sketch that could lead to solution e.g. $y = 0.88^x$ and $y = 0.3$ or M1 for $1240 \times 0.82 \times 0.88^n = 310$ or $\text{their } 894.78 \times 0.88^n = 310$ or at least 2 trials or suitable graph e.g. $y = 0.88^x$
6(a)	919	2	M1 for at least 5 correct midpoints soi
6(b)	[4], 10, 22, 40, 56, 76, [80]	1	
6(c)	Correct curve or polygon and correct points plotted (870, 4) (890, 10) (900, 22) (920, 40) (940, 56) (950, 76) (1000, 80)	3	M1 for at least 6 horizontal plots correct M1FT for at least 6 vertical plots correct
6(d)(i)	920	1	
6(d)(ii)	42 to 47	2	M1FT for [UQ =] 941 to 944 or [LQ =] 897 to 899
6(e)	72.5	2	B1 for $18+16+20+4$ or $80 - 22$ soi (58) or M1 for $(80 - \text{their } 22)/80 \times 100$
7(a)(i)	118	1	
7(a)(ii)	298 cao	1	
7(b)	680 or 680.1 to 680.2	2	M1 for $420^2 + 535^2$
7(c)	427 or 427.3 to 427.4	3	M2 for $[CD =] \sqrt{420^2 + 750^2 - 2 \times 420 \times 750 \times \cos 28}$ or M1 for $[CD^2] = 420^2 + 750^2 - 2 \times 420 \times 750 \times \cos 28$
7(d)	186000 or 186200 to 186300	3	M1 for area $ABC = 0.5 \times 420 \times 535$ M1 for area $ACD = 0.5 \times 420 \times 750 \times \sin 28$

Question	Answer	Marks	Partial Marks
7(e)	145 or 145 to 146 nfw	4	M2 for $\sin ACD = \frac{750 \times \sin 28}{\text{their } CD}$ or M1 for $\frac{750}{\sin ACD} = \frac{\text{their } CD}{\sin 28}$ And M1 for $360 - 90 - (180 - \text{their acute } C)$ OR M2 for $\cos ACD = \frac{420^2 + 427.367^2 - 750^2}{2 \times 420 \times 427.367}$ or M1 for $750^2 = 420^2 + 427.37^2 - 2 \times 420 \times 427.37 \cos C$ And M1 for $360 - 90 - \text{their obtuse } C$
8(a)	[A =] 11, 13, 17, 19	1	
8(b)		2	B1 for at least 8 elements correct
8(c)	10, 14, 16, 20	1	FT their Venn diagram
8(d)	6	1	FT their Venn diagram
9(a)	55	1	
	$109 - 9n$ oe	2	M1 for $k - 9n$
9(b)	-2	1	
	$(-2)^{7-n}$ oe	2	M1 for $c(-0.5)^k$ or $c \div (-2)^k$

Question	Answer	Marks	Partial Marks
9(c)	84	1	
	$2n^2 + 3n - 6$	2	M1 for any 3 term quadratic or for 2nd differences of 4
10(a)	-84	3	M1 for correct substitution B1 for answer 84
10(b)	$\frac{10y}{3}$ or $3\frac{1}{3}y$ or 3.3 (or 3.33 or 3.333...)y final answer	2	B1 for $\frac{20xy}{6x}$ or $\frac{10xy}{3x}$ or $\frac{20y}{6}$ or $\frac{5y \times 2}{3}$ or correct answer seen
10(c)(i)	$5b(3a - 5c)$ final answer	2	M1 for $b(15a - 25c)$ or $5(3ab - 5bc)$ or correct answer seen
10(c)(ii)	$2x^2y^3(3y^2 - 8x)$ final answer	2	M1 for $x^2y^3(6y^2 - 16x)$ or $2y^3(3x^2y^2 - 8x^3)$ or $2x^2(3y^5 - 8xy^3)$ or better i.e. answers which are correct and have only one common factor left inside brackets e.g. $2x^2y(3y^4 - 8xy^2)$ or correct answer seen
10(c)(iii)	$(2c - 3)(3d + 1)$ final answer	2	M1 for $2c(3d + 1) - 3(3d + 1)$ or $3d(2c - 3) +$ $2c - 3$ or correct answer seen
10(d)	$[x =] \frac{a+2}{3a^2+6a+2}$ oe final answer	4	M1 for correctly eliminating fractions M1 for correctly expanding brackets M1 for correctly collecting all terms in x on one side and other terms on other side of equation M1 for correctly isolating x by factorising and dividing Max 3 marks only if final answer is incorrect
10(e)	$-2 < x < 0.5$ final answer	3	M2 for -2 and 0.5 SOI or M1 for correct graph(s) sketched or M1 for $\frac{1-2x}{2+x} > 0$ oe or B1 for 0.5 soi
11(a)	$(1/3) \times 9^2 \times h$ or $\frac{1}{3}81h$ or $27h$ oe	1	
11(b)	$\frac{10}{h} = \frac{a}{9}, \frac{9}{h} = \frac{a}{10} \left[a = \frac{90}{h} \right]$ oe	1	
11(c)(i)	$[V =] \frac{1}{3}9^2h - \frac{1}{3}a^2 \times 10$ oe isw	1	FT their (a)

Question	Answer	Marks	Partial Marks
11(c)(ii)	$[V =] \frac{1}{3} b^2 (h-1)$ oe isw	1	
11(c)(iii)	$\frac{1}{3} 9^2 h - \frac{1}{3} \left(\frac{90}{h} \right)^2 \times 10 = \frac{1}{3} \left(\frac{h-1}{h} \right)^2 9^2 (h-1)$ oe	M2	M1 for $\frac{h-1}{h} = \frac{b}{9}$ oe
	$h - \frac{1000}{h^2} = \frac{(h-1)^3}{h^2}, \quad h^3 - 1000 = (h-1)^3$	A1	No errors or omissions seen
11(c)(iv)	18.8 or 18.75 to 18.76 cao	2	B1 for 18.8 or 18.75 to 18.76 and negative root as final answers M1 for [quadratic/cubic]sketch(es)
12(a)(i)	0	1	
12(a)(ii)	$\frac{7}{44}$ oe	2	M1 for $\frac{7}{12} \times \frac{6}{11} \times \frac{5}{10}$
12(a)(iii)	$\frac{21}{55}$ oe	3	M2 for $\frac{7}{12} \times \frac{6}{11} \times \frac{4}{10} \times 3$ oe or M1 for $\frac{7}{12} \times \frac{6}{11} \times \frac{4}{10}$ oe
12(b)	420	1	FT (<i>their</i> part (a)(ii)) $\times 2640$

Additional Guidance

5 (a) (i) allow $\frac{3}{5} = 0.6$ then $0.6 \times 1240 = 744$

allow $\frac{5x}{12} = 1240$ then $5x = 14880$ then $x = \frac{14880}{5} = 2976$ then $\frac{3}{12} = \frac{1}{4}$ then $\frac{1}{4} \times 2976 = 744$

but do not allow $\frac{5x}{12} = 1240$ then $5x = 14880$ then $x = 2976$ then $\frac{3}{12} = \frac{1}{4}$ then $\frac{1}{4} \times 2976 = 744$
(division by 5 not seen)

5 (b),(c) all parts use of e.g. $1 - 20\%$, do not allow M marks but correct answers can score.

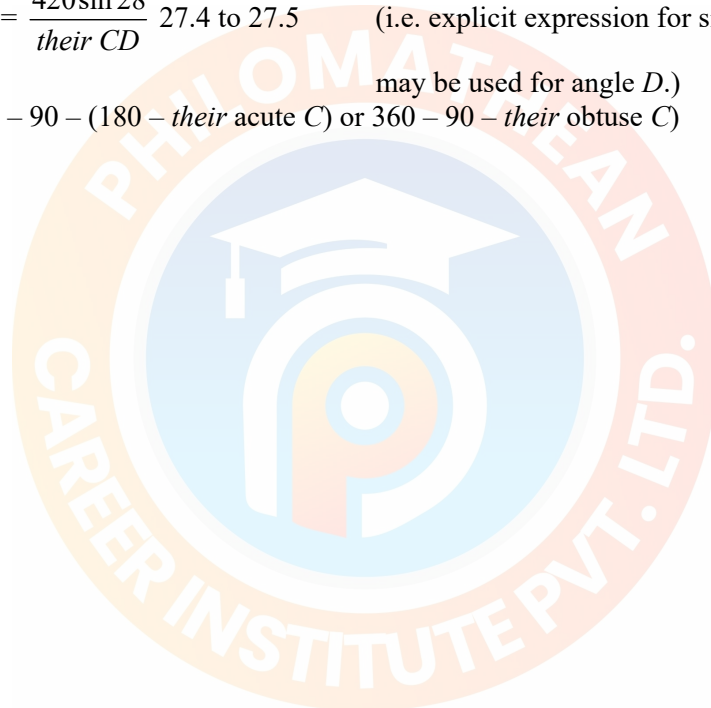
7 (e) e.g. Finding angle D first

B2 for $[C =] 180 - 28 - (27.4 \text{ to } 27.5)$ (124.5 to 124.6)

or M1 for $\sin D = \frac{420 \sin 28}{\text{their } CD}$ 27.4 to 27.5 (i.e. explicit expression for $\sin D$) (note: cosine rule

may be used for angle D .)

and M1 for $360 - 90 - (180 - \text{their acute } C)$ or $360 - 90 - \text{their obtuse } C$





Cambridge IGCSE™

CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/43

Paper 4 (Extended)

May/June 2022

MARK SCHEME

Maximum Mark: 120

Published

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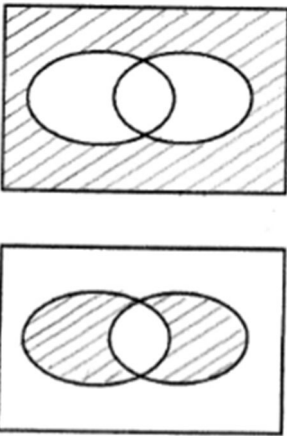
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Abbreviations

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- cao correct answer only
- dep dependent
- FT follow through after error
- isw ignore subsequent working
- nfwf not from wrong working
- oe or equivalent
- rot rounded or truncated
- SC Special Case
- soi seen or implied

Question	Answer	Marks	Partial Marks
1(a)	2621.93	2	M1 for $2500 \times \left(1 + \frac{1.6}{100}\right)^3$ oe
1(b)	1.75	3	M2 for $2500 \times \frac{r}{100} \times 5 = 2718.75 - 2500$ oe or M1 for $2500 \times \frac{r}{100} \times 5$ or $\frac{2718.75 - 2500}{5}$ (43.75) or 8.75 seen
1(c)	31	4	B3 for 30.55 to 30.56 OR M3 for $n \log\left(1 + \frac{1.55}{100}\right) = \log\left(\frac{4000}{2500}\right)$ oe or correct trials as far as 30 and 31 or good sketch indicating value between 30 and 31 or M2 for $\left(1 + \frac{1.55}{100}\right)^n = \frac{4000}{2500}$ oe or at least 3 correct trials or sketch that could lead to solution e.g. $y = 1.0155^x$ and $y = \frac{4000}{2500}$ or M1 for $2500 \times \left(1 + \frac{1.55}{100}\right)^n = 4000$ soi. or at least 2 correct trials
2(a)	7.48	2	M1 for evidence of mid-values
2(b)	6.5 7.5	1	
2(c)(i)	Correct cf curve through 6 points	4	B3 for curve through 5 correct points or B2 for curve through 4 correct points or correct cfs [9], 27, 54, 73, 89, 100 or B1 for curve through 4 correct points or 3, 4 or 5 cfs
2(c)(ii)	7.2 to 7.4	1	FT Only from increasing diagram
2(c)(iii)	2.0 to 2.4	2	B1FT for $lq = 6.3$ to 6.5 or $uq = 8.5$ to 8.7 FT only from increasing diagram
2(c)(iv)	34 to 38	2	B1 FT for 62 to 66 seen
3(a)	Triangle at (1, 0), (–5, 0), (–5, –3)	2	B1 for enlargement with centre (4, 3) wrong scale factor or enlargement scale factor 3 with wrong centre.
3(b)(i)	Translation $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$	2	B1 for each

Question	Answer	Marks	Partial Marks
3(b)(ii)	Stretch Invariant line $y = 4$ [Factor] 2	3	B1 for each
3(c)	e.g. (7, 2) $x = 5.5$ (6, 3) $x = 4.5$ (8, 1) $x = 6.5$ (9, 0) $x = 7.5$ (5, 4) $x = 3.5$ (4, 5) $x = 2.5$ (3, 6) $x = 1.5$	2	B1 for point on $x + y = 9$ and $x = k$
4(a)	2.5 oe	2	M1 for $4x = 7 + 3$ oe
4(b)	-6.95	2	M1 for correct substitution or B1 for 13.9 seen
4(c)	correctly equating one set of coefficients	M1	Allow one incorrect number or making x or y the subject of 1 equation
	Correct method to eliminate one variable	M1	e.g. Adding, subtracting substitution
	$x = 5, y = 2$	A2	A1 for either nfw. If 0 scored, SC1 for a pair of numbers that satisfy either equation or for correct solutions with no working.
4(d)	$\frac{4x}{y(x-2)}$ oe final answer	4	B1 for $2x(x+2)$ B1 for $(x-2)(x+2)$ M1 for inverting and changing sign to multiplication at any stage
5(a)	Correct Sketch 	2	B1 for any cubic
5(b)	$(-1.29, 7.3[0])$	2	-1.291 to -1.290... 7.303... B1 for each
5(c)	Rotational Order 2 (0, 3)	3	B1 for each

Question	Answer	Marks	Partial Marks
5(d)	–2.49 or –2.491 to –2.490 0.657 or 0.6566... 1.83 or 1.834...	3	B1 for each If 0 scored SC1 for –2.5, 0.66 and 1.8
5(e)(i)	Correct Sketch with y intercept below the y -intercept of $f(x)$	2	B1 for any U shaped quadratic
5(e)(ii)	–1.48 or –1.481... 0.311 or 0.3111... 2.17 or 2.170...	3	B1 for each If 0 scored, SC1 for –1.5, 0.31 and 2.2
6(a)	$\frac{2}{3}\sqrt{12^2 - 6^2}$ oe	M3	M2 for $12^2 - 6^2$ or M1 for attempt at Pythagoras e.g. $BD^2 + 6^2 = 12^2$
	6.9282...	A1	
6(b)	204 or 203.6 to 203.7	4	M1 for $\sqrt{12^2 - 6.928^2}$ M1 for $0.5 \times 12 \times 12 \times \sin 60$ or $0.5 \times \text{their } BD \times 12$ M1 for $\frac{1}{3} \times \text{their } 62.35 \times \text{their } 9.798$ dependent on use of Pythagoras and not BD or BO .
7(a)	Correct shading 	2	B1 for each
7(b)(i)	$\frac{4}{7}$ oe	1	
7(b)(ii)	$\frac{9}{35}$ oe	2	M1 for $\frac{3}{7} \times \frac{6}{10}$ oe
7(b)(iii)	$\frac{22}{45}$ oe	3	M2 for $\frac{3}{7} \times \frac{4}{10} + \frac{4}{7} \times \frac{5}{9}$ oe or M1 for one of above products
8(a)	140	1	

Question	Answer	Marks	Partial Marks
8(b)	$360 - (120 + 125)$ or $60 + 55$ or $180 + 55 - 120$	1	
8(c)(i)	178 or 177.5...	3	M2 for $\sqrt{((\text{their}140)^2 + 65^2 - 2 \times (\text{their}140) \times 65 \times \cos 115)}$ OR M1 for $(\text{their}140)^2 + 65^2 - 2 \times (\text{their}140) \times 65 \times \cos 115$
8(c)(ii)	254 or 255 or 254.3 to 254.5...	4	M2 for $\sin[C] = \frac{\text{their}140 \sin 115}{\text{their}178}$ oe or M1 for $\frac{\sin[C]}{\text{their}140} = \frac{\sin 115}{\text{their}178}$ oe A1 for 45.18 to 45.63 M1 for $360 - 60 - \text{their } C$ oe calculated as answer
8(d)	20.3 or 20.26 to 20.31...	3	M2 for $\frac{\text{their}140 + 65 + \text{their}178}{7 + 3.6 + \frac{\text{their}178}{21.5}}$ or M1 for $\frac{\text{their}178}{21.5}$ or for clear indication of $\frac{\text{total distance}}{\text{total time}}$
9(a)(i)	14	1	
9(a)(ii)	-46	2	B1 for $f(16)$ or M1 for $2 - 3(x + 1)^2$ soi
9(a)(iii)	$-\frac{2}{3}$ oe	2	M1 for $2 - 3x = 4$ or $\frac{2 - x}{3}$
9(a)(iv)	100	2	M1 for $\log x = 2$ or 10^x
9(b)	$\frac{3}{5}$ oe	3	M2 for $1 = 5(2 - 3x)$ oe or M1 for $\frac{1}{2 - 3x} [=5]$
9(c)	$9x^2 - 18x + 9$	3	M1 for $(2 - 3x + 1)^2$ M1FT for $3^2 - 3 \times 3x - 3 \times 3x + (3x)^2$
9(d)	$\frac{2 - 10^y}{3}$ oe	3	M2 for $10^y = 2 - 3x$ or M1 for y or $h(x) = \log(2 - 3x)$ or for 10^y seen
10(a)(i)	62	1	
10(a)(ii)	54	1	
10(a)(iii)	36	1	FT 90 – <i>their BAD</i>

Question	Answer	Marks	Partial Marks
10(a)(iv)	126	1	
10(a)(v)	126	1	
10(b)	$[a =] 36$ $[b =] 12$	5	B3 for $36\sqrt{3}$ or M1 for $[AP =] 6\tan 60$ oe $(6\sqrt{3}) (10.392\dots)$ M1 for $2 \times \frac{1}{2} \times 6 \times \text{their } 6\sqrt{3}$ oe $(62.35\dots)$ B2 for 12π or M1 for $\frac{120}{360} \times \pi \times 6^2 (37.699\dots)$
11(a)	$\frac{400}{x}$	1	
11(b)	$\frac{400}{x-2} - \frac{400}{x} = 10$ oe	M1	
	$400x - 400(x-2) = 10x(x-2)$	M1	FT clearing fractions. Dep on equation in the right form
	$-400x + 800$ or $400x - 800$ and $10x^2 - 20x$	M1	FT Expansion of both brackets of correct form
	Completion to $x^2 - 2x - 80 = 0$ with no errors or omissions	A1	
11(c)	22 min 13 sec	4	B2 for $x = 10$ isw other value of x or M1 for $(x-10)(x+8)$ or $\frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-80)}}{2 \times 1}$ or sketch of parabola with one positive zero and one negative zero M1 for $\frac{400}{\text{their } 10 + \text{their } 10 - 2}$



Cambridge IGCSE™

CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/41

Paper 4 (Extended)

May/June 2023

MARK SCHEME

Maximum Mark: 120

Published

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- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
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2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation 'dep' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

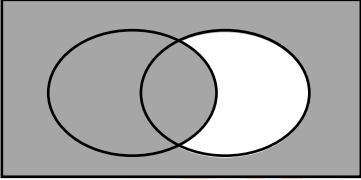
awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfw	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	5	1	
1(a)(ii)	5	1	
1(a)(iii)	6	1	
1(a)(iv)	1.5	2	B1 for [u.q. =] 6.5 or [l.q. =] 5
1(a)(v)	5.6 or $5\frac{3}{5}$	2	M1 for $(1 \times 3 + 3 \times 4 + 8 \times 5 + 7 \times 6 + 7 \times 5 + 1 \times 8) \div 25$
1(b)	72	2	M1 for $\frac{5}{25} [\times 360]$ or $\frac{360}{25} [\times 5]$
1(c)	12	2	M1 for $\frac{8}{3} [\times 4.5]$ or $\frac{4.5}{3} [\times 8]$
2(a)(i)	75	2	M1 for $5^2 [\times 3]$
2(a)(ii)	268 or 268.0 to 268.1...	2	M1 for $\frac{4}{3} \times \pi \times 4^3$
2(b)	2.52 or 2.522 to 2.523...	2	M1 for $\pi \times r^2 \times 6 = 120$ or better
2(c)	7.42 or 7.422 to 7.423	3	M1 for $\frac{1}{3} \times \pi \times r^2 \times 6 = 120$ oe or better M1 for $6^2 + (\text{their } r)^2$ or better
3(a)	Correct sketch 	2	B1 for correct shape but inaccurate or different domain
3(b)	90, 270	2	B1 for each –1 if y cords (0) included
3(c)(i)	60, 120, 240, 300	2	B1 for two or three correct with no extras or four correct with extras –1 if y cords (0.5) included
3(c)(ii)	$60 < x < 120$, $240 < x < 300$	2	B1 for each
3(c)(iii)	Correct areas shaded, i.e. below $y = 0.5$ and above $y = f(x)$	1	
3(d)	$0 [< k <] 1$	1	

Question	Answer	Marks	Partial Marks
4(a)(i)	792 or 792.35	2	M1 for $650 \times \left(1 + \frac{2}{100}\right)^{10}$
4(a)(ii)	22	4	B3 for 21.8 or 21.75... OR M3 for $n \log\left(1 + \frac{2}{100}\right) = \log\left(\frac{1000}{650}\right)$ oe or sketch indicating value between 21 and 22 or correct trials as far as 21 and 22 or M2 for $\left(1 + \frac{2}{100}\right)^n = \frac{1000}{650}$ oe or graph which could lead to solution, e.g. $y = 1.02^x$ and $y = \frac{1000}{650}$ or at least 3 correct trials with $n > 10$ or M1 for $650 \left(1 + \frac{2}{100}\right)^n = 1000$ oe or at least 2 correct trials
4(b)	573	2	M1 for $(...) \times \left(1 + \frac{3}{100}\right)^2 = 607.90$ oe
4(c)	1.2[0] or 1.199 to 1.200	3	M2 for $\sqrt[18]{\frac{247.9}{200}}$ or M1 for $200 \times [...]^{18} = 247.90$ oe or better
5(a)(i)	4	1	
5(a)(ii)	(0, 7)	1	
5(b)(i)	8.94 or 8.944...	3	M2 for $(5 - 1)^2 + (11 - 3)^2$ oe soi by $4^2 + 8^2$ or M1 for $(5 - 1)$ or $(11 - 3)$ or $(1 - 5)$ or $(3 - 11)$ soi by ± 4 or ± 8

Question	Answer	Marks	Partial Marks
5(b)(ii)	$-2x + 17$	5	B1 for (7, 3) M1 for gradient = $\frac{5-1}{11-3}$ oe M1 for perp gradient = $-\frac{1}{\text{their } \frac{1}{2}} = m$ M1 for $\text{their } 3 = \text{their } m \times \text{their } 7 + c$ oe
6(a)	6	1	
6(b)	100	2	M1 for $h((2+1)^2)$ oe or $((x+1)^2 + 1)^2$
6(c)	$4 - 2x$ or $2(2 - x)$ final answer	2	M1 for $3 - 2x + 1$
6(d)	$\frac{3-x}{2}$ oe final answer	2	M1 for $y + 2x = 3$ or for $\frac{y}{2} = \frac{3}{2} - x$ or for $x = 3 - 2y$ or $y - 3 = -2x$ or $\frac{3-y}{2}$ oe
6(e)	3.73 or 3.732... or $2 + \sqrt{3}$	2	M1 j(75) oe
7(a)(i)	26.6 or 26.56 to 26.57	2	M1 for $\tan = \frac{6}{12}$ oe
7(a)(ii)	13.4 or 13.41 to 13.42	2	M1 for $6^2 + 12^2$ oe
7(a)(iii)	56.4 to 56.5	3	M2 for $2 \times 10 + 2 \times 12 + \frac{2 \times \text{their } CMN}{360} \times 2 \times \pi \times \text{their } MC$ or M1 for $\frac{2 \times \text{their } CMN}{360} \times 2 \times \pi \times \text{their } MC$ oe
7(a)(iv)	203 to 205	5	M1 for $\frac{2 \times \text{their } CMN}{360} \times \pi \times (\text{their } MC)^2$ M2 for $\frac{1}{2} \times 12 \times \sqrt{10^2 - 6^2}$ oe or M1 for $10^2 - 6^2$ B1 for 144 or 72 or 36

Question	Answer	Marks	Partial Marks
7(b)	248	3	M2 for $558 \times \left(\sqrt[3]{\frac{240}{810}} \right)^2$ oe or M1 for $\left(\sqrt[3]{\frac{240}{810}} \right)^2$ or $\left(\sqrt[3]{\frac{810}{240}} \right)^2$ or for $\left(\frac{810}{240} \right)^2 = \left(\frac{558}{area} \right)^3$ oe
8(a)	$2t + c = 1470$	B1	
	$3t + 2c = 2480$	B1	
	Correctly eliminating one variable	M1	
	460	B1	
8(b)(i)	$\frac{969}{x} = \frac{969}{x+6} + 2$ oe	M1	
	$969(x+6) = 969x + 2x(x+6)$ oe	M1	dep fractions cleared from equation with two algebraic fractions with linear denominators
	Leading to $x^2 + 6x - 2907 [= 0]$	A1	With at least one step and no errors or omissions
8(b)(ii)	-57 and 51	B2	M1 for $(x + 57)(x - 51)$ or $\frac{-6 \pm \sqrt{(-6)^2 - 4(1)(-2907)}}{2 \times 1}$ or sketch of parabola (+ve x^2) with one positive x intercept and one negative
8(b)(iii)	19	B1	FT $\frac{969}{their\ 51}$ FT only if <i>their</i> 51 and answer are positive integers
9(a)	235	2	M1 for $180 + 55$ or for $360 - 125$ or for $270 - 35$ or for 35 or 55 or 125 or 145 correctly indicated at C.
9(b)	55.1 or 55.06...	2	M1 for $\frac{1}{2} \times 12 \times 16 \sin 35$ oe

Question	Answer	Marks	Partial Marks
9(c)	40.4 or 40.41...	4	M2 for $[\sin C =] \frac{5 \sin 118}{12}$ or M1 for $\frac{12}{\sin 118} = \frac{5}{\sin C}$ oe M1 dep for $180 - 118 - \text{their } C$ dependent on sine rule used to find angle.
9(d)	15.5 or 15.51... nfw	3	M2 for $\sqrt{5^2 + 16^2 - 2 \times 5 \times 16 \cos(35 + \text{their } A)}$ or M1 for $5^2 + 16^2 - 2 \times 5 \times 16 \cos(35 + \text{their } A)$ A1 for 241 or 240.6 to 240.7...
10(a)(i)		1	
10(a)(ii)	$(A \cup B) \cap C'$ oe	1	
10(b)(i)	10	1	
10(b)(ii)	18	1	
10(b)(iii)	$\frac{8}{20}$ oe	1	
10(b)(iv)	$\frac{33}{95}$ oe	2	M1 for $\frac{n}{20} \times \frac{n-1}{19}, n < 20$
10(b)(v)(a)	$\frac{4}{13}$	2	M1 for $\frac{n}{14} \times \frac{n-1}{13}, n < 14$
10(b)(v)(b)	$\frac{48}{91}$	3	M2 for $\frac{6}{14} \times \frac{8}{13} + \frac{8}{14} \times \frac{6}{13}$ oe or M1 for $[2 \times] \frac{n}{14} \times \frac{14-n}{13}$ oe $n < 14$
11(a)(i)	2.1×10^{-5}	1	
11(a)(ii)	3.431×10^{-17}	1	
11(a)(iii)	8×10^{-101}	2	B1 for 0.8×10^{-100} seen
11(a)(iv)	$2.5 \times 10^{2p+1}$	2	B1 for 25×10^{2p} or $2.5 \times 10 \times 10^{2p}$ seen
11(b)	logy or $\log_{10} y$ final answer	1	

Question	Answer	Marks	Partial Marks
11(c)	1.36 or 1.356... or $\frac{\log 14}{\log 7}$ or $\log_7 14$ final answer	1	
11(d)	$\frac{10x^3}{\sqrt{w}}$ or $\frac{10x^3}{w^{\frac{1}{2}}}$ or $10x^3w^{-\frac{1}{2}}$ or $\frac{10x^3\sqrt{w}}{w}$ final answer	4	M1 log10 soi M1 for $\log w^{\frac{1}{2}}$ or $\log \sqrt{w}$ or $\log x^3$ M1 for correct use of $\log p - \log q = \frac{p}{q}$ or correct use of $\log p + \log q = \log pq$





Cambridge IGCSE™

CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/42

Paper 4 (Extended)

May/June 2023

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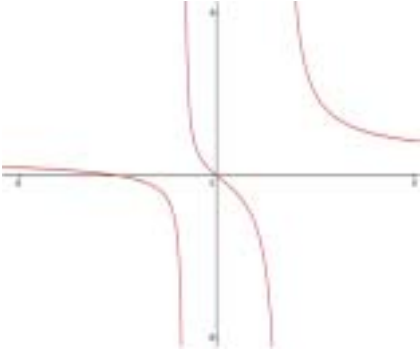
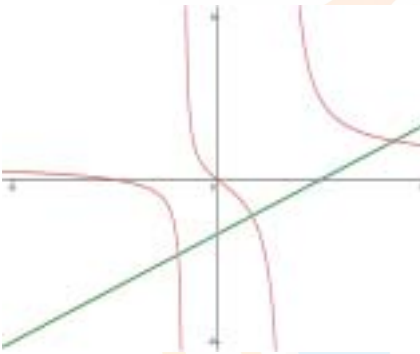
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Abbreviations

- awrt answers which round to
- cao correct answer only
- dep dependent
- FT follow through after error
- isw ignore subsequent working
- nfwf not from wrong working
- oe or equivalent
- rot rounded or truncated
- SC Special Case
- soi seen or implied

Question	Answer	Marks	Partial Marks
1(a)	2	1	
	$20 - 3n$ oe final answer	2	M1 for $k - 3n$ or for correct answer seen but then spoiled
1(b)	$\frac{6}{7}$	1	
	$\frac{n}{n+1}$ oe final answer	1	
1(c)	128	1	
	2^{n+1} oe final answer	2	M1 for 2^{n+k} oe
1(d)	213	1	
	$n^3 - 3$ oe final answer	2	B1 for any cubic seen or M1 for third differences = 6 or for correct answer seen but then spoiled
2(a)	4.5 nfw	2	M1 for $P\left(1 - \frac{4}{100}\right) = 4.32$ oe or better
2(b)	3.52 or 3.522... nfw	2	M1 for $4.32\left(1 - \frac{4}{100}\right)^5$ oe
2(c)	2039 nfw	4	B3 for answer 18.9 or 18.86 to 18.87 or 19 nfw OR M3 for $n \log\left(1 - \frac{4}{100}\right) = \log\left(\frac{2}{4.32}\right)$ oe or good sketch indicating value between 18 and 19 or correct trials as far as 18 and 19 or M2 for $\left(1 - \frac{4}{100}\right)^n = \frac{2}{4.32}$ oe or sketch that could lead to solution or at least 3 correct trials or M1 for $4.32 \times \left(1 - \frac{4}{100}\right)^n = 2$ soi. or at least 2 correct trials
3(a)	Triangle at (1, -6), (2, -4), (4, -4)	2	B1 for reflection in $x = -1$ or $y = k$
3(b)	Triangle at (-4, 7), (-3, 5), (-1, 5)	2	B1 for translation $\begin{pmatrix} -5 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 3 \end{pmatrix}$

Question	Answer	Marks	Partial Marks
3(c)	Rotation 90° [anticlockwise] oe [Centre] origin or (0, 0)	3	B1 for each
3(d)	Stretch [Factor] 2 x-axis (or $y = 0$) invariant	3	B1 for each
4(a)	Correct cumulative frequency curve	4	B2 for 5 or 6 correct heights or B1 for 6, 24, 90, 130, 148, 160 soi B1 for points plotted at right hand end of interval.
4(b)(i)	66 to 69	1	FT <i>their</i> curve
4(b)(ii)	10 to 15	2	B1 for [lq] = 62 to 64 or [uq] = 74 to 77 FT <i>their</i> curve for B1
4(c)	61 to 63 nfw	3	B2 for 34 soi or M1 for 160×0.6 oe soi by 96
5(a)(i)	72	1	
5(a)(ii)	$\frac{5}{\sin\left(\frac{1}{2}\text{their}72\right)}$ oe	M2	M1 for $\sin\left(\frac{1}{2}\text{their}72\right) = \frac{5}{OD}$ oe
	8.506 to 8.507	A1	
5(a)(iii)	172 or 172.0 to 172.2	2	M1 for $\frac{1}{2} \times 8.51^2 \times \sin(\text{their } 72)$ oe or $\frac{1}{2} \times 8.51 \times (10 \text{ or } 5) \sin 54$ oe or $\frac{1}{2} \times (10 \text{ or } 5) \times 5 \tan 54$ oe
5(b)(i)	15.9 or 15.86...	3	M2 for $18^2 - 8.51^2$ or M1 for $VO^2 + 8.51^2 = 18^2$
5(b)(ii)	909 to 913	2	M1 for $\frac{1}{3} \times (\text{their}172) \times (\text{their}15.9)$
5(b)(iii)	11.8 or 11.79 to 11.82	3	M2 for $10 \times \sqrt[3]{\frac{1500}{\text{their}(\mathbf{b})(\mathbf{ii})}}$ oe or M1 for $\sqrt[3]{\frac{1500}{\text{their}(\mathbf{b})(\mathbf{ii})}}$ or $\sqrt[3]{\frac{\text{their}(\mathbf{b})(\mathbf{ii})}{1500}}$ or $\frac{\text{their}(\mathbf{b})(\mathbf{ii})}{1500} = \left(\frac{10}{x}\right)^3$ oe

Question	Answer	Marks	Partial Marks
6(a)	Correct Sketch 	3	B1 for each branch correct
6(b)	$x = 2, x = -1$	2	B1 for each
6(c)	$-3, 0$	2	B1 for each
6(d)(i)	Correct Sketch 	1	
6(d)(ii)	-1.16 or $-1.162...$ 1 5.16 or $5.162...$	3	B1 for each
6(d)(iii)	$-1.16 < x < -1$ $1 < x < 2$ $x > 5.16$	3	B1FT from <i>their (d)(ii)</i> and <i>their (b)</i> for each. FT dep on answers to (d)(ii) that lead to three equivalent inequalities Same accuracy as (d)(ii)
7(a)	$y = \frac{1}{2}x + 6$ oe final answer	3	B2 for $\frac{1}{2}x + 6$ OR M1 for $\frac{10-2}{8-(-8)}$ oe M1 for substituting $(-8, 2)$ or $(8, 10)$ into $y = (\text{their } m)x + c$ oe
7(b)	$\frac{1}{2} \times 4 + 6 = 8$ oe	1	

Question	Answer	Marks	Partial Marks
7(c)	$y = -2x + 16$ oe final answer	3	B2 for $-2x + 16$ OR M1 for $\text{grad} = -\frac{1}{\text{their } \frac{1}{2}}$ M1 for substituting (4, 8) into $y = (\text{their } m)x + c$ oe, $\text{their } m \neq \text{their } \frac{1}{2}$
7(d)(i)	(6, 4)	2	B1 for each coordinate
7(d)(ii)	Kite	1	
7(d)(iii)	17.9 or 17.88 to 17.89 or $8\sqrt{5}$ oe	2	M1 for $(8 - (-8))^2 + (10 - 2)^2$
7(d)(iv)	80 or 79.5 to 80.5	3	M2 for $\frac{1}{2} \times \text{their (d)(iii)} \times \text{their } BD$ or $2 \times \frac{1}{2} \times \text{their (d)(iii)} \times \text{their } BN$ oe i.e. a correct method for the area of $ABCD$. or B1 for $[BN =] 4.47$ or $4.472\dots$ or $2\sqrt{5}$ oe or $[BD =] 8.94$ or $8.944\dots$ or $4\sqrt{5}$ oe or M1 for a correct method for the area of one of the triangles in $ABCD$.
8(a)	40.2 or 40.17 to 40.19	5	B1 for 27 M2 for $\sqrt{30^2 + (\text{their } 27)^2 - 2 \times 30 \times (\text{their } 27) \times \cos 140}$ oe or M1 for $30^2 + (\text{their } 27)^2 - 2 \times 30 \times (\text{their } 27) \times \cos 140$ M1 for $\text{their } 53.57 \div \text{their time from } B \text{ to } C$
8(b)	[0]18.9 or [0]18.89 to [0]18.91	3	M2 for $\frac{\text{their } 27 \sin 140}{\text{their } 53.57}$ oe or M1 for $\frac{\sin \theta}{\text{their } 27} = \frac{\sin 140}{\text{their } 53.57}$
9(a)	$\frac{1}{10}$ oe	1	
9(b)	$\frac{2}{15}$ oe	2	M1 for $\frac{4}{10} \times \frac{3}{9}$

Question	Answer	Marks	Partial Marks
9(c)	$\frac{1}{40}$ oe	3	M2 for $k \times \frac{1}{10} \times \frac{3}{9} \times \frac{1}{8}$ oe, $k = 3, 4, 5$ or 6 or M1 for $\frac{1}{10} \times \frac{3}{9} \times \frac{1}{8}$ oe If 0 scored SC1 for indicating 6 possibilities
9(d)	5	3	M2 for $\left(\frac{4}{10}\right)^{n-1} \times \left(\frac{6}{10}\right) = \frac{48}{3125}$ oe or M1 for $\left(\frac{4}{10}\right)^k \times \left(\frac{6}{10}\right)$, $k \geq 2$ oe
10(a)	$7x - 11y$ Final answer	2	B1 for $7x - ky$ or $kx - 11y$ k not zero
10(b)	$x^2 + 2x$ Final answer	1	
10(c)	$(5b + 4c)(2a - 3b)$ Final answer	2	M1 for $2a(5b + 4c) - 3b(5b + 4c)$ or $5b(2a - 3b) + 4c(2a - 3b)$ or correct answer seen but spoiled
10(d)(i)	$2(x - 3) - 5(2x + 1) = 3(2x + 1)(x - 3)$ oe or better	M1	
	$2x - 6 - 10x - 5$ or better	B1	
	$[3](2x^2 - 6x + x - 3)$ oe or better	B1	
	completion to $6x^2 - 7x + 2 [= 0]$	A1	at least one step with no errors or omissions
10(d)(ii)	$(2x - 1)(3x - 2) [= 0]$ or sketch of parabola showing two positive solutions or $\frac{-(-7) \pm \sqrt{(-7)^2 - 4(6)(2)}}{2 \times 6}$	M2	M1 for pair of brackets giving two terms correct or sketch of any parabola for +ve x^2 or correct formula with $\frac{-(-7)}{2 \times 6}$ or $\sqrt{(-7)^2 - 4(6)(2)}$ seen
	$\frac{1}{2}, \frac{2}{3}$ oe	B1	
11(a)	1	1	
11(b)	-2	3	B2 for $-6x = 12$ oe or better or M1 for $2(1 - 3x) + 5 = 19$
11(c)	$\frac{1-x}{3}$ oe Final answer	2	M1 for $x = 1 - 3y$ or $y + 3x = 1$ or $\frac{y}{3} = \frac{1}{3} - x$ or $y - 1 = -3x$

Question	Answer	Marks	Partial Marks
11(d)	$\frac{1-5y}{2y+3}$ oe Final answer	3	M1 for $y(2x+5) = 1-3x$ oe M1FT dep for $2xy+3x = 1-5y$ dependent on 4 term equation with 2 terms in x . M1FT for factorising and dividing to form $\frac{a+by}{c+dy}$ Max 2 marks if final answer is incorrect.





Cambridge IGCSE™

CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/43

Paper 4 (Extended)

May/June 2023

MARK SCHEME

Maximum Mark: 120

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **9** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

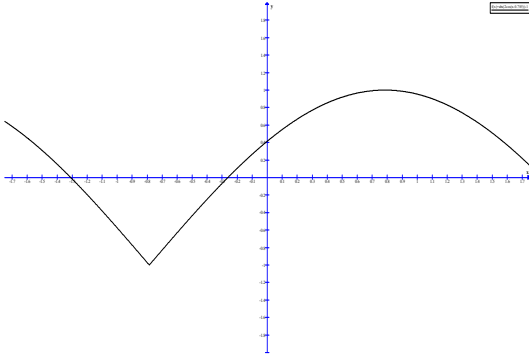
When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation 'dep' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

- awrt answers which round to
- cao correct answer only
- dep dependent
- FT follow through after error
- isw ignore subsequent working
- nfwf not from wrong working
- oe or equivalent
- rot rounded or truncated
- SC Special Case
- soi seen or implied

Question	Answer	Marks	Partial Marks
1(a)	Correct triangle (-5, 5) (-5, 7) (-6, 7)	2	B1 for translation by $\begin{pmatrix} -6 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 4 \end{pmatrix}$
1(b)	Rotation 90 clockwise oe Centre (2, 0)	3	B1 for each
1(c)	Correct triangle (-5, 1) (-5, 3) (-4, 3)	2	B1 for correct reflection in $x = k$ or in $y = -2$
1(d)	Correct triangle (-1, -1) (0, -3) (-1, -3)	2	B1 for correct enlargement, incorrect centre
2(a)	3.6 oe	2	M1 for $\frac{1500}{25}$ or $\frac{1.5}{25}$ or figs $15 \div \frac{25}{60}$
2(b)	75	2	M1 for $\frac{40}{32}$
2(c)	12	3	M1 for $51.5 \div 20.6$ oe M1 for $10 \div (their150 - 25 - their(b))[\times 60]$ oe
3(a)	57.5	1	
3(b)	53.8 or 53.83...	1	
3(c)	$0.791x + 13.2$	2	M1 for $0.791x + k$ or $kx + 13.2$ or $0.79x + 13$
3(d)(i)	53 or 52.7 to 52.8	1	FT their part (b)
3(d)(ii)	18 or 17.90 to 17.95	1	FT their part (b)
3(e)	50, within range or 50 as 6 is not in range	1	
4(a)	67.2[0]	2	M1 for $42 \div 5[\times k]$ where k is 8 or 3 or 1
4(b)(i)	143.84 cao	2	M1 for $124 \times (1 + \frac{16}{100})$ oe or B1 for 19.84 or 143.84 seen
4(b)(ii)	94.5 [0]	2	M1 for $(1 - \frac{2}{9})x = 73.5$ oe
4(c)	300	2	M1 for $2000 \times [0].03 \times 5$ oe

Question	Answer	Marks	Partial Marks
4(d)	19.51	3	M2 for $400 \times \left(1 + \frac{1.6}{100}\right)^3 - 400$ oe or M1 for $400 \times \left(1 + \frac{1.6}{100}\right)^3$ oe
4(e)(i)	$1000 + \frac{1000 \times x \times 18}{100}$ oe	M1	
	$1000 \left(1 + \frac{0.7x}{100}\right)^{18}$ oe	M1	
4(e)(ii)	5.76 or 5.756...	3	M2 for sketch of $[y =] \left(1 + \frac{18x}{100}\right)$ and $[y =] \left(1 + \frac{0.7x}{100}\right)^{18}$ oe with distinct curve and line with clear point of intersection or M1 for sketch of $[y =] \left(1 + \frac{18x}{100}\right)$ or $[y =] \left(1 + \frac{0.7x}{100}\right)^{18}$ oe
5(a)	$\cos ACB = \frac{55^2 + 64^2 - 50^2}{2 \times 55 \times 64}$	M2	M1 for $50^2 = 55^2 + 64^2 - 2 \times 55 \times 64 \times \cos ACB$
	$[ACB =] 48.97... [= 49.0]$	A1	
5(b)	$\sin CAB = \frac{55 \sin 49}{50}$	M2	M1 for $\frac{55}{\sin CAB} = \frac{50}{\sin 49}$ oe
	56.1 or 56.07 to 56.12	B1	
5(c)	1330 or 1327 to 1328...	2	M1 for $0.5 \times 64 \times 55 \times \sin 49$ oe
5(d)	331.9 to 333	2	FT their 5(c) $\div 4$ M1 for 0.5^2 or 2^2 or $\frac{1}{2} \times 32 \times 27.5 \sin 49.0$ oe
6(a)	4.5 oe	2	B1 for $7x - 3x = 13 + 5$ oe

Question	Answer	Marks	Partial Marks
6(b)	$\frac{15}{14}$ oe	3	B1 for $8x - 12 = 3 - 6x$ oe M1 for correctly collecting terms in an equation
6(c)	$\frac{2}{3}, -2$ oe	3	B2 for $3x + 2 = \pm 4$ oe or for $[3](3x - 2)(x + 2)[= 0]$ oe or for $\frac{-4 \pm \sqrt{4(3)(-4)}}{2(3)}$ oe or M1 for $(3x + 2)^2 = 8 \times 2$ oe
6(d)	0.35 – 2.85	3	B2 for –2.851 to –2.850 and 0.350 to 0.351 OR M2 for correct sketch indicating both roots or for $\frac{-5 \pm \sqrt{5^2 - 4(2)(-2)}}{2(2)}$ or M1 for $2x^2 + 5x - 2 [= 0]$ or $-2x^2 - 5x + 2 [= 0]$
6(e)	$[x =] 10y^4$	3	M1 for $\log y^4$ B1 for $1 = \log 10$
6(f)	$\frac{1}{4}$ oe	4	B3 for $n = 3$ or M2 for $2\frac{n}{12} = \frac{n+6}{18}$ or for $12p = 36p - 6$ oe or M1 for $p = \frac{n}{12}$ or $2p = \frac{n+6}{18}$ or for $\frac{n}{12}$ and $\frac{n+6}{18}$ seen
7(a)		3	Correct graph B1 for 1 max in first quadrant B1 for V shape at approx $x = -45$

Question	Answer	Marks	Partial Marks
7(b)	-75, -15	2	B1 for each
7(c)	(45, 1)	1	
7(d)(i)	-64.8 or -64.76... -17.9 or -17.92... 88.8 or 88.78....	3	B1 for each or for -65, -18, 89 If 0 scored SC1 for sketch of $y = 0.005x$
7(d)(ii)	$x < -64.8$ $-17.9 < x < 88.8$	2	FT B1 for one correct
8(a)	Correct method to eliminate one variable	M1	
	$[x =] 1$ $[y =] -2$	A2	If 0 scored, SC1 for answers that satisfy one equation
8(b)(i)	-5	1	
8(b)(ii)	$9x + 4$ final answer	2	M1 for $3(3x+1)+1$ oe
8(b)(iii)	1	3	M2 for $2(3x+1)-3=5$ oe or M1 for $\frac{1}{2(3x+1)-3} = \frac{1}{5}$ oe OR M2 for $f(x) = 4$ oe or M1 for $\frac{1}{2f(x)-3} = \frac{1}{5}$
9(a)(i)	$\frac{5}{80}$ oe	1	
9(a)(ii)	$\frac{28}{80}$ oe	1	
9(b)	$\frac{17}{36}$ oe	2	M1 for $\frac{k}{36}$ where $k < 36$ or $\frac{17}{m}$ where $17 < m < 80$
9(c)(i)	$\left[\frac{12}{44} \times \frac{16}{43} \times \frac{8}{42} \times 6 \right] = \frac{384}{3311}$ oe	1	

Question	Answer	Marks	Partial Marks
9(c)(ii)	$\frac{3648}{9933}$ or $\frac{1216}{3311}$ oe or 0.367 or 0.3672 to 0.3673	3	M2 for $\left(\frac{12}{44} \times \frac{16}{43} \times \frac{8}{42} + \frac{12}{44} \times \frac{16}{43} \times \frac{8}{42} + \frac{12}{44} \times \frac{8}{43} \times \frac{8}{42} + \frac{16}{44} \times \frac{8}{43} \times \frac{8}{42} \right) [\times k < 6]$ or M1 for any two of these products seen
10(a)	156	1	
10(b)	12	2	M1 for $\frac{180 - \text{their}(a)}{2}$
10(c)	102	1	
10(d)	29	3	M2 for $x + x + \text{their} 102 + 20 = 180$ oe or M1 for indicating angle $CDE =$ angle DAC
11(a)	5440	2	M1 for $20 \times 34 \times 8$ or $20 \times 34 \times 16$
11(b)	8.39 or 8.394...	4	B3 for 0.394 or 0.394... OR M3 for $q = \frac{\frac{4}{3}\pi(4)^3 [+ \text{their}(a)]}{680}$ or M2 for $(20 \times 34)q = \frac{4}{3}\pi(4)^3 [\text{their}(a)]$ or M1 for $\frac{4}{3}\pi(4)^3$
11(c)(i)	6.67 or 6.666 to 6.667	3	M2 for $\frac{20}{3}$ and $\frac{34}{5}$ or M1 for $\frac{20}{3}$ or $\frac{34}{5}$ or recognition of 3 by 5

Question	Answer	Marks	Partial Marks
11(c)(ii)	14.5 or 14.53 to 14.55	3	B2 for 6.53 to 6.55 OR M2 for $p = \frac{15 \times (\text{their}(\mathbf{c})(\mathbf{i}))^3 [+ \text{their}(\mathbf{a})]}{20 \times 34}$ or M1 for $20 \times 34 p = 15 \times (\text{their}(\mathbf{c})(\mathbf{i}))^3 [+ \text{their}(\mathbf{a})]$
12	61.5 or 61.46....	6	B1 for AB or $BC = 10$ M2 for $CD = \frac{\text{their } BC}{\sin 30}$ oe or M1 for $\sin 30 = \frac{\text{their } BC}{CD}$ oe M2 for $BD = \frac{\text{their } BC}{\tan 30}$ oe or M1 for $\tan 30 = \frac{\text{their } BC}{BD}$ oe

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CAMBRIDGE INTERNATIONAL MATHEMATICS**0607/41**

Paper 4 (Extended)

May/June 2024**MARK SCHEME**Maximum Mark: 120

Published

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Generic Marking Principles

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- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

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Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

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- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mathematics-Specific Marking Principles

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

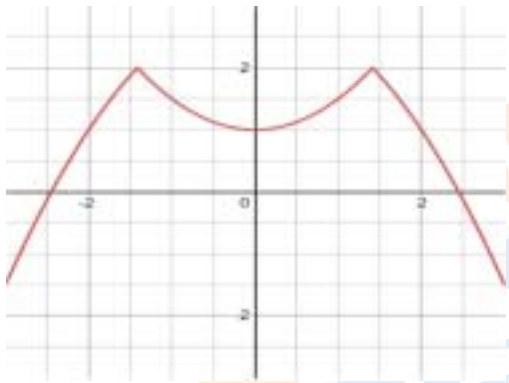
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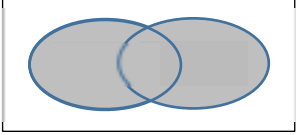
Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfww	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	−4	2	M1 for $3x = -14 + 2$ or $x - \frac{2}{3} = \frac{-14}{3}$
1(a)(ii)	1.5 or $\frac{15}{10}$ oe	2	M1 for $7x + 3x = 26 - 11$ oe or better
1(b)	Correctly equating coefficients or sketch of one equation with negative slope and y intercept	M1	If 0 scored, SC1 for answers that satisfy one equation. SC1 for two correct answers with no working
	Correct method to eliminate one variable or sketch of other equation with negative slope and y intercept	M1	
	$x = -1.5$ oe	A1	
	$y = -2.5$ oe	A1	
1(c)	$x > 4$ and $x < -5$ Mark final answer	4	M2 for correct sketch showing both answers or M1 for appropriate sketch of $y = 2x + 1 $ OR M2 for $2x + 1 > 9$ oe and $2x + 1 < -9$ oe or M1 for either correct inequality or for $2x + 1 = 9$ and $2x + 1 = -9$ oe B1 for $x > 4$ or $x < -5$ Mark final answer
2(a)(i)	47.4 or 47.375	2	M1 for at least 4 correct mid-points soi
2(a)(ii)(a)	[7], 20, 40, 72, [100]	1	
In parts (a)(ii)(b) and (a)(ii)(c) , marks can only be earned with an increasing curve			
(a)(ii)(b)	Correct curve (20, 7), (35, 20), (40, 40), (60, 72), (80, 100)	3	B1 for horizontal plot correct B1FT for at least 4 vertical plots correct
(a)(ii)(c)	24 to 28	2	M1FT for [UQ =] 61 to 63 or [LQ =] 35 to 37
2(b)	44	3	M2 for $10 \times 25 + \frac{(20+k)}{2} \times 15 + \frac{(k+80)}{2} \times 10$ [= 27 × 50] oe or M1 for $\frac{(20+k)}{2}$ or $\frac{(k+80)}{2}$
3(a)	Correct triangle (−5, 0), (−5, −3), (−4, −3)	2	B1 for translation $\begin{pmatrix} 2 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -6 \end{pmatrix}$

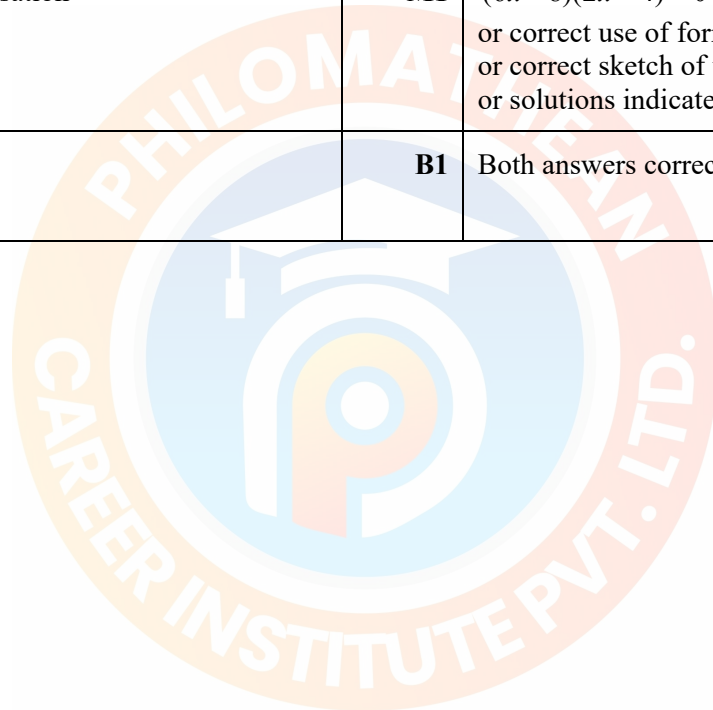
Question	Answer	Marks	Partial Marks
3(b)	Translation $\begin{pmatrix} -2 \\ 6 \end{pmatrix}$	2	B1 for each
3(c)	Correct triangle (3, 6), (3, 7), (6, 7)	2	B1 for correct rotation, incorrect centre or for correct rotation 90 anticlockwise
3(d)	Correct triangle (3, -6), (3, -7), (6, -7)	2	B1 for correct size and orientation, wrong position or reflection in $y = -x$
3(e)	Rotation 90 anticlockwise oe [Centre] (4, -2)	3	B1 for each
4(a)	73.92	2	M1 for $84 \times \frac{(100-12)}{100}$ soi or B1 for 10.08
4(b)	1280	2	M1 for $x \times \frac{100-25}{100} = 960$ oe
4(c)(i)	732.09	2	M1 for 600×1.051^4 oe
4(c)(ii)	11	4	B3 for 10.3 or 10.26 to 10.27 or M3 for $n \log 1.051 = \log \frac{1000}{600}$ or good sketch indicating value between 10 and 11 or correct trials reaching 10 and 11 or M2 for $1.051^n = \frac{1000}{600}$ oe or sketch that could lead to solution e.g. $y = 1.051^x$, $y = 1.67$ or at least 3 correct trials with $n > 4$ or M1 for $600 \times 1.051^n = 1000$ oe or suitable graph e.g. $y = 1.051^x$ or at least 2 trials with $n > 4$
4(d)(i)	75	2	M1 for $[120 \times] \frac{5}{5+3}$ oe
4(d)(ii)	$3[k](1+0.2) : 5[k](1-0.2)$	M2	M1 for $5[k](1-0.2)$ or $3[k](1+0.2)$ or for <i>their</i> (d)(i) $(1-0.2)$
	Leading to 9:10	A1	No errors seen
5(a)	trapezium	1	

Question	Answer	Marks	Partial Marks
5(b)	Angle between tangent and radius oe	1	
5(c)(i)	119	1	
5(c)(ii)	22	1	
5(c)(iii)	39	2	M1 for $180 - \text{their (i)} - \text{their (ii)}$
5(c)(iv)	29	2	B1 for 61 or 122 at centre (may be on diagram)
6(a)	Correct sketch 	3	B2 if peaks below $y = 1$ (by eye) or rounded at peaks not cusps or outside branches convex from below. or B1 for graph symmetrical about y-axis and in all 4 quadrants
6(b)	4.9[0] or 4.898 to 4.899	2	B1 for 2.45 or -2.45 or 2.449... or $-2.449...$ seen
6(c)	-2.24 or $-2.236...$ 2.24 or $2.236...$	2	B1 for each or for both values seen
6(d)	(0, 1)	1	
6(e)	$k = 2$ $k < 1$	2	B1 for each
7(a)(i)	0	1	
7(a)(ii)	$\frac{3}{5}$ oe	1	
7(a)(iii)	$\frac{6}{25}$ oe	3	M2 for $\left(\frac{1}{5} \times \frac{1}{5}\right) + \left(\frac{2}{5} \times \frac{1}{5}\right) + \left(\frac{1}{5} \times \frac{1}{5}\right) + \left(\frac{1}{5} \times \frac{2}{5}\right)$ or correct sample space showing all 6 points or M1 for 2 correct products or correct sample space showing at least 3 points or list of all correct pairs If 0 scored, SC1 for $\frac{12}{25}$

Question	Answer	Marks	Partial Marks
7(a)(iv)	$\frac{6}{25}$	3	M2 for $\left(\frac{2}{5} \times \frac{2}{5}\right) + \left(\frac{1}{5} \times \frac{1}{5}\right) + \left(\frac{1}{5} \times \frac{1}{5}\right)$ or correct sample space showing all 6 points or M1 for 2 correct products or correct sample space showing at least 3 or for listing pairs that sum to 6
7(b)(i)		1	
7(b)(ii)	$A \cap B'$	1	
7(b)(iii)	8	1	
8(a)	$[BC =] \sqrt{8^2 + 7^2 - 2 \times 8 \times 7 \times \cos 73.4}$	M2	or M1 for $[BC^2] = 8^2 + 7^2 - 2 \times 8 \times 7 \times \cos 73.4$ M1 for 81.[00...]
	9.00[0...]	A1	
8(b)	$[\sin \dots] = \frac{8 \times \sin 73.4}{9}$	M2	M1 for $\frac{8}{\sin BCA} = \frac{9}{\sin 73.4}$ oe
	58.4 or 58.41...	B1	
8(c)	26.8 or 26.83...	2	M1 for $\frac{1}{2} \times 7 \times 8 \times \sin 73.4$ oe
8(d)	1.64 or 1.641 to 1.642	3	M1 for area sector $BCD = \pi \times 9 \times 9 \times \frac{36}{360}$ M1 for area triangle $BCD = 0.5 \times 9 \times 9 \times \sin 36$
8(e)	29.7 or 29.65 to 29.66 NFWW	2	M1 for $\frac{36}{360} \times 2\pi \times 9$ oe
9(a)	74 900 or 74 929 to 74 941.1	3	M2 for $\frac{1}{3} \times \pi \times 40^2 \times \sqrt{60^2 - 40^2}$ oe or M1 for $60^2 - 40^2 [=2000]$ oe
9(b)	$\pi \times 40^2 + \pi \times 40 \times 60$ $= 4000\pi$ with no errors	M2	M1 for $\pi \times 40^2$ or $\pi \times 40 \times 60$

Question	Answer	Marks	Partial Marks
9(c)	Ratio areas = $4000\pi : 1000\pi$ implies Ratio sides = 2 : 1 oe	M1	
	[r=] $40 \times 0.5 = 20$ oe	A1	
	ALTERNATIVE $40 \times \sqrt{\frac{1000\pi}{4000\pi}}$ oe = 20 with no errors	(M2)	M1 for $\sqrt{\frac{1000\pi}{4000\pi}}$ or $\sqrt{\frac{4000\pi}{1000\pi}}$ oe or $\left(\frac{40}{x}\right)^2 = \frac{4000\pi}{1000\pi}$ oe
9(d)	11 900 or 11 930 to 11 940 or 3800π	3	M2 for $\pi \times 40^2 + \pi \times 20^2 + \pi \times 60 \times 40 - \pi \times 30 \times 20$ or M1 for $\pi \times 60 \times 40 - \pi \times 30 \times 20$ If 0 scored, SC1 for 3400π or 10 700 or 10 680 to 10 681.4...
10(a)(i)	12	3	M1 for $y = \frac{k}{\sqrt{x+1}}$ oe A1 for $k = 36$ OR M2 for $18 \div \sqrt{\frac{8+1}{3+1}}$ oe or M1 for $\frac{y}{18} = \sqrt{\frac{3+1}{8+1}}$ oe
10(a)(ii)	575	2	M1 for $\sqrt{x+1} = \frac{\text{their } k}{1.5}$ or better
10(b)	18 NFWW	3	M1 for $18 = c \times \sqrt{3+1}$ oe or better M1 for $wy = \text{their } 9\sqrt{x+1} \times \frac{\text{their } 36}{\sqrt{x+1}}$ oe
11(a)	11	1	
11(b)	-2	2	M1 for $3x = -7 + 1$
11(c)	$\frac{5-x}{2}$ oe final answer	2	M1 for $x = 5 - 2y$ or $\frac{y}{2} = \frac{5}{2} - x$ or $2x = 5 - y$

Question	Answer	Marks	Partial Marks
11(d)	$[h(f(x))] = \frac{1}{2(3x-1)-3}$	M1	
	$(5-2x)(6x-5)=7$	A1	All further FTs dep on second stage in correct form $(5-2x)(ax+b)=k$ where a , and b are integers or sketch of rectangular hyperbola
	Correct expansion of brackets	M1	$30x-25-12x^2+10x [=7]$ or sketch of straight line with negative gradient
	Correct rearrangement to 3 term quadratic on one side	M1	$12x^2-40x+32=0$ oe or graphs intersecting twice in 1st quadrant
	Correct factorisation	M1	$(6x-8)(2x-4)=0$ oe or correct use of formula or correct sketch of the quadratic or solutions indicated at points of intersection
	$2, \frac{4}{3}$ oe	B1	Both answers correct



Cambridge IGCSE™

CAMBRIDGE INTERNATIONAL MATHEMATICS**0607/42**

Paper 4 (Extended)

May/June 2024**MARK SCHEME**Maximum Mark: 120

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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This document consists of **9** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

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- the specific skills defined in the mark scheme or in the generic level descriptors for the question
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GENERIC MARKING PRINCIPLE 2:

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Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

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Mathematics-Specific Marking Principles

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
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- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

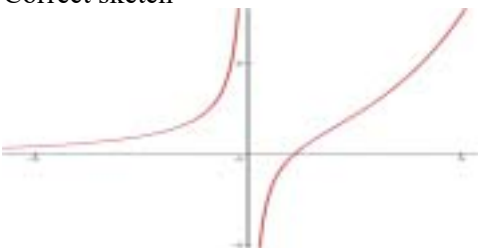
Types of mark

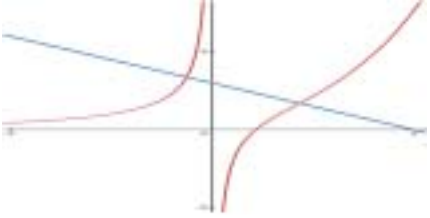
- M** Method marks, awarded for a valid method applied to the problem.
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- B** Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation 'dep' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt	answers which round to
cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
nfww	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	17	3	M1 for $\frac{1}{2} \times 8 \times 7$ M1 for $476 \div \text{their area of triangle}$
1(b)(i)	62.88	1	
1(b)(ii)	2400	3	M2 for $6 \times (\sqrt[3]{8000})^2$ oe or M1 for $\sqrt[3]{8000}$ oe
1(b)(iii)	44 cao	3	M2 for $8000 \div \left(\frac{4}{3}\pi 3.5^3\right)$ or M1 for $\frac{4}{3}\pi 3.5^3$
2(a)	$\frac{210}{7} \times 8$ oe	M1	
2(b)(i)	50.4[0]	1	
2(b)(ii)	47.5 or 47.54...	2	M1 for $\frac{75.50 + \text{their(i)}}{240} \times 100$ or $\frac{240 - (75.50 + \text{their(i)})}{240}$
2(c)	175	2	M1 for $X \times \frac{100 - 20}{100} = 140$ oe or better
2(d)	1.63 [: 1]	2	M1 for $(240 - 75.50 - \text{their (b)(i)}) : (210 - 140)$ or $\frac{8}{7} \times 47.5 : 33.3$ oe
3(a)(i)	2.1 oe	1	
3(a)(ii)	2.5 oe	1	
3(a)(iii)	10	1	
3(a)(iv)	4.47 to 4.48	2	B1 for 7.24 or 7.236... or for 2.76 or 2.763 to 2.764 or $5 + \sqrt{5}$ or $5 - \sqrt{5}$ seen
3(b)(i)	Correct sketch 	3	B2 for both branches correct but joined or with excessive feathering or curl-backs or B1 for one correct branch

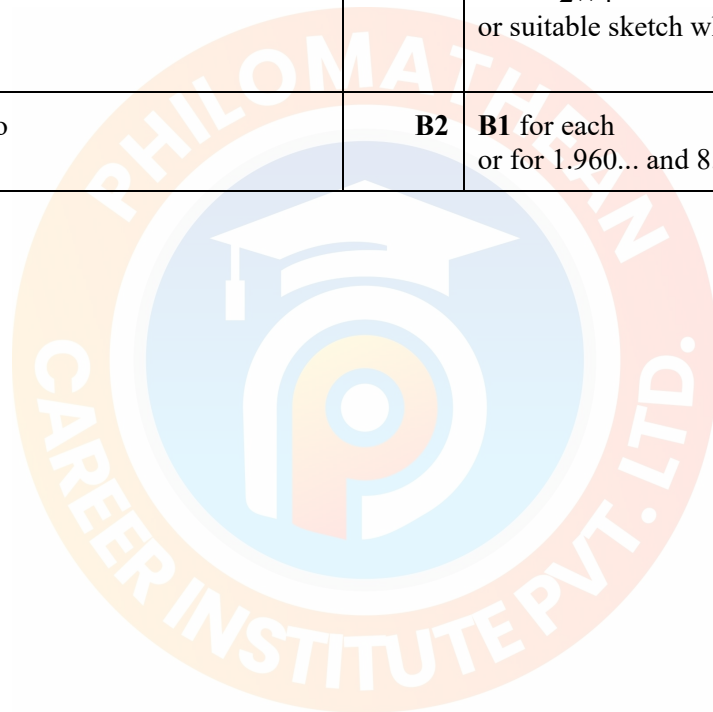
Question	Answer	Marks	Partial Marks
3(b)(ii)	$x = 0$ $y = 0$	2	B1 for each
3(b)(iii)	$k > 0$ cao	1	
3(b)(iv)(a)	Correct sketch 	2	B1 for negative gradient and positive y-intercept or B1 for passing through (3, 0)
3(b)(iv)(b)	-0.382 or -0.3824 to -0.3823 $1.3[0]$ or $1.302...$	2	B1 for each or B1 for both correct values used in an inequality in x only
4(a)	15	3	M2 for $\frac{6 \times 1 + 6 \times 2 + 3 \times 3 + 6 \times 4 + x \times 5 + 4 \times 6}{6 + 6 + 3 + 6 + x + 4} [=3.75]$ or better or M1 for $6 \times 1 + 6 \times 2 + 3 \times 3 + 6 \times 4 + x \times 5 + 4 \times 6$ or better or for $3.75 \times (6 + 6 + 3 + 6 + x + 4)$ or better
4(b)(i)	12.8 or 12.78 to 12.79	2	M1 for at least correct four mid-values soi
4(b)(ii)(a)	7, 28, 51, 67, 70	2	B1 for 3 correct
	In (b), (c) and (d) marks can only be earned with an increasing curve or polygon at least as far as (25, 67 FT)		
4(b)(ii)(b)	Correct curve	3	B1 FT for 5 points with correct heights B1 for 5 points with correct t values
4(b)(ii)(c)	11 to 12	1	FT
4(b)(ii)(d)	26, 27, 28, 29 or 30 cao	2	B1 for 40 to 44 seen or B1FT for <i>their</i> reading from $t = 13$

Question	Answer	Marks	Partial Marks
5(a)	62.4 or 62.35...	2	M1 for $\frac{1}{2} \times 12 \times 12 \times \sin 60$ oe
5(b)	286 or 286.4 to 286.5	3	M2 for $\pi \times (\text{their radius})^2$ or M1 for $2\pi r = 60$ or better
5(c)	277 or 276.9 to 277.0	4	M3 for $[10 \times] \frac{1}{2} \times 6 \times \text{their height}$ oe or $[10 \times] \frac{1}{2} \times \left(\frac{3}{\sin 18} \right)^2 \times \sin 36$ or M2 for $3 \tan 72$ or $\frac{3}{\sin 18}$ oe or M1 for angles 72 or 36 or 18 or 144 seen
6(a)	11 cao	4	B3 for 10.6 or 10.62 to 10.63 OR or M3 for $n \log \left(1 + \frac{2.5}{100} \right) = \log \left(\frac{130}{100} \right)$ oe or good sketch indicating value between 10 and 11 or correct trials reaching 10 and 11 or M2 for $\left(1 + \frac{2.5}{100} \right)^n = \frac{130}{100}$ oe or suitable graph with $n > 1$ or at least 3 correct trials or M1 for $[...] \times \left(1 + \frac{2.5}{100} \right)^n = \frac{130}{100} \times [...]$ oe soi by at least 2 trials with $n > 1$
6(b)	3154 cao	4	B3 for 3153.7 to 3153.8 OR M3 for $25215 \times 1.025^3 - \frac{25215}{1.025^2}$ oe OR M1 for $X \times \left(1 + \frac{2.5}{100} \right)^2 = 25215$ M1 for $25215 \times \left(1 + \frac{2.5}{100} \right)^3$ or for <i>their</i> $24000 \times \left(1 + \frac{2.5}{100} \right)^5$ oe provided <i>their</i> 2027 amount is greater than 25215

Question	Answer	Marks	Partial Marks
7(a)	$-\frac{39}{16}$ oe or -2.44	3	B1 for $[63 =] 24 - 16a$ M1 for <i>their</i> $16a = \text{their } 24 - 63$ oe or better
7(b)	Correctly equating coefficients	M1	Allow 1 arithmetic slip
	Correct method to eliminate one variable	M1	Allow 1 further arithmetic slip
	$[p =] \frac{1}{2}$ oe $[q =] -\frac{1}{4}$ oe	B1	If 0 scored, SC1 for answers that satisfy one equation
7(c)(i)	$(c + 7)(c - 8)$ oe	2	B1 for $(c + a)(c + b)$ where $ab = -56$ or $a + b = -1$ or $c(c + 7) - 8(c + 7)$ or $c(c - 8) + 7(c - 8)$
7(c)(ii)	$-7, 8$	1	FT <i>their (i)</i>
8(a)(i)	Translation $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$	2	B1 for each
8(a)(ii)	Enlargement (or reduction) [scale factor] $\frac{1}{2}$ oe [centre] (5, 1)	3	B1 for each
8(b)(i)	(2, -5)	1	
8(b)(ii)	8.6[0] or 8.602...	2	M1 for $5^2 + (-7)^2$ or $5^2 + 7^2$ or better
8(c)	$[y =] 3x + 8$	3	M1 for $\frac{11 - -1}{1 - -3} (m)$ oe M1 for substituting $(-3, -1)$ or $(1, 11)$ into $y = (\text{their } m)x + c$ oe
9(a)(i)	-7	1	
9(a)(ii)	4.084×10^6 cao	3	B2 for 4084 101 or $4.084\ 101 \times 10^6$ or $4.0841[0] \times 10^6$ or answer 4.08×10^6 or M1 for $(3 + 2 \times 9)^5$ If 0 scored, SC1 for <i>their</i> 5 or more figure answer in standard form and corrected to 4 sf. or for 4084000 seen
9(a)(iii)	$4x^2 + 12x + 10$ final answer	3	M1 for $(3 + 2x)^2 + 1$ B1 for $[(3 + 2x)^2 =] 9 + 6x + 6x + 4x^2$

Question	Answer	Marks	Partial Marks
9(a)(iv)	$\frac{x-3}{2}$ oe final answer	2	M1 for $x = 3 + 2y$ or $y - 3 = 2x$ or $\frac{y}{2} = \frac{3}{2} - x$
9(a)(v)	$-1 \leq h(x) \leq 32$	2	B1 for $-1 \leq h(x) \leq k$ or $k \leq h(x) \leq 32$ or -1 and 32 evaluated
9(b)(i)	500	2	M1 for $2x = 10^3$
9(b)(ii)	$\frac{10^x}{2}$ oe final answer	2	M1 for $2x = 10^y$ or $x = \log(2y)$
9(b)(iii)	$4x^3$ final answer	2	M1 for $[3\log(2x) =] \log(2x)^3$ oe
10	For all parts accept decimals or percentages with the usual rules for 3sf Do not penalise incorrect cancelling or converting Do not accept ratios or words		
10(a)(i)	$\frac{5}{12}$	1	
10(a)(ii)	25	1	FT $60 \times \text{their(a)(i)}$ but must be an integer
10(b)(i)	$\frac{1}{11}$ oe	2	M1 for $\frac{4}{12} \times \frac{3}{11}$
10(b)(ii)	$\frac{10}{33}$ oe	3	M2 for $\frac{5}{12} \times \frac{4}{11} \times 2$ oe or M1 for $\frac{5}{12} \times \frac{4}{11}$ oe If 0 scored, SC1 for $\frac{5}{18}$ oe
10(c)	$\frac{25}{72}$ oe	3	M2 for $\frac{5}{12} \times \frac{5}{12} + \frac{4}{12} \times \frac{4}{12} + \frac{3}{12} \times \frac{3}{12}$ oe or M1 for two of these products If 0 scored, SC1 for $\frac{19}{66}$ oe
11(a)	$\frac{3x+2y}{3x}$ or $1 + \frac{2y}{3x}$ final answer	3	B1 for $(3x+2y)(3x-2y)$ isw B1 for $3x(3x-2y)$ isw

Question	Answer	Marks	Partial Marks
11(b)(i)	$5(4 - x) - 7(2x - 3) = 2(2x - 3)(4 - x)$ or better	M1	Correctly clearing fractions
	$[(2x - 3)(4 - x)] = 8x - 12 - 2x^2 + 3x$ or $2[(2x - 3)(4 - x)] = 16x - 24 - 4x^2 + 6x$	B1	
	Leading to $4x^2 - 41x + 65 = 0$ with no errors or omissions	A1	
11(b)(ii)	$\frac{-(-41) \pm \sqrt{(-41)^2 - 4 \times 4 \times 65}}{2 \times 4}$ or sketch with both answers indicated	M2	M1 for $\sqrt{(-41)^2 - 4 \times 4 \times 65}$ or $\frac{-(-41) + \text{or} - \sqrt{p}}{2 \times 4}$ or suitable sketch which would lead to answers
	1.96 , 8.29 cao	B2	B1 for each or for 1.960... and 8.289 to 8.290



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CAMBRIDGE INTERNATIONAL MATHEMATICS**0607/43**

Paper 4 (Extended)

May/June 2024**MARK SCHEME**Maximum Mark: 120

Published

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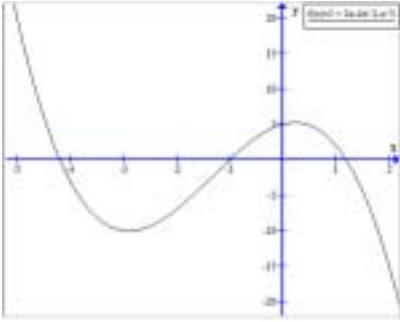
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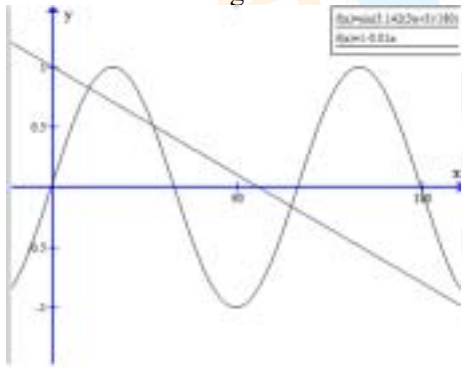
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isw	ignore subsequent working
nfww	not from wrong working
oe	or equivalent
rot	rounded or truncated
SC	Special Case
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	0	1	
1(a)(ii)	2	1	
1(a)(iii)	6	1	
1(a)(iv)	3	1	
1(a)(v)	2.175	2	M1 for $[0 \times 26] + 1 \times 20 + 2 \times 23 \dots$
1(b)	14	3	M2 for $\frac{[0 \times 18] + 1 \times 31 + 2 \times 27 + 3 \times 18 + 4 \times n + 5 \times 12 + 6 \times 5}{18 + 31 + 27 + 18 + n + 12 + 5} = 2.28$ oe or M1 for $[0 \times 18] + 1 \times 31 + 2 \times 27 + 3 \times 18 + 4 \times n + 5 \times 12 + 6 \times 5$ or for $(18 + 31 + 27 + 18 + n + 12 + 5) \times 2.28$
2(a)	2400	2	M1 for $\frac{3000}{5}$
2(b)	2.8	3	M2 for $\frac{3000 \times r \times 10}{100} = 840$ oe or M1 for $\frac{3000 \times r \times 10}{100}$ or B1 for [1 year interest] = 84
2(c)	2.6[0]	3	M2 for $\sqrt[8]{\frac{7367.67}{6000}}$ oe or M1 for $6000(\dots)^8 = 7367.67$
2(d)	23 cao	4	B3 for 22.7 or 22.72 to 22.73 OR M3 $n \log \left(1 + \frac{1.8}{100} \right) = \log \left(\frac{6000}{4000} \right)$ oe or good sketch indicating value between 22 and 23 or correct trials reaching 22 and 23 or M2 for $\left(1 + \frac{1.8}{100} \right)^n = \frac{6000}{4000}$ oe or suitable graph with $n > 1$ or at least 3 correct trials or M1 for $4000 \times \left(1 + \frac{1.8}{100} \right)^n = 6000$ oe soi by at least 2 correct trials with $n > 1$

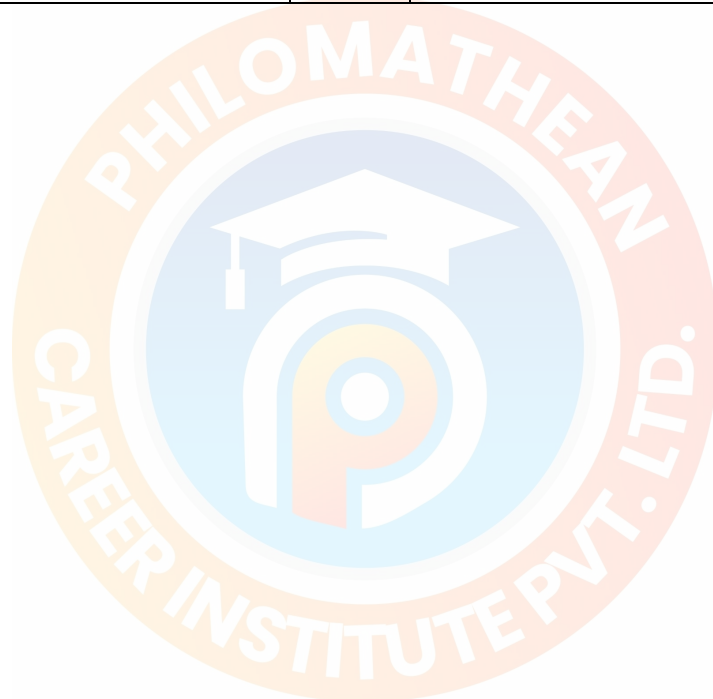
Question	Answer	Marks	Partial Marks
3	For all parts accept decimals or percentages with the usual rules for 3sf. Do not penalise incorrect cancelling or converting. Do not accept ratios or words		
3(a)(i)	$\frac{2}{3}$ oe	1	
3(a)(ii)	$\frac{1}{2}$ oe	1	
3(b)(i)	$\frac{1}{36}$ oe	2	M1 for $\frac{1}{6} \times \frac{1}{6}$
3(b)(ii)	$\frac{35}{36}$ oe	1	FT 1 – <i>their</i> (b)(i)
3(c)(i)	0.52 oe 0.72 oe 0.16 oe Correctly placed	2	B1 for one correctly placed
3(c)(ii)	0.4368 oe	2	M1 for <i>their</i> 0.52×0.84 oe
4(a)	$-\frac{2}{3}$ oe $\frac{8}{3}$ or $2\frac{2}{3}$ oe	3	B2 for one correct or M1 for correctly isolating y oe
4(b)	gradient = $\frac{3}{2}$	M1	FT 1 ÷ <i>their</i> $-\frac{2}{3}$
	substituting (2, 10) into $y = \text{their } m + c$	M1	FT <i>their</i> $m \neq -\frac{2}{3}$
	completing to $2y - 3x = 14$ with at least one line of working and no errors	A1	
4(c)	Correctly equating coefficients or sketch of one equation with positive slope and positive y -intercept	M1	If 0 scored, SC1 for correct answer with no working
	Correct method to eliminate one variable or sketch of other equation with negative slope and positive y -intercept	M1	
	$x = -2$ in correct answer space	A1	
	$y = 4$ in correct answer space	A1	

Question	Answer	Marks	Partial Marks
5(a)	Correct sketch 	2	With minimum in 3rd quadrant and maximum in 1st quadrant B1 for any cubic with negative x^3
5(b)	−4.19 or −4.193 to −4.192 −1 1.19 or 1.192 to 1.193	3	B1 for each or B1 for −1 and B1 for −4.2 and 1.2
5(c)	(−2.9[0], −10.1) or (−2.897 to −2.896, −10.05...)	2	B1 for each coordinate
5(d)	<i>their</i> −2.9[0] < a < 0.23[0]	2	−2.896 to 2.897, 0.2301... B1 for 0.23[0] seen or <i>their</i> −2.9[0] < a < k
5(e)	Integer ≤ -11 or ≥ 6	1	
6(a)(i)	Triangle at (1, −2), (4, −2), (3, −3)	2	B1 for reflection in $y = -x$ or correct size and orientation
6(a)(ii)	Rotation 90° clockwise oe [centre] (4, 1)	3	B1 for each
6(a)(iii)	Enlargement [Scale factor] −2 [centre] (−1, 3)	3	B1 for each
6(b)(i)	Translation $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$	2	B1 for each
6(b)(ii)	Stretch $y = 1$ [invariant] factor $\frac{1}{3}$	3	B1 for each
7(a)	$\pi d = 37 - 12 - 12$ oe	M1	
	4.137 to 4.138...	A1	

Question	Answer	Marks	Partial Marks
7(b)	40.3 or 40.4 or 40.33 to 40.36...	3	M2 for $2 \times \sin^{-1} \frac{4.14}{12}$ oe or $\cos ACB = \frac{12^2 + 12^2 - (2 \times 4.14)^2}{2 \times 12 \times 12}$ or M1 for $\sin(\dots) = \frac{4.14}{12}$ or $(2 \times 4.14)^2 = 12^2 + 12^2 - 2 \times 12 \times 12 \times \cos ACB$
7(c)	60[.0] or 60.1 or 60.01 to 60.13	3	M1 for $\frac{1}{2} \times 12 \times 12 \times \sin(\text{their } b)$ oe M1 for $2 \times \frac{1}{2} \times \pi \left(\frac{4.14}{2} \right)^2$ oe
7(d)	28.2 or 28.3 or 28.23 to 28.26	3	M2 for $37 \times \sqrt{\frac{35}{\text{their } c}}$ oe or M1 for $\sqrt{\frac{35}{\text{their } c}}$ or $\sqrt{\frac{\text{their } c}{35}}$ or $\frac{35}{\text{their } c} = \left(\frac{p}{37} \right)^2$
8(a)	$h^2 = 0.8^2 - 0.35^2 - 0.35^2$ oe	M3	M2 for $0.8^2 = 0.35^2 + 0.35^2 + h^2$ oe or M1 for $0.35^2 + 0.35^2$ or $0.7^2 + 0.7^2$ oe
	0.6284 to 0.6285	A1	
8(b)	165 or 165.0 to 165.2	5	B4 for 262 or 262.1 to 262.2... OR M1 for $6 \times 4.9 \times 4.9$ M1 for $\pi \times 1.2^2 \times 25$ M1 for $\frac{1}{3} \times 0.7 \times 0.7 \times 0.628$ [$\times 49$] M1 for at least 1 of <i>their</i> volumes $\times 0.63$
9(a)(i)	$-\frac{2}{7}$	2	M1 for $2x + 5x = 1 - 3$ or better
9a)(ii)	-5 -1	2	B1 for each or M1 for $x + 3 = \pm 2$
9(b)	$3x^2y^2(2x - y)$ Final answer	2	B1 for correctly extracting 2 or more factors
9(c)	$\frac{x-31}{(2x+3)(x-5)}$ or $\frac{x-31}{2x^2-7x-15}$ Final answer	3	B1 for $5(x-5) - 2(2x+3)$ oe ISW B1 for denominator $(2x+3)(x-5)$ oe

Question	Answer	Marks	Partial Marks
9(d)	$\frac{-3 \pm \sqrt{3^2 - 4 \times 2 \times -7}}{2 \times 2}$ or suitable sketch(es) with both answers indicated	M2	M1 for $\sqrt{3^2 - 4 \times 2 \times -7}$ or M1 for $\frac{-3 + \sqrt{p}}{2 \times 2}$ or $\frac{-3 - \sqrt{p}}{2 \times 2}$ Denominator must be shown as 2×2 to earn the second M1 but a denominator of 4 is condoned for M2
	1.27 and -2.77 cao	B1	
10(a)	$3\frac{1}{2}$ oe	1	
10(b)	6	2	M1 for $5 - \frac{1}{2}x = 2$
10(c)	$3\frac{1}{2} - 1\frac{1}{2}x$ or $\frac{7-3x}{2}$ oe Final answer	2	M1 for $5 - \frac{1}{2}(3(x+1))$ oe
10(d)	$\frac{x-3}{3}$ oe Final answer	2	M1 for $x = 3(y+1)$ or $x+1 = \frac{y}{3}$ or $y-3 = 3x$
10(e)	1	2	B1 for h(90) or M1 for $\sin(3(x+1))$ oe
10(f)	$\sin(3(x+1))$ soi	1	
	Correct sketches e.g. 	2	or a single graph of $h(g(x)) - 1 + 0.01x$ B1 for each graph
	17.5 or 17.52... 48.7 or 48.71... 115.9 or 115.94...	2	B1 for 1 correct.
11(a)	$[\cos B] = \frac{72^2 + 85^2 - 102^2}{2 \times 72 \times 85}$	M2	M1 for $102^2 = 72^2 + 85^2 - 2 \times 72 \times 85 \times \cos[B]$
	80.57...	A1	
11(b)	319.4	2	B1 for 40.6 or 139.4

Question	Answer	Marks	Partial Marks
11(c)	14 17	6	M2 for $\sin [A] = \frac{85 \sin 80.6}{102}$ or $\cos [A] = \frac{72^2 + 102^2 - 85^2}{2 \times 72 \times 102}$ or M1 for $\frac{85}{\sin A} = \frac{102}{\sin 80.6}$ or $85^2 = 72^2 + 102^2 - 2 \times 72 \times 102 \cos A$ M1 for $[AP =] 72 \cos \text{their } A$ oe M1 for $[\text{time} =] \text{their } AP \div 32$ M1 adding <i>their</i> time to 13 00 If 0 scored, SC1 for showing BP on diagram with right angle correctly placed on AC



MARK SCHEME for the October/November 2012 series

0607 CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/04

Paper 4 (Extended), maximum raw mark 120

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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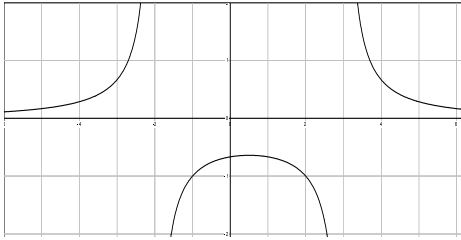
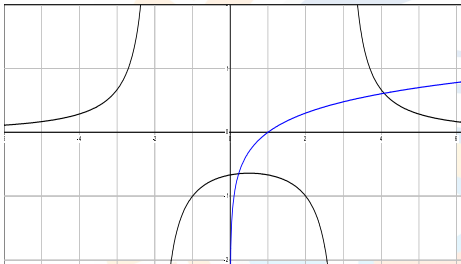
Page 2	Mark Scheme	Syllabus	Paper
	IGCSE – October/November 2012	0607	04

1	(a)	50	1	
	(b)	2	1	
	(c)	1.88 o.e.	1	Seen and not spoiled
	(d)	3	1	
	(e)	6	1	
	(f)	1	1	
2	(a) (i)	1.5 o.e.	1	
	(ii)	$200 \div (3 + 2) \times 3$ o.e.	1	If work backwards M1 for $3:2 = 120:80$ and for $120 + 80 = 200$, either order. Allow 5 for $3 + 2$.
	(iii)	129.6(0) final answer	2	M1 for $\frac{120 \times 4 \times 2}{100}$ o.e. (9.6)
	(iv)	86.44 (or 86.4(0) or 86.444 to 86.445)	2	M1 for $80(1.0395)^2$ o.e. not spoiled
	(v)	$1.0395^2 = 1.08056\dots$ i.e. 8.056.. interest > 8% o.e.	2	Any full and accurate explanation – will often use values from earlier working. Must compare interest with interest or amount with amount. If 0 , M1 for method but lacking accuracy or full details and methods may be seen in (iii) and/or (iv). Use of different principals 0 (unless finding interest or amount as percentage of each principal)
	(b) (i)	19 440	2	M1 for $24\,000 \times 0.9^2$ o.e. Allow 19 400 full marks
	(ii)	9	M1 A1	M1 for $24000 \times 0.9^n = 10000$ o.e. including repeated multiplication by 0.9 8.31 or 8.309... or 10330 to 10331 or 9298 imply M1 SC1 for answer 9 without working or without wrong working
3	(a) (i)	1947	4	M1 for $\frac{2}{3}\pi 4.8^3$, M1 for $\pi 4.8^2.23.7$, A1 for 1947. ... or 1950, B1 for <i>their</i> volume rounded to nearest cubic centimetre.
	(ii)	0.001947 (0.00195 or 0.001947...)	1 FT	FT <i>their</i> (i) $\div 100^3$
	(iii)	1.6[0] (1.596 to 1.599)	1 FT	FT <i>their</i> (ii) $\times 820$

Page 3	Mark Scheme	Syllabus	Paper
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(b)	1.40 www	5	M1 for $2\pi 4.8^2$ (144.7 to 144.8 or 145) M1 for $\pi 9.6 \times 23.7$ o.e. (714.7 to 714.9 or 715) M1 for $\pi 4.8^2$ (72.38 to 72.40) not subtracted M1 for $\times 0.15$ and $\div 100$ 1.4 or 1.397 to 1.400... implies M4 figs 14 or 1397 to 1400... or total surface area = 931.4 to 932.4 or 296.64π or 296.6... π or 297π implies M3
4 (a)	72	2	M1 for $360 \div \text{their}$ (180 – 175) (not 175 or negative) or for $\frac{180(n-2)}{n} = 175$ o.e.
(b) (i)	58	3	B1 for $x = 32$, M1 for $0.5(180 - 2 \text{ their } x)$ Allow on diagram
(ii)	Clear explanation using correct vocabulary supported by values in working or on diagram. allied o.e. angles not 180° , alternate angles not equal, corresponding angles not equal etc. e.g $74 + 96 \neq 180$, $74 \neq 64$ etc.	2 FT	FT x only B1 for values of angles being used stated or seen in diagram.
(c) (i)	75	1	Allow on diagram B1 for angle $CAB = 27$. Allow on diagram.
(ii)	12	3	B1 for angle OAB or angle $OBA = 15$. Allow on diagram
5 (a)	16.9 (16.87...)	2	M1 for $0.5 \times 7 \times 7.5 \sin 40$ Any other method must be complete Must see method if grads or radians used.
(b)	4.98 (4.981....)	3	M1 for $7^2 + 7.5^2 - 2 \times 7 \times 7.5 \cos 40$ A1 for 24.81 to 24.82... or 24.8 Must see method if grads or radians used.

Page 4	Mark Scheme	Syllabus	Paper
	IGCSE – October/November 2012	0607	04

6	(a)		5	B1 for branch approx to left of $x = -2$, correct shape B1 for branch approx to right of $x = 3$, correct shape B1 for branch approx between $x = -2$, $x = 3$ correct shape B1dependent if outside branches approach x-axis from above B1dependent if middle branch below x-axis Allow touching x-axis at ends Pen – 1 if branches joined
	(b)	$x = -2, x = 3, y = 0$	3	B1 B1 B1
	(c)	$y \leq -0.64$ $y > 0$	3	M1 for finding max point, implied by -0.64. condone $<$ Allow $f(x)$ or x for y and ignore inclusion of -2 and/or 2 condone $\geq$
	(d)	$y > 0$	1	Condone $\geq$
	(e) (i)		2	B1 for correct shape cutting x-axis B1dependent for nothing to left of y-axis
	(ii)	0.225 (0.2249 to 0.2250), 4.08 (4.078...)	2	B1 B1
	(iii)	4.08 (4.078...)	1 FT	B1 FT <i>their</i> relevant root from (e)(ii)
7	(a) (i)	$\in$	1	Lists can be in any order
	(ii)	$\subset$ or $\subseteq$	1	
	(iii)	ϕ or $\{ \}$	1	
	(iv)	$\cup$	1	
	(b) (i)	t, u, v, w, x	1	
	(ii)	t, w	1	
	(iii)	l, m	1	
	(iv)	n, t, u, w, y	1	

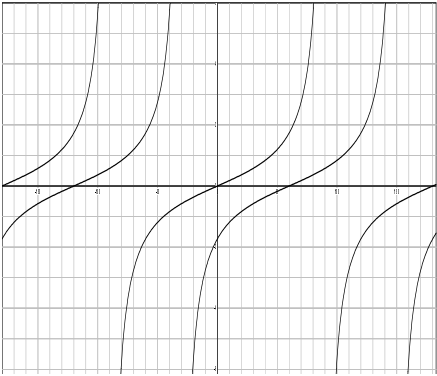
Page 5	Mark Scheme	Syllabus	Paper
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8	(a)		
	(i)		1
	(ii)		1
	(b)	(0.75, 1.75) o.e.	1
	(c)	0.375 o.e.	2 FT
	(d)	$y = -x + 2.5$ o.e. (e.g. $2x + 2y = 5$) cao	3 FT
			M1 for $0.5 \times (1) \times \text{their } 0.75$ o.e. FT their x-coordinate only FT their (b) B1 for gradient $= -1$, implied by $y = -x + c$ M1 for correct use of <i>their</i> (0.75, 1.75) in linear equation e.g. $\frac{y - \text{their } 1.75}{x - \text{their } 0.75} = -1$ or <i>their</i> 1.75 $= -1(\text{their } 0.75) + c$
9	(a)	330 (330.125, 330.1, 330.12, 330.13)	2
	(b)	4 correct widths Heights 0.065, 0.19, 1.66, 1.4	1 3
			B2 for 3 correct, B1 for 2 correct. Accuracy – touching line of 1.4 and $0.05 \leq h < 0.1$, $0.15 < h \leq 0.2$, $1.65 \leq h < 1.7$ i.e. only touching nearest horizontal line. Condone freehand If no diagram, SC2 for 4 correct frequency densities.
10	(a)	-4.37 ($-4.372\dots$), 1.37 ($1.372\dots$) or $\frac{-3 \pm \sqrt{33}}{2}$ o.e. Mark final answer	M1 B1B1
	(b)	$x \leq -4.37$ ($-4.372\dots$), $x \geq 1.37$ ($1.372\dots$)	2 FT
			Full method e.g. graph showing intersections with x-axis or full explicit formula correctly applied No working can only score B1B1 FT only if outside parts of a parabola. Condone $<$, $>$. Allow in words if clear. If B0, SC1 for region shown on sketch

Page 6	Mark Scheme	Syllabus	Paper
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11	(a)	19	2	B1 for $[g(2)] = 2^2 + 2 + 2$ soi e.g. $f(8)$
	(b)	$4x^2 + 14x + 14$ o.e. final answer	3	M1 for $(2x + 3)^2 + (2x + 3) + 2$ soi B1 for $(2x + 3)^2 = 4x^2 + 6x + 6x + 9$ soi
	(c)	$\frac{x-3}{2}$ o.e. final answer	2	M1 for swapping x and y or $y - 3 = 2x$ or $\frac{y}{2} = x + \frac{3}{2}$ i.e. correct first step
	(d) (i)	13	1	
	(ii)	-3	2	M1 for $2(2x + 3) + 3 = 2x + 3$ or $f(x) = x$ or $2x + 3 = x$
12	(a) (i)	Reflection only, $y = -x$ o.e.	2	Extra transformations invalidate all marks
	(ii)	Stretch only, y -axis o.e. invariant, (factor) 3	3	B1 B1 B1 Extra transformations invalidate all marks
	(b)	Correct rotation	2	SC1 for rotation clockwise 90° about other point or 90° anti-clockwise about $(1, -1)$
13	(a) (i)	$\frac{10}{x+3}$	1	
	(ii)	$\frac{10}{x+3} + \frac{4}{x} = 1$ o.e. $10x + 4(x+3) = x(x+3)$ or $10x + 4x + 12 = x^2 + 3x$ o.e. $x^2 - 11x - 12 = 0$	M1	Final equation reached with at least 1 intermediate step with brackets or 5 terms without any errors or omissions
			E2	E1 if one error or omission but still at least 1 intermediate step with brackets or 5 terms
	(b)	$(x-12)(x+1)$	2	SC1 for $(x+a)(x+b)$ where $ab = -12$ or $a+b = -11$ isw solutions
	(c)	40	2 FT	FT $10 \div (\text{a positive } x+3) \times 60$ but x from <i>their</i> factors . M1 for $10 \div (\text{a positive } x+3) \times 60$ but must be correct from (b) If two positive roots, allow either. If only negative roots M0

Page 7	Mark Scheme	Syllabus	Paper
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14 (a) (i)		2	Translated by approx 60° to right B1 for translation of middle branch approx. 60° to right
	(ii) Translation only $\begin{pmatrix} 60 \\ 0 \end{pmatrix}$ o.e.	B1 B1	B 's independent Allow in words e.g. $60^{(o)}$ to right
(b)	$-120^{(o)}, 60^{(o)}$ final answers	2	– 1 each incorrect extra but isw any answers outside domain SC1 for $(-120, \sqrt{3})$ and $(60, \sqrt{3})$ o.e.

MARK SCHEME for the October/November 2013 series

0607 CAMBRIDGE INTERNATIONAL MATHEMATICS

0607/04

Paper 4 (Extended), maximum raw mark 120

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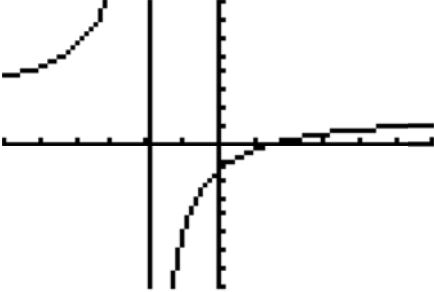
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Page 2	Mark Scheme	Syllabus	Paper
	IGCSE – October/November 2013	0607	04

1	(a) (i)	5272.65 (allow 5270, 5272 to 5273)	2	M1 for 8000×0.92^5 oe
	(ii)	4 (allow 3.31, 3.312 to 3.313) nfw	2	M1 for $8000 \times 0.92^n = 4000$ oe or SC1 for 9 or 8.31 or 8.312 to 8.313
	(b) (i)	72.3 (72.30 to 72.31)	2	M1 for $235 \div 3.25$ oe
	(ii)	8.38 (8.382 to 8.383)	1	
2	(a) (i)	Triangle at (1, -1), (4, -1), (4, -2)	2	SC1 for reflection in y-axis
	(ii)	Triangle at (-1, -1), (-1, -4), (-2, -4)	2 FT	FT SC case only SC1 for anti-clockwise rotation of 90° about (0, 0)
	(iii)	Reflection $y = -x$ oe	B1FT B1FT	FT the transformation FT full description B's independent but both marks lost if more than one transformation stated
	(b)	Enlargement (or reduction) (0, 2) [factor] 0.5	B1 B1 B1	B's independent but all 3 marks lost if more than one transformation stated No ratios
3	(a)	147 nfw	4	B3 for $[A =] 31.9$ to 32.1 nfw or M2 for $[\cos \text{ angle } A =]$ $\frac{346^2 + 493^2 - 271^2}{2 \times 346 \times 493}$ oe or M1 for correct implicit expression with angle A B1 FT 179 – their angle A
	(b)	4.52 (4.519 to 4.520)	3	M2 for $0.5 \times 4.93 \times 3.46 \times \sin(\text{their } A)$ oe e.g. $0.5 \times 493 \times 346 \times \sin(\text{their } A) \div 100^2$ or use of Hero's formula or M1 for scale correctly applied or correct use of $0.5ab \sin C$ or correct use of Hero's formula figs 4519 to 4520 imply M1

Page 3	Mark Scheme	Syllabus	Paper
	IGCSE – October/November 2013	0607	04

4	(a) (i) 7.21 (7.211..) or $2\sqrt{13}$ (ii) 653 (653.2 to 653.5...) or 208π (b) (i) 317.1 to 317.2... (ii) 185 (185.3 to 185.5)	3 2FT 2 3	M2 for $\sqrt{14^2 - 12^2}$ or M1 for $r^2 + 12^2 = 14^2$ oe FT <i>their</i> (a)(i) M1 for $\frac{1}{3}\pi(\text{their}(a)(i))^2(12)$ M1 for $\pi(\text{their}(a)(i))(14)$ M2 for $\frac{\text{their}(b)(i)}{\pi(14)^2} \times 360$ oe or M1 for $\frac{\text{their}(b)(i)}{\pi(14)^2}$ oe or correct implicit statement e.g. $\frac{x}{360} \times \pi \times 14^2 = 317$ or 317.1 to 317.2
5	(a) (i) 20 (ii) 16 (iii) 9 (iv) 29 (v) 180 (b) (i) 60, 50 (ii) 20.125 (or 20.1 or 20.12 to 20.13) (iii) 2.67 (2.666 to 2.667) oe 12 5	1 1 1 1 2 1, 1 2FT 1 1FT 1FT	M1 for 20 indicated e.g. on y-axis or SC1 for answer of 20 FT <i>their</i> (b)(i) only if answers add to 110 M1 for at least 3 mid-values seen or implied FT <i>their</i> (b)(i) FT <i>their</i> (b)(i)
6	(a)  (b) -1.5 oe (c) 1.5 oe (d) $x = -2, y = 2$	3 1 1 1, 1	M1 for reasonable rectangular hyperbola shape A1 for asymptotes approximately $x = -2$ and $y = 2$ (sol) A1 for x-intersection positive and y-intersection negative Do not allow co-ordinates Do not allow co-ordinates

Page 4	Mark Scheme	Syllabus	Paper
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(e)	$-1.5 \leq f(x) \leq 1.3$ oe	2	Strict inequality at either end or both ends scores only 1 Allow in words but “between -1.5 and 1.3 ” scores only 1 B1 for -1.5 and 1.3 seen or for $f(x) \geq -1.5$ or for $f(x) \leq 1.3$
(f) (i)	Reasonable $y = 3 - x$ added to sketch	1	
(ii)	-3.54 ($-3.541\dots$), 2.54 ($2.541\dots$)	1, 1	
(iii)	$2x - 3 = (x + 2)(3 - x)$ $[(x + 2)(3 - x)] = 3x - x^2 + 6 - 2x$ $x^2 + x - 9 = 0$	M1 B1 E1	Allow $2x - 3 = 3(x - 2) - x(x - 2)$ or $2x - 3 = x(3 - x) + 2(3 - x)$ Allow $x + 6 - x^2$ No errors or omissions
(iv)	37	2	M1 for $b^2 - 4ac = 1^2 - 4(1)(-9)$ seen or $(x + \frac{1}{2})^2 - \frac{1}{4} = 9$ or better
7 (a)	5.66 (5.656 to 5.657) or $4\sqrt{2}$	3	M2 for $\sqrt{(5-1)^2 + (6-2)^2}$ oe or better or M1 for $5 - 1$ and $6 - 2$ (or $2 - 6$) soi
(b)	$x + y = 7$ oe	3	M1 for gradient = $\frac{2-6}{5-1}$ oe M1 for using $(1, 6)$ or $(5, 2)$ in $y = mx + c$ oe
(c) (i)	$y = x$	2 FT	M1 for gradient = $\frac{-1}{\text{their gradient in (b)}}$
(ii)	$(3.5, 3.5)$ oe cao	1	
8 (a)	$25 - 4n$ oe	2	M1 for answer of $-4n + c$
(b)	$3 \times 2^{n-1}$ oe	2	M1 for 3×2^q seen and with no other terms
(c)	$\frac{n^2}{n+3}$ oe	2	B1 for fraction with either numerator or denominator correct
(d)	$n^3 - n$ oe	4	M3 for comparing sequence with values of n^3 or $an^3 + bn^2 + cn + d$ with 4 values of n substituted correctly oe or M2 for attempting cubic expression oe or listing values of n^3 or M1 for reaching equal third differences

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9	(a)	$\frac{5}{6}, \frac{1}{6}, \frac{9}{10}, \frac{1}{10}, \frac{3}{10}, \frac{7}{10}$ oe all correctly placed	3	B1 for each pair correctly placed
	(b)	$\frac{48}{60}$ oe ($\frac{16}{20}$, 0.8 etc.)	3	isw any cancelling or converting M2 for $\frac{5}{6} \times \frac{9}{10} + \frac{1}{6} \times \frac{3}{10}$ or M1 for one of the products by itself
	(c)	Fine weather but Alex does not go to the beach	1	
10	(a)	$x + 3x + 6x = 180$ or $10x = 180$ $x = 18$ angles in the same segment oe	B1 B1 B1	Allow angles subtended by the same arc or same chord
	(b) (i)	similar	1	No alternatives
	(ii)	3[.00] or 2.990 to 3.002	2	M1 for $\frac{8.55}{9.23} = \frac{2.78}{BX}$ oe allow s.f = 1.08 or 1.079 to 1.080
	(iii)	0.86	2	M1 for $\left(\frac{8.55}{9.23}\right)^2$ oe (implied by 0.857 to 0.859 or 1.16 to 1.17) or $\frac{0.5 \times 2.78 \times 8.55 \sin 54}{0.5 \times \text{their } BX \times 9.23 \sin 54}$ $\left(\frac{9.61476}{11.2008}\right)$
11	(a)		2	M1 for shape A1 for through (1, 0) and positive y-values approx. double those on log x graph

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(b)	$\log(x^5) = \log(16)$ or $x^5 = 16$ or $\log x^3 = \log\left(\frac{16}{x^2}\right)$ or $x^3 = \frac{16}{x^2}$ or appropriate sketch	M2	M1 for using a rule of logarithms once correctly
	1.74 (1.741...) or $\sqrt[5]{16}$ or $2^{0.8}$ oe	B1	
(c)	$y \log 5 = \log 100$ or $y = \log_5 100$ or $\frac{\log 100}{\log 5}$ or sketch 2.861	M1	e.g. for sketch $y = 5^x$ with $y = 100$
		B2	B1 for 2.86 or 2.8613 to 2.8614
12 (a)	$10x^2 + \frac{1}{2}\pi x^2$ oe final answer	2	B1 for $10x^2$ or $\frac{1}{2}\pi x^2$ seen
(b)	$A = x^2(10 + \frac{1}{2}\pi)$ or $2A = x^2(20 + \pi)$	3	M1 for correctly taking x^2 as a factor from two terms, one containing π
	$x^2 = \frac{A}{10 + \frac{1}{2}\pi}$ or $\frac{2A}{20 + \pi}$ $\sqrt{\frac{A}{10 + \frac{1}{2}\pi}}$ or $\sqrt{\frac{2A}{20 + \pi}}$ final answer		M1 for correct division by other factor which has two terms and no x in it M1 for correct square root to give x
(c)	4.16 (4.157 to 4.158) cao	B1	
13 (a) (i)	$(2x+1)(x-1)$	2	SC1 for $(ax+1)(bx-1)$ where $ab = 2$ or $b-a = -1$ or for answer $x = -\frac{1}{2}$, $x = 1$ but only from factors
(ii)	$\frac{8x+5}{(2x+1)(x-1)}$ oe final answer	3	B2 for $8x+5$ seen or M1 for $x-1+4(2x^2-x-1)$ or better seen e.g. $1+4(2x+1)$ B1 for denominator $(2x+1)(x-1)$ oe in final answer
	$\frac{p-5q}{1-t}$ oe nfw final answer	4	B1 for $(p+5q)(p-5q)$ B2 for $(p+5q)(1-t)$ or B1 for $p+5q-t(p+5q)$ or $p(1-t)+5q(1-t)$